TCS CORPORATE WEBSITE

# Architecture Guide

| | |
|---|---|
| **Role / audience** | Architecture — Solution architects, technical leads, developers, AI agents |
| **Version** | 1.0 |
| **Status** | Draft for review |
| **Owner** | Solution Architect (proposed) |
| **Last validated** | 2026-11-08 |
| **Documents** | 11 |
| **Classification** | INTERNAL — classification to be confirmed |

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# Architecture Guide

| Attribute | Value |
|---|---|
| Document ID | ARC-000 |
| Audience | Solution architects, technical leads, developers, AI agents |
| Purpose | Entry point of the architecture package: the system, its delivery streams, the implementation status of every capability and where each architecture view is described |
| Scope | The TCS Corporate Website on AEM Sites as a Cloud Service with Universal Editor (xwalk) and Edge Delivery Services; the repository sradha-tcs/tcs-corp (main) |
| Prerequisites | None |
| Owner | Solution Architect (proposed) |
| Approver | Product Owner (to be confirmed) |
| Version | 1.0 |
| Status | Draft for review |
| Last updated | 8 November 2026 |
| Last validated | 8 November 2026 |
| Source files | AGENTS.md, scripts/scripts.js, utils/site.js, paths.json, helix-query.yaml, component-models.json |
| Related documents | Solution architecture, Content architecture, Technical architecture, Integration architecture, Security architecture, Deployment architecture, Performance architecture, Localization architecture, Architecture decisions, Risks and open items |
| Related AEM paths | /content/tcs-corp/language-masters/en, /content/tcs-corp/language-master/en, /content/tcs-corp/design-block-master/en, /content/tcs-corp/us/en … /content/tcs-corp/kr/ko-kr |

## 1. System context

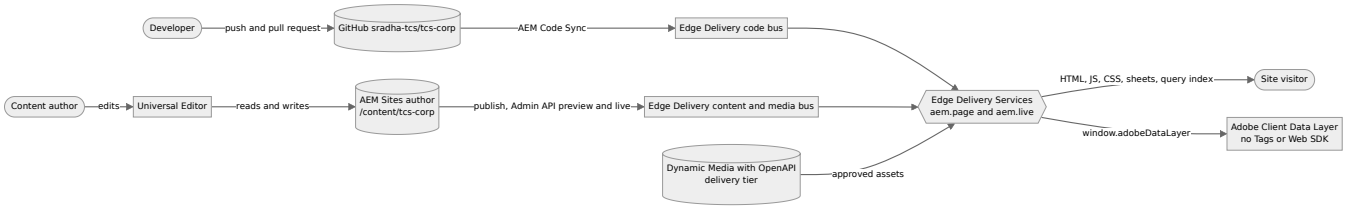



*Diagram: system context (source: diagrams/system-context.mmd).*

## 2. Delivery streams

| Stream | Source | Delivered by | Environments |
|---|---|---|---|
| Code | GitHub sradha-tcs/tcs-corp (feature branches, main) | AEM Code Sync | <branch>--tcs-corp--sradha-tcs.aem.page / .aem.live |

| Stream | Source | Delivered by | Environments |
|---|---|---|---|
| Content | AEM Sites author /content/tcs-corp (Universal Editor) | Publish from AEM (technical account) or the Admin API | main--tcs-corp--sradha-tcs.aem.page (preview), .aem.live (live) |
| Configuration | Site configuration (Config Service: public.json paths, metadata source), fstab.yaml at onboarding only | Admin API /config | all branches |
| Assets | AEM Assets of the Sites environment (/content/dam/tcs-corp) | Media bus (images published with pages) or Dynamic Media with OpenAPI (approved assets, KB-D-067) | delivery tier of the environment |

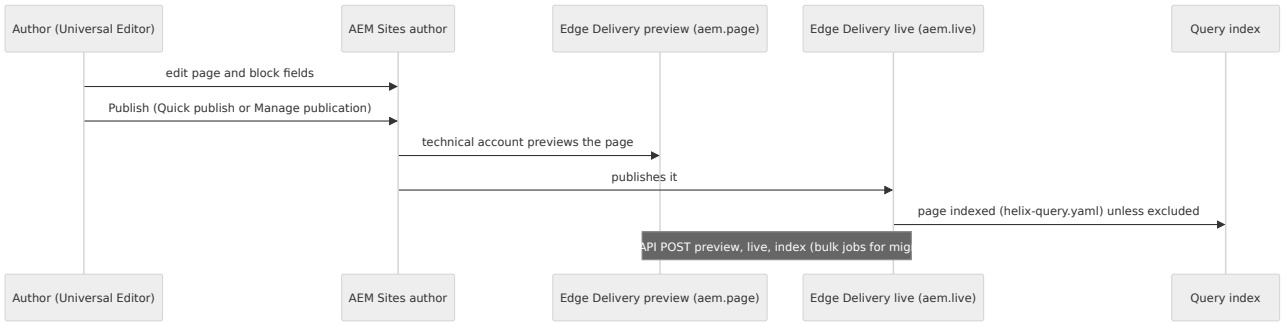



*Diagram: authoring to delivery flow (source: diagrams/delivery-flow.mmd).*

## 3. Site roots

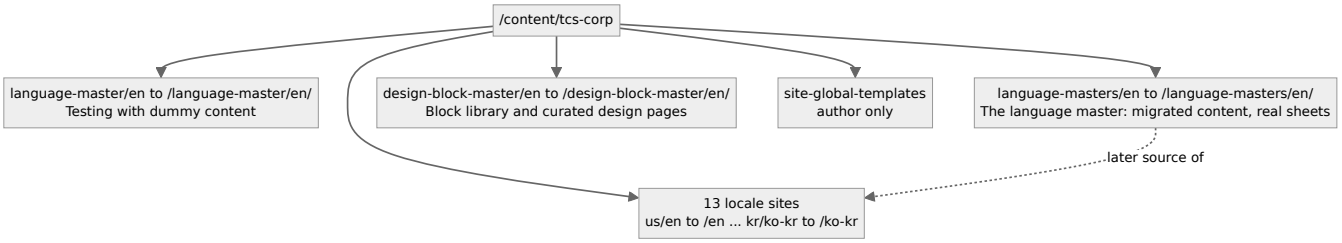



*Diagram: site roots under /content/tcs-corp (source: diagrams/site-roots.mmd).*

## 4. Implementation status

Status vocabulary: Implemented, Partially implemented, Configured but not validated, Planned, Not implemented, Blocked, To be confirmed.

| Capability | Status | Evidence |
|---|---|---|
| Universal Editor authoring (xwalk) of 32 blocks and 5 default-content components | Implemented | component-models.json, component-definition.json, component-filters.json |
| Edge Delivery Services rendering, three-phase loading, module preloading | Implemented | scripts/scripts.js, scripts/module-graph.js, head.html |
| 13 locale sites by path | Implemented (code and path mapping); content Partially implemented | utils/site.js LOCALES, paths.json |
| Three review roots /language-master/en, /language-masters/en, /design-block-master/en (KB-D-068) | Implemented | utils/site.js REVIEW_ROOTS, paths.json, site configuration |

| Capability | Status | Evidence |
|---|---|---|
| Query index and listing (List, Filter List, Cards from metadata) | Implemented | helix-query.yaml, utils/query-index.js, blocks/list, blocks/filter-list |
| Placeholders, taxonomy, data sheets, redirects sheets per locale | Implemented (reading); redirects Not applied by the platform | utils/placeholders.js, utils/taxonomy.js, utils/data.js; redirects need a CDN rule (KB-D-055, on hold) |
| Adobe Client Data Layer (page, components, interactions) | Implemented | decorators/data-layer-decorator.js, utils/analytics.js |
| Adobe Tags, Web SDK, Analytics, CJA, Target, consent platform | Not implemented | decision log; AGENTS.md |
| Dynamic Media with OpenAPI delivery of all assets (KB-D-067) | Partially implemented: rendering Implemented, page rewrite Configured but not validated, asset picker Planned (PR #200 open) | decorators/responsive-image-decorator.js, utils/media.js, aem-sites.sh on the delivery branch |
| Separate AEM Assets environment (KB-D-064) | Not implemented (not adopted) | decision log KB-D-064 |
| Forms (Form Variations) submission back end | Not implemented (no back end connected) | blocks/form-variations; finding F-01 |
| Personalization | Not implemented | docs: personalization (knowledge base) |
| CDN redirects / root redirect / to /en/ | Planned (on hold) | KB-D-055 |
| Production custom domain and Adobe Managed CDN | To be confirmed | not recorded in the repository |
| Monitoring and alerting beyond Edge Delivery RUM | To be confirmed | not recorded in the repository |

## 5. Architecture views


- Solution architecture
- Content architecture
- Technical architecture
- Integration architecture
- Security architecture
- Deployment architecture
- Performance architecture
- Localization architecture
- Architecture decisions
- Risks and open items


## 6. Page rendering lifecycle

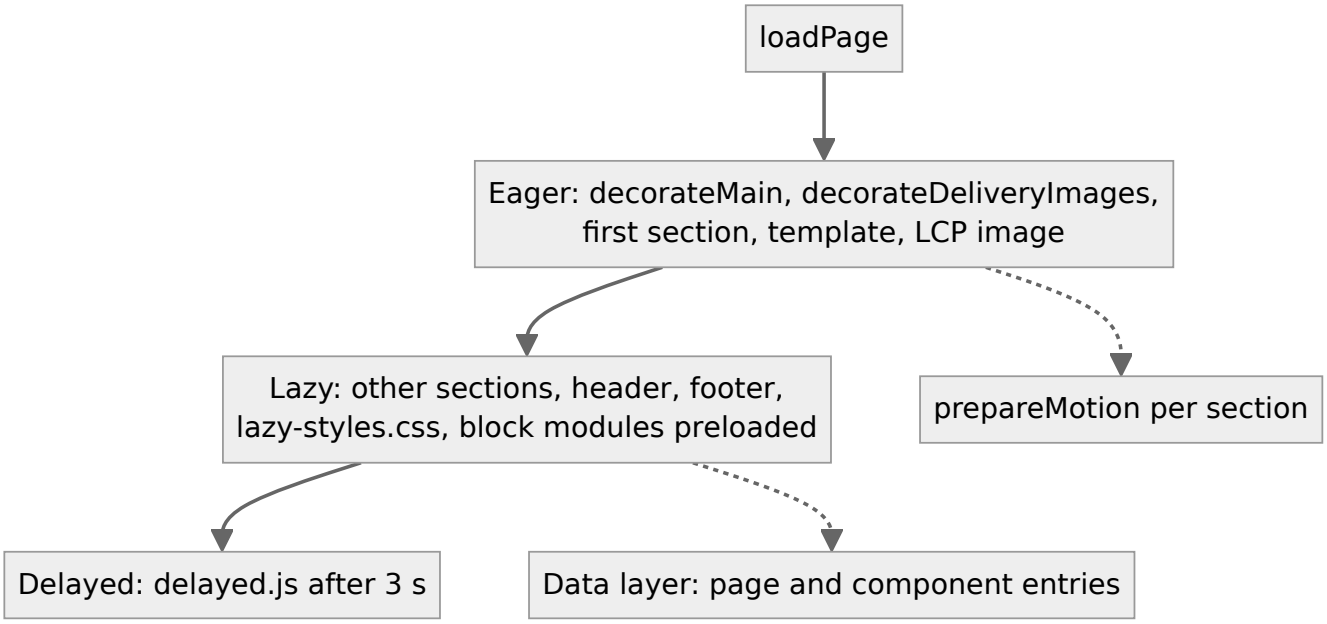



*Diagram: three-phase page loading (source: diagrams/page-loading.mmd).*

## 7. Screenshots

No screenshot belongs to this document (it describes code, configuration or process). Evidence screenshots are listed in the screenshot register.

## 8. Known limitations

- Production domain, CDN and monitoring are not recorded in the repository (To be confirmed).

## 9. Open items

- Confirm the production domain and CDN (owner: Solution Architect, to be confirmed).
- Approve or reject the asset picker PR #200 after the Extension Manager set-up.

## 10. Change history

| Version | Date | Author | Change |
|---|---|---|---|
| 1.0 | 8 November 2026 | Delivery team (documentation program) | First version of the role-specific documentation (master documentation prompt), generated from the repository and the published reference pages |

# Architecture Decisions

| Attribute | Value |
|---|---|
| Document ID | ARC-009 |
| Audience | Solution architects, technical leads, developers, AI agents |
| Purpose | The decision log (KB-D-xxx) as maintained in the code repository knowledge base |
| Scope | — |
| Prerequisites | — |
| Owner | Solution Architect (proposed) |
| Approver | Product Owner (to be confirmed) |
| Version | 1.0 |
| Status | Draft for review |
| Last updated | 8 November 2026 |
| Last validated | 8 November 2026 |
| Source files | docs/handover/knowledge-base/decision-log.md |
| Related documents | — |
| Related AEM paths | — |

The authoritative decision log is docs/handover/knowledge-base/decision-log.md (code repository); this copy is generated from it.

## 1. How to Use This Log

- **Status:** Approved (implemented and recorded), Proposed (open, not implemented), Superseded (replaced by a later entry, named in the row). Entries derived by the knowledge base authors, not decided by the project, are listed separately under Recommendations (not approved).
- **Date:** the date of the handover revision or the merge date of the pull request that recorded the decision; "Not recorded" when the source has none.
- **References:** "rev 2.xx" is a row of the handover's Revision History, "F-xx" a finding and "O-xx" an open item of its Part 16, "PR #nn" a merged pull request of `sradha-tcs/tcs-corp`.
- **Adding a decision:** append the next id (`KB-D-0nn`), status Proposed until the user approves it; when it replaces an entry, set the old one to Superseded and name the new id. Never rewrite an Approved entry's history.

## 2. Site, Content and Configuration

| Decision ID | Date | Area | Context | Decision | Alternatives | Impact | References | Status |
|---|---|---|---|---|---|---|---|---|
| KB-D-001 | 11 October 2026 | Code structure | Blocks repeated buttons, links, headings, images and property readers | Shared behaviour lives in single-purpose modules: `decorators/` build or decorate elements, `utils/` build no UI; decorators may import utils, never the reverse | Per-block implementations (replaced) | Blocks import shared modules; new shared behaviour goes to these folders with their READMEs | PR #23; AGENTS.md "Shared code"; decorators/README.md, utils/README.md | Approved |
| KB-D-002 | 11 October 2026 | Site data | TCS Custom Properties held site data outside the standard Edge Delivery sources | Site data comes from standard sources: page properties and bulk metadata, placeholders, taxonomy and per-locale data sheets | TCS Custom Properties (replaced) | `utils/metadata.js`, `placeholders.js`, `taxonomy.js`, `data.js`; all UI text from placeholders | PR #24; AGENTS.md "Site structure and site data" | Approved |
| KB-D-003 | 11 October 2026 | Site structure | 13 markets and languages | 13 locale sites (`/content/tcs-corp/<country>/<lang>`), each with references, error pages, block library, sheets and data, published at language-only paths (`/en`, `/de-de`, ...); Site Global Templates selected by Page Type | Not recorded | `utils/site.js` LOCALES; templates in `templates/` | PR #25, PR #26; AGENTS.md; templates/README.md | Approved |
| KB-D-004 | 11 October 2026 | Configuration | Locale mappings and the code's locale list could drift | `paths.json` is generated from LOCALES (`npm run build:paths`); every locale sheet and the taxonomy page are mapped | A hand-maintained `paths.json` (replaced) | After a locale change: regenerate, commit both, update the site configuration's `public.json` paths (open item O-04) | PR #22, PR #27; tools/paths/generate-paths.mjs | Approved |
| KB-D-005 | 11 October 2026 | Redirects | The platform applies only a root `/redirects.json` | Redirects are kept per locale (`/{locale}/redirects.json`), by the user's choice; List links follow them, the platform doesn't apply them | A single root `/redirects.json` (applied by the platform) | Redirects, including `/` → `/en/`, need another mechanism such as CDN rules (F-36, O-03) | AGENTS.md "Site structure and site data" (added in the commit of PR #27); handover 3.8 | Approved |
| KB-D-006 | 11 October 2026 | Navigation | Sections of the site need their own header and footer | Navigation, Footer and 404 Page are inherited like page properties: the page's value, else the nearest parent page's up to Home, else the locale's `references/nav` / `references/footer` / `error-pages/404`; cached per session | Not recorded | `utils/inherited-metadata.js`; changes show in a new session (F-35) | PR #29; AGENTS.md | Approved |
| KB-D-007 | 12 October 2026 | Site structure | The locale root needed a Home page | Each locale page is published as its Home (`/de-de/` serves `/de-de/index`) and `paths.json` stays in sync with the site configuration | Not recorded | Folder mappings before Home mappings in `paths.json` | PR #32 | Approved |
| KB-D-008 | 12 October 2026 | SEO | No production domain is configured | The sitemap uses the aem.live address as its origin until a production domain is configured | Not recorded | Replace the origin in `helix-sitemap.yaml` at go-live (O-03) | PR #33; helix-sitemap.yaml | Approved |

| Decision ID | Date | Area | Context | Decision | Alternatives | Impact | References | Status |
|---|---|---|---|---|---|---|---|---|
| KB-D-009 | Not recorded | List | Lists could be built from Content Fragments | Content Fragment lists aren't supported in List: lists are built from pages (query index) or List Items | Content Fragment lists | Use pages with metadata, or Cards | Handover 16.4 Not Implemented, 6.16 | Approved |
| KB-D-010 | 17 October 2026 | Language master | The language master had to be reviewed before its rollout to /en | Temporary review publication at `/language-master/en/`: its own lists, references, sheets, inheritance up to its root page (`REVIEW_ROOTS`, `REVIEW_SHEETS`, `REVIEW_PAGES` in `utils/site.js`; `REVIEW` in `tools/paths/generate-paths.mjs`) | Not recorded | Temporary: must be removed after the rollout: unpublish, remove the code and paths, regenerate `paths.json`, re-apply the configuration (F-88) | rev 2.11, rev 2.14; F-88, F-91; PR #91–#102 | Approved |
| KB-D-011 | 20 October 2026 | Block library | Block library pages are test content | Block library pages stay out of the query index (by design): the locale block libraries are excluded from publishing in `paths.json` and from the index in `helix-query.yaml` (`/*/block-library`); the language master's library is published for the review with noindex (rev 2.20) | Not recorded | Lists don't show library pages; the index pattern doesn't match `/language-master/en/block-library` (requires validation) | rev 2.20, rev 2.35; helix-query.yaml, paths.json | Approved |

## 3. Authoring Model and Universal Editor

| Decision ID | Date | Area | Context | Decision | Alternatives | Impact | References | Status |
|---|---|---|---|---|---|---|---|---|
| KB-D-012 | 13 October 2026 | Additional tab | Authors needed anchors, hiding and spacing on blocks | Every block model except header and footer ends with an Additional tab (Anchor ID, Hide block, Padding / Margin as `classes_*` fields) applied by `decorators/block-options-decorator.js` | Not recorded | Shared classes in `styles/styles.css`; copy the tab into new blocks | PR #61; AGENTS.md "Additional tab" | Approved |
| KB-D-013 | 17 October 2026 | Universal Editor dialogs | Universal Editor collapses tabs that don't fit on one line into a dropdown (List, Filter List, Separator, Video and others did) | At most three short tabs per dialog, the last one Additional; fields and stored values unchanged | The earlier 4–7 tab dialogs (replaced) | The tab names in the handover's Part 6 are those before 2.8 (F-84, F-107) | rev 2.8; F-84; PR #88 | Approved |

| Decision ID | Date | Area | Context | Decision | Alternatives | Impact | References | Status |
|---|---|---|---|---|---|---|---|---|
| KB-D-065 | 8 November 2026 | Language master path | The language master under review lived at `/content/tcs-corp/language-master/en`, published at `/language-master/en/` (REVIEW_ROOTS, `paths.json`, `helix-query.yaml`); the product owner moved it to the AEM convention `language-masters` | The language master is `/content/tcs-corp/language-masters/en`, published at **`/language-masters/en/`**: `REVIEW_ROOTS = ['/language-masters/en']` (`utils/site.js`), the `paths.json` mappings (`npm run build:paths`), the query index exclusions; the migration's page packages 1.3.0 are below the new root. Transition (product owner, 8 November 2026): both roots are served until the tree is moved and edam-main is merged to main: `REVIEW_ROOTS` lists `/language-master/en` and `/language-masters/en`, `paths.json` and the site configuration map both, the query index excludes the references, data and block library of both; the old root is removed afterwards | Keep `language-master` | On author: move the tree (Sites console > Move, references adjusted); the site configuration's `public.json` paths must be updated by an administrator; the pages published at `/language-master/en/` stay live until unpublished (redirects: on hold, redirects-cdn.md) | `utils/site.js`, `paths.json`, `helix-query.yaml`, delivery branch `migrated-page-content` (`consolidated/packages-1.3.0-RENAME-REPORT.md`) | Approved (product owner, 8 November 2026) |
| KB-D-066 | 8 November 2026 | Project documents | The project documents (the handover Markdown, PDF and its 593 screenshots, the author / developer / admin guides, the design proposals, the audit, animation and block documentation reports) lived in the code repository under `docs/` (88 MB, 698 files) | Approved (product owner): they move to the branch `migrated-page-content-edam` under `docs/` (same paths, https://github.com/sradha-tcs/tcs-corp/tree/migrated-page-content-edam/docs); only this knowledge base stays in the code (AGENTS.md makes it the starting point of every task). Links in the code repository point to the branch; the block READMEs embed the screenshots from it (`tools/docs/handover-screenshots.json`, refreshed with `HANDOVER=<handover file> npm run build:docs`); handover revisions are made on the branch | Keep everything in `docs/`; move the knowledge base too | Smaller code repository; the handover and reports are versioned with the migration deliverables; a documentation change in a code PR names the branch change | AGENTS.md, README.md | Approved (product owner, 8 November 2026) |

| Decision ID | Date | Area | Context | Decision | Alternatives | Impact | References | Status |
|---|---|---|---|---|---|---|---|---|
| KB-D-068 | 8 November 2026 | Site roots | Two language master roots were served during the move (KB-D-065); the product owner gave each tree a purpose and added a third | Approved (product owner): three review roots, each a site of its own with its references, error pages and sheets: `/content/tcs-corp/language-master/en` (testing: page functionality with dummy content, content limitations), `/content/tcs-corp/language-masters/en` (the language master: the migrated tcs.com content, validated with its real sheet data; later the source of the global site and the regions and languages, MSM or separate content to be decided once the content is final, as tcs.com's regions have no language master content today) and `/content/tcs-corp/design-block-master/en` (the block library and the curated design pages); `REVIEW_ROOTS`, `paths.json` (48 mappings) and the site configuration map all three, the query index leaves out their references, data and block libraries; the block READMEs' library examples and usage refer to `/design-block-master/en` | One root (KB-D-065) | Content, tests and library validations kept apart; the old root is no longer removed | AGENTS.md, content-architecture.md | Approved (product owner, 8 November 2026) |
| KB-D-069 | 8 November 2026 | Documentation | The project documentation was one consolidated handover (Markdown 2.53, PDF 2.44) plus guides and reports on the documents branch | Approved (product owner): a role-specific documentation set generated from the repository and the published reference pages (master documentation prompt) replaces it in `docs/handover/` of `migrated-page-content-edam`: 01 architecture, 02 development (a folder per block and curated page), 03 authoring (a guide per component), 04 business analysis (inferred EPICs, features, stories), 05 QA (test cases, execution results with evidence), 06 operations, 07 administration and governance (Dynamic Media set-up), 08 release and DevOps, 09 knowledge base (humans, agents, LLM retrieval, machine-readable), 10 training, 11 audit and remediation; Markdown with paired PDFs; its generator in `docs-tools/` on that branch; every code change updates it; the knowledge base of this repository stays the source for code work; the former documents are kept in the tag `archive/docs-before-role-docs-2026-11-08` | Keep the consolidated handover | Each role reads its own documents; documentation follows the code through the generator; the handover PDF no longer lags | AGENTS.md, README.md | Approved (product owner, 8 November 2026) |
| KB-D-014 | 17 October 2026 | List, Filter List | Universal Editor didn't save `multi: true` aem-content fields (`pages`, `featuredPages`) | Pages as composite multifields (`pageItems`, `featuredItems`, a Page per item) | Multi-value aem-content fields (still read) | Superseded for new content by item children (KB-D-015); the multifields stay in the dialogs and are still read | rev 2.9; F-86; PR #89 | Superseded (KB-D-015) |

| Decision ID | Date | Area | Context | Decision | Alternatives | Impact | References | Status |
|---|---|---|---|---|---|---|---|---|
| KB-D-015 | 17 October 2026 | Content model | Composite multifields written by content packages weren't rendered on published pages (CTAs, stats, links, filters empty) | Repeating content is item children added with Add (List Items, Cards, `hero-cta`, `page-header-cta`, `teaser-link`, `teaser-stat`, `testimonial-stat`, `anchor-list-item`, `filter-list-tag`, `filter-list-category`, `filter-list-page`, `footer-link`); `itemRows` (`utils/cells.js`) finds them | Composite multifields (`container`, `multi: true`), still read for content authored before | New repeating fields must be item models; tests in `test/items/` | rev 2.10; F-87; PR #90; AGENTS.md "Item children" | Approved |
| KB-D-016 | 17 October 2026 | Editor experience | Items just added had no content and seemed not to appear | In Universal Editor an item without content is a dashed placeholder (`itemEditorHint`, `decorators/editor-item-decorator.js`); on AEM author (no query index) lists say their pages show on the published page | Not recorded | Placeholders `itemEditorHint`, `listEditorPages`, `filterListEditor` | rev 2.8, rev 2.10, rev 2.12; F-85, F-89 | Approved |
| KB-D-017 | 19 October 2026 | Delivery rules | The code and local preview assumed markup AEM doesn't deliver (forms empty, tables and accordions with empty rows, no video poster) | Code and the library preview assume what AEM really delivers: tab rows, `name, classes` first cells, no grouped fields in key-value blocks, collapse suffixes, template values stored | The earlier assumptions (replaced) | Rules in AGENTS.md; options AEM doesn't deliver removed (Columns Additional tab, Teaser personalization, Hero and Table List Motion Type, Quote / Testimonial Style, Separator Auto / Custom) | rev 2.27; F-101; PR #118; docs/audit/BLOCK-ECOSYSTEM-AUDIT.md | Approved |
| KB-D-018 | 20 October 2026 | Delivery rules | The re-audit found item names lower-cased, sample text on every new item, Lottie alt as link text, DAM JSON / video 404s | Case-sensitive values get their own field (Form Variations `fieldName`); no sample content in templates or model defaults; non-image DAM assets are published as an operations step | Not recorded | Content migrations and the publication step (package 1.0.9) | rev 2.29; F-102; PR #120; docs/audit/REAUDIT-2026-10-20.md | Approved |

| Decision ID | Date | Area | Context | Decision | Alternatives | Impact | References | Status |
|---|---|---|---|---|---|---|---|---|
| KB-D-019 | 20 October 2026 | Section containers | AEM doesn't deliver a section's `classes_*` fields, so the Additional tab of Tabs Container, Columns Container and Gen Carousel had no effect | Their Additional tab fields are `container*` (delivered as data attributes) applied by `scripts.js` `applySectionBlockOptions` | `classes_*` on sections (not delivered) | `.eslintrc.js` `max-cells` limits per container | rev 2.30; F-102; PR #121; AGENTS.md | Approved |
| KB-D-020 | 20 October 2026 | Field naming | `hideTitle` would collapse into the title (Title suffix rule) | The List Item's hide toggle is `titleHidden`, the name the Teaser uses; values that must be read avoid the Text, Title, Type, Alt and MimeType suffixes (`buttonVariant`) | `hideTitle` | Same naming in List, Teaser and Testimonial models | rev 2.34; F-105; PR #125 | Approved |

## 4. Components

| Decision ID | Date | Area | Context | Decision | Alternatives | Impact | References | Status |
|---|---|---|---|---|---|---|---|---|
| KB-D-021 | 13 October 2026 | Teaser, Hero | The "Lotto" image type is a Lottie animation | The Lottie player (lottie-web light) is vendored in `scripts/lib/lottie` and loaded once per page by `utils/lottie.js`, with a pause / play button | Not recorded | No external script dependency; Lottie JSON must be published as a DAM asset (KB-D-018) | PR #52; utils/lottie.js | Approved |
| KB-D-022 | 14 October 2026 | Cards | Authors needed hand-made cards with the List's look | The Cards block is the manual authoring alternative to List: each Card holds its own content, with the List's presentation and shared code (`models/_card-list.json`, `utils/card-list-config.js`, `decorators/card-list-decorator.js`, `styles/card-list.css`) | Not recorded | Cards made before the redesign have no block model and changed look (F-34, F-41, F-42; proposal KB-D-046...051) | PR #62; rev 1.1; blocks/cards/README.md | Approved |
| KB-D-023 | 16 October 2026 | Header | Figma's desktop header is a logo bar | Desktop (>= 1200px) shows the Figma logo bar only (TCS left, TATA right), a product decision; desktop navigation comes from Conversation Canvas blocks; tablet and mobile open the menu full screen one level at a time | Show the tablet menu on desktop by changing the 1200px breakpoint (recorded in F-72 as the way to revisit) | A page without a Canvas has no desktop navigation (F-72, P2) | rev 2.2; F-64, F-72; PR #81 | Approved |

| Decision ID | Date | Area | Context | Decision | Alternatives | Impact | References | Status |
|---|---|---|---|---|---|---|---|---|
| KB-D-024 | 16 October 2026 | Conversation Canvas | Several Canvases per page (header, body, footer) floated independently | One floating journey for all Canvases of a page (`canvas-journey.js`, pure and unit tested; `canvas-controller.js` applies it); the header announces its menu with `header:menu` and the journey hides while it is open; header and Canvas don't import each other | The PR's "Don't float" boolean (replaced by the Floating journey field) | No Canvas logic in `header.js` or `footer.js` | rev 2.3; F-61 to F-63, F-73; PR #71; AGENTS.md "Floating Canvas" | Approved |
| KB-D-025 | 16 October 2026 | Header | `/en/home` named an unpublished Navigation page and showed no header | The header falls back to the locale's `references/nav` when the configured Navigation page can't be loaded | Not recorded | Header always renders | rev 2.6; F-80; PR #83 | Approved |
| KB-D-026 | 17 October 2026 | List, Cards | The option class `teaser` collided with the Teaser block's styles | Display as teaser adds `display-teaser`; option classes must not be block names | `teaser` (replaced) | Lists next to a Teaser render | rev 2.7; F-83; PR #87 | Approved |
| KB-D-027 | 18 October 2026 | List | Pages needed their own card look | Card style and Card colour precedence: the List Item's value, else the List's (other than Mixed / Default), else the page's `card-style` / `card-theme` metadata, else automatic | Not recorded | New bulk metadata columns indexed as `cardStyle`, `cardTheme` | rev 2.16; PR #104; blocks/list/README.md | Approved |
| KB-D-028 | 19 October 2026 | Columns Container | Edge Delivery has no blocks inside blocks; the Columns block holds default content only | Columns are a section container (Columns Container, like Tabs Container and Gen Carousel) whose columns hold default content and any block except Page Header, Footer and TCS Header; Columns per row (Automatic, 1–4) | Blocks inside the Columns block (not possible) | New content uses the container; the Columns block stays for existing content | rev 2.22, rev 2.23; PR #112, PR #113; blocks/columns-container/README.md | Approved |
| KB-D-029 | 19 October 2026 | Columns Container | AEM renders a block with the class of its template name: the "Column" item came out as `column` | The column item's template name is "Column Break"; columns are plain wrappers, so editor updates of a component in a column work in place; `column` blocks authored before are read as Column Breaks | "Column" (replaced) | `test/models/block-names.test.mjs` checks template names against block folders | rev 2.24; F-99; PR #114 | Approved |
| KB-D-030 | 19 October 2026 | Columns Container | Column widths only applied to rows of two columns | Column widths removed: columns are of equal width | Column widths (rev 2.22, removed) | Simpler Layout tab | rev 2.26; PR #117 | Approved |
| KB-D-031 | 19 October 2026 | Button | AEM's built-in Button delivers only its link and Primary / Secondary (as bold / italic): Views, Icon, Alignment and Tertiary never reached the page | The Button authors add is the Button Block (`blocks/button-block`, title "Button", template name "Button Block"), a key-value block whose every field is delivered (`buttonVariant` for the type) | The built-in Button (now "Button (basic)") | Buttons authored before keep Primary / Secondary | rev 2.26; F-100; PR #117; AGENTS.md "Button" | Approved |

| Decision ID | Date | Area | Context | Decision | Alternatives | Impact | References | Status |
|---|---|---|---|---|---|---|---|---|
| KB-D-032 | 20 October 2026 | Columns, Button | The built-in Columns and Button were still offered next to their replacements | The built-in Columns and Button are no longer offered in any filter; their definitions stay so content authored before remains editable; inside a Tabs Container a tab's Column Breaks lay its content out in columns (Tab field Columns per row) | Keep offering both | Library content migrated (package 1.0.8); `blocks/columns` is Compatibility-only | rev 2.28; PR #119; blocks/columns/README.md | Approved |
| KB-D-033 | 20 October 2026 | Teaser | Title Type changed only the level, Title size the size; BG Color added padding; Image position Right only from 900px | One Title Type sets level and size (Title size removed); Image position Left / Right / Top / Bottom, side by side from 600px; BG Color is only the colour (spacing from the Additional tab); Buttons side by side; List arrow on every link | Title size and the BG Color padding (removed) | Content action: Teasers that relied on Title size or the padding (F-103) | rev 2.32; F-103; PR #123 | Approved |
| KB-D-034 | 20 October 2026 | Backgrounds | Section offered only Highlight (grey); dark bands needed a block with its own colour | Section Style is one choice of Default, Grey, Dark, Dark Green; Teaser BG Color, Columns Container `containerBgColor` and Section Style share labels, classes (`bg-grey`, `bg-dark`, `bg-dark-green`) and tokens and set only the colour; Highlight no longer offered | Highlight only (F-21; sections authored with it keep their grey band) | Shared token `--color-bg-dark-green` | rev 2.32, rev 2.33; F-21, F-104; PR #124; AGENTS.md "Backgrounds and theme" | Approved |
| KB-D-035 | 20 October 2026 | Theme | Content inside dark backgrounds could be unreadable (contrast down to 1.05:1) | Theme contract of the dark Section and Columns Container backgrounds: light text, link and focus tokens, underlined links in running text, inverted buttons; Grey Teasers and Columns Containers inside reset the tokens | Not recorded | New blocks use the semantic tokens so they follow | rev 2.34; F-105; PR #125; styles/styles.css | Approved |
| KB-D-036 | 20 October 2026 | List Item | The request's "Link Item" needed per-item display control | The "Link Item" is the List Item (decided with the product owner); its Display tab adds Hide Title (`titleHidden`), Show Description, Show Image, Show Eyebrow, Show Label, Show Action Label (on by default) | Not recorded | Items authored before are unchanged | rev 2.34; F-105; PR #125; docs/audit/DIALOG-OPTIONS-2026-10-20.md | Approved |
| KB-D-037 | 19 October 2026 | Accessibility | Secondary text, brand text and fill tokens failed WCAG 1.4.3 | Contrast tokens changed to pass WCAG 1.4.3, subject to design sign-off | The earlier tokens (F-27, F-28) | Design sign-off is open (O-14) | rev 2.27; F-28; O-14 | Approved |

## 5. Animation, Performance and Loading

| Decision ID | Date | Area | Context | Decision | Alternatives | Impact | References | Status |
|---|---|---|---|---|---|---|---|---|
| KB-D-038 | 17 October 2026 | Page loading | The footer appeared late on throttled phones | The footer loads alongside the main sections (hidden until they have loaded); the header loads nav and placeholders in parallel; shared list data starts with the lazy phase | Loading the footer after all sections (replaced) | Footer 1.2–2.8 s earlier on a throttled phone (as recorded) | rev 2.13; F-90; PR #100 | Approved |

| Decision ID | Date | Area | Context | Decision | Alternatives | Impact | References | Status |
|---|---|---|---|---|---|---|---|---|
| KB-D-039 | 18 October 2026 | Header layout shift | The header and its Canvas pushed the page down when they arrived (CLS 0.14–0.25) | The header reserves its space until it loads (`--header-reserved` or the height remembered per breakpoint, `utils/header-space.js`); from rev 2.21 the reserve is the logo bar and the header Canvas row lies on the bottom of the hero (two zones) | Not recorded | CLS below 0.01 (as recorded) | rev 2.15, rev 2.21; F-82, F-93, F-98; PR #103, PR #92 | Approved |
| KB-D-040 | 18 October 2026 | Animation | Authors needed block motion without hurting loading | Additional tab Animation Type (content) and Background animation, Motion Speed and Animation Trigger (rev 2.19); run once in view, never in Universal Editor edit mode or for reduced motion; Preview plays them (rev 2.20); `utils/motion.js`, `decorators/motion-decorator.js`, `styles/motion.css` | Hero and Table List Motion Type (removed from the dialogs in rev 2.27; values authored before play as the new animations) | Background animation delays the LCP image above the fold (F-94) | rev 2.18–2.20; F-94 to F-97; PR #106–#108; docs/animation/README.md | Approved |
| KB-D-041 | 18 October 2026 | Skeletons | Query index data held back sections | Blocks built from the query index render through `renderWhenLoaded`: data later than 150ms shows a block-shaped skeleton | Waiting for the data | Small shift below the block when content is taller (F-95, accepted) | rev 2.18; F-95; decorators/skeleton-decorator.js | Approved |
| KB-D-042 | 20 October 2026 | Module loading | Without a build step, modules loaded level by level (up to 4 round trips before the first section, up to 7 more per block) | Module preloading: `head.html` modulepreload links for the `scripts.js` modules and `preloadBlockModules` from the generated `scripts/module-graph.js` (`npm run build:modules`, checked by `npm test`); no forced layouts while decorating | Not recorded (the project keeps "no build step") | Every import change requires `npm run build:modules` | rev 2.36; F-106; PR #127; AGENTS.md "Module preloading" | Approved |
| KB-D-043 | 20 October 2026 | Fonts | Text reflowed when Inter replaced an unscaled system font on Android / Linux (CLS 0.33) | Fallback faces scaled to Inter per system font (`inter-fallback`, `inter-fallback-roboto`, `inter-fallback-dejavu`) | A single Arial-based fallback (replaced) | Keep the faces when changing the font | rev 2.36; F-106; PR #128; styles/styles.css | Approved |

## 6. Documentation and Process

| Decision ID | Date | Area | Context | Decision | Alternatives | Impact | References | Status |
|---|---|---|---|---|---|---|---|---|
| KB-D-044 | 21 October 2026 | Block documentation | The handover's component chapters lagged behind the code | Every block folder has a README in the structure of `blocks/README.md`; its dialog fields, screenshots, files and library usage are generated (`npm run build:docs`) and validated by `npm test`; the READMEs are current where Part 6 differs | Part 6 only | Part 6 needs an alignment pass (F-107, open) | rev 2.37; F-107; PR #129; docs/blocks/BLOCK-DOCUMENTATION-REPORT.md | Approved |
| KB-D-045 | Not recorded | Publishing | Code changes must be reviewable on real content | Feature branch, PR with a feature preview test URL, `gh pr checks` (lint, Code Sync, PageSpeed), human review and merge; Code Sync deploys `main` | Not recorded | PRs without a test URL fail the PageSpeed check | AGENTS.md "Publishing Process"; handover 12.2, 13.15 | Approved |

# 7. Proposed Decisions

The Cards and Card refactoring proposal (14 October 2026, "For review — decisions D-1 to D-6 open") lists six product owner decisions; the proposal's recommendation is quoted, nothing is implemented.

| Decision ID | Date | Area | Context | Decision | Alternatives | Impact | References | Status |
|---|---|---|---|---|---|---|---|---|
| KB-D-046 | 14 October 2026 | Cards (D-1) | Whether Universal Editor offers Card in the + of a new key-value Cards block (open item O-20) | Proposed: confirm by test before Phase 0 | Not recorded | Blocks Phase 0 of the proposal | cards-card-refactoring.md section 21 | Proposed |
| KB-D-047 | 14 October 2026 | Cards (D-2) | Cards made before the redesign changed look (F-41) | Proposed: restore the original look (Classic) for those blocks, Figma design for new blocks | Move them to the Figma design | Content migration of G1 instances | Same, section 21; F-41 | Proposed |
| KB-D-048 | 14 October 2026 | Cards (D-3) | Overlap with List and Filter List | Proposed: dynamic sources in Cards through shared code; review in 6 months whether List can be retired | Keep Cards manual only | Overlapping listing blocks | Same, section 21 | Proposed |
| KB-D-049 | 14 October 2026 | Cards (D-4) | The Card item's cell limit | Proposed: raise the Card cell limit to 22 | Merge the tracking attributes into one field | `.eslintrc.js max-cells` | Same, section 21 | Proposed |
| KB-D-050 | 14 October 2026 | Cards (D-5) | Personalization of cards | Proposed: attributes now; behaviour when Target is connected | Hide / show by audience | Requires a personalization architecture decision (none implemented) | Same, section 21 | Proposed |

| Decision ID | Date | Area | Context | Decision | Alternatives | Impact | References | Status |
|---|---|---|---|---|---|---|---|---|
| KB-D-051 | 14 October 2026 | Cards (D-6) | Filtering in Cards | Proposed: client-side filtering in Cards for up to 50 cards; Filter List beyond | Filter List for every filtered set | Not recorded | Same, section 21 | Proposed |
| KB-D-052 | 22 October 2026 | Header over the first block | KB-D-039 (the header reserves its space, CLS below 0.01) assumes the layout is known before the header loads; an author option on the TCS Header (on the nav page, loaded in the lazy phase) isn't | Proposed: the TCS Header's Header over the first block (Auto: over an opening light Hero, as before; Always: over any first block; Never: the page starts below the header), at every width (desktop only at first), as <html data-header-overlay>; remembered per browser (utils/header-space.js, header-overlay-v1) and applied before the header loads on the next pages, as KB-D-039's header height | A page metadata column read in the head (no first-page shift, set per page); automatic only (the Hero rule) | A visitor's first page whose mode isn't Auto shifts once when the header arrives (measured locally: CLS 0.086 for Never over a light Hero, 0 on the next page); a block other than the Hero needs its own top space under the logos (Padding top, --header-overlay-space) | blocks/tcs-header, styles/styles.css, blocks/hero, templates/home | Proposed |
| KB-D-053 | 22 October 2026 | Conversation Canvas journey options | KB-D-024's Floating journey field offered Floats and docks, Docks only and Decorative for every Canvas Style: on Primary the field did nothing or hid the dock for good, and Decorative let the floating row pass over a Canvas in place (two rows on screen) | Proposed: three behaviours only, Primary (sticky: floats from the top of the page to the footer), Floats and docks, Stays in place (nofloat, the former Docks only; decorative is read the same way); Floating journey and Quick actions shown only for Default / Secondary, Navigation style and BG Style only for Primary (dialog conditions); while a row floats it is the only Canvas row in use (the others are inert placeholders), with none floating every row is in use; Canvases with Hide block or not shown aren't active | Keep Decorative as a separate option | No existing content uses Docks only, Decorative or Primary on /en (scan of the published pages and the block library, 22 October 2026) | blocks/conversation-canvas; canvas-journey.js focused; README "Floating journey: the three behaviours" | Proposed |

| Decision ID | Date | Area | Context | Decision | Alternatives | Impact | References | Status |
|---|---|---|---|---|---|---|---|---|
| KB-D-0054 | 22 October 2026 | Media contacts | The Press Release template appended "Media contacts" (Media Contacts sheet) at the end of every press release in the browser: not in the content tree, not authorable, always after the last section | Proposed (requested by the user): a **Media Contacts** block (`blocks/media-contacts`: Title, Title Type, Additional tab) renders the sheet's contacts where it is authored, grouped by designation and region; the template only adds the Page Header. `mailto:` / `tel:` clicks are recorded in the data layer by kind (email / phone), never with the address (`utils/privacy.js` `contactLinkKind`) | Keep the template section and add only the grouping; the template inserts the block when none is authored | Press releases that relied on the template show no contacts until the block is added (the block library press release template page, the Site Global Templates press-release page in AEM); `decorators/contact-list-decorator.js` now exports `buildContactGroups` (replaces `buildContactList`) | blocks/media-contacts, templates/press-release, decorators/contact-list-decorator.js, decorators/data-layer-decorator.js | Proposed |
| KB-D-0058 | 23 October 2026 | List, List Item | KB-D-036 gave the List Item Show Action Label (on by default) on top of the List's Show call to action, and Hide Title kept the action label; authors asked for the call to action to be decided per List Item only | Proposed (requested by the user): the List has no Show call to action (removed from its dialog; a value saved on Lists authored before is ignored); a List Item shows its call to action (`p.list-card-action`, the Call to action label, else Explore / Download) only with its own Show Action Label on (off by default), each item independently, rendered or left out by the existing card list code (`showsCta` in `utils/card-list-config.js`); with Hide Title on and no call to action the title stays visible as the link. The Cards block keeps its Show call to action. List Item rows delivered without their tab cells (19 or 13 cells) are read field by field | Keep the List toggle with an item override | Existing Lists lose their calls to action until each List Item's Show Action Label is turned on; cards from the query index (Child pages, Tagged pages) have no List Item and show none | Changes KB-D-036; blocks/list/README.md; branch `feature/list-item-action-label-refactor` | Proposed |

| Decision ID | Date | Area | Context | Decision | Alternatives | Impact | References | Status |
|---|---|---|---|---|---|---|---|---|
| KB-D-059 | 29 October 2026 | List Show call to action: Child pages with Panel cards | KB-D-058 removed the List's Show call to action, so cards from the query index (no List Item) never showed a call to action; the request: a Show call to action for Child pages with Panel cards only | Proposed (requested by the user): the List's own Show call to action is in the dialog only for List source Child pages, Display as teaser on and Card style Panel, and shows or hides Explore on every fetched card; for every other source and card style a value saved before isn't read and the cards keep their call to action (`ctaToggleApplies`); List Items keep KB-D-058 (their Show Action Label alone decides); the Cards block keeps its toggle | Keep KB-D-058's no call to action on the other query-index cards | Query-index cards other than Child pages + Panel show their default call to action again (Explore on panels, Download on reports), whatever was saved | blocks/list (_list.json, list-data.js), utils/card-list-config.js | Proposed |
| KB-D-060 | 29 October 2026 | List Item Show Action Label on by default | KB-D-058 made a List Item's Show Action Label off by default, so a newly added List Item (Fixed list, e.g. Panel cards) showed no Explore call to action until the author turned it on | Proposed (requested by the user): Show Action Label (List Item Display tab) is on by default (model value `true`; `itemDisplay` reads it on unless authored `false`); Off renders no action element and no default label; each item decides for its own card only, the List's Show call to action never overrides it (KB-D-058, KB-D-059 otherwise unchanged). The toggle is on every List Item: Universal Editor item conditions can't read the List's source or Card style, and List Items render only in a Fixed list | Keep KB-D-058's off by default | List Items never changed (no value stored) show their call to action again: Explore on panels, Download on reports, the authored label otherwise; items with Show Action Label authored off are unchanged | blocks/list (_list.json, list-data.js) | Proposed |

| Decision ID | Date | Area | Context | Decision | Alternatives | Impact | References | Status |
|---|---|---|---|---|---|---|---|---|
| KB-D-061 | 29 October 2026 | Form Variations Additional tab | Commit 9aae6f7 ("block changes", 27 October 2026) rewrote Form Variations and removed its Additional tab fields: Hide block, Padding / Margin and the animation fields (Animation Type, Motion Speed, Animation Trigger, Background animation). AGENTS.md requires the Additional tab on every block but header and footer, and `test/motion/motion.test.mjs` expects the animation fields on Form Variations (2 failing tests since) | Proposed: restore the Additional tab of Form Variations (the fields of the other blocks, Background animation included: the block has an authored background) | Keep the block without the tab: then AGENTS.md, the Animation section, `CONTENT_TARGETS` / `BACKGROUNDS` and the motion tests drop Form Variations | Restoring: forms get Hide block, spacing and animation again; keeping: the rule gets an exception | 9aae6f7; test/motion | Proposed |
| KB-D-055 | 24 October 2026 | Redirects at the CDN | The platform applies only a root `/redirects.json`; the redirects are kept per locale (KB-D-005), so nothing redirects: `/` returns 404 (F-36) and the old tcs.com URLs of the 3,191 migrated pages have no locale prefix | Proposed: the production CDN applies the rules of redirects-cdn.md: `/` → `/en/`, the Redirects sheet rows of the path's locale (read from the published sheets, refreshed every 5 minutes), old paths without a locale → the default locale + the path in lower case; decisions R-D1 (302 until go-live), R-D2 (`/` always `/en/`), R-D3 (pattern rule only for the migrated paths), R-D4 (at the CDN) open | A root `/redirects.json` (exact paths only, replaces the per-locale sheets: a change to KB-D-005); redirects in the browser from the 404 page (no 301 for search engines) | Authors keep maintaining one sheet per locale; the CDN Owner implements the rules once; O-03 (domain, CDN) must be resolved first | F-36; O-03; handover 3.8 (rev 2.44) | Proposed |

| Decision ID | Date | Area | Context | Decision | Alternatives | Impact | References | Status |
|---|---|---|---|---|---|---|---|---|
| KB-D-056 | 24 October 2026 | Author host in configuration files | AGENTS.md allows no AEM program / environment ids or author hosts in any file; two files need the author host to work: `fstab.yaml` (the content source: preview and publish read the pages from it) and `.migration/project.json` (managed by the workspace tooling, which can't be changed or untracked) | Proposed: both files are a recorded exception to the rule; no other file may hold the host or the ids (the handover since 2.43, F-112; the block README and knowledge base validators) | Move the content source to the site configuration (`content.source`) and remove `fstab.yaml` (needs the Site Administrator and a test of preview and publish); keep the rule without exceptions (not possible while the files are needed) | AGENTS.md's privacy rule gets the exception once approved | F-112; rev 2.44 | Proposed |
| KB-D-062 | 3 November 2026 | Hero Secondary CTA colours in the CTA List | hero.css styled the Hero's Secondary CTAs from block classes (secondary-bg-light-grey, secondary-bg-black-40, secondary-border-black, secondary-border-white) that no field set; a Style tab select for the whole Hero was proposed, but the colours are wanted per CTA in the CTA List. This conflicts with the AGENTS.md rule "item children instead of composite multifields" (the CTA List is a composite multifield kept for Heroes authored before; CTA items are the current way to add CTAs) | Proposed: the CTA List gets Primary CTA BG Color (`ctaPrimaryBackground`: Blue-cyan (azure), #216AC3, `primary-bg-azure`, Primary only; no Primary border colour), Secondary CTA BG Color (`ctaBackground`), Secondary CTA Border color (`ctaBorder`) and, for every CTA view, CTA Text color (`ctaTextColor`: Default (Black) `cta-text-black`, the field default, or White `cta-text-white`: the chosen colour applies whatever the CTA view and background; Figma: white text on a black background), labelled "(Secondary only)" and always shown (conditions inside multi containers aren't supported by Universal Editor: `var` resolves absolutely, not per row); `hero.js` applies them only to a Secondary CTA, treats their values as CTA values and adds them to that CTA's button as classes, `hero.css` styles the button; no Style tab fields | The same fields on the CTA item (`hero-cta`, follows the rule); a Style tab select for all Secondary CTAs of the Hero | Only CTAs in the CTA List can have the colours, CTA items can't; composite multifields didn't render from content packages (AGENTS.md); authors see the two fields on Default and Primary CTAs too, where they have no effect | blocks/hero/README.md, blocks/hero/hero.js, blocks/hero/hero.css, blocks/hero/_hero.json | Proposed |

| Decision ID | Date | Area | Context | Decision | Alternatives | Impact | References | Status |
|---|---|---|---|---|---|---|---|---|
| KB-D-063 | 29 October 2026 | Image Collage block | The Teaser has a Layout Image collage with Collage image items (`teaser-collage-image`, PR #167); a separate Image Collage block (`blocks/image-collage`, items `image-collage-item`) was requested for the same design. AGENTS.md: reuse before create (extend an existing block) | Proposed: add the Image Collage block next to the Teaser layout; decide later which one authors use and deprecate the other (content using it would move) | Keep only the Teaser layout; keep only the new block and remove the Teaser layout | Two ways to author the same design until one is retired; the code and styles are duplicated | blocks/image-collage/README.md, blocks/teaser/README.md | Proposed |
| KB-D-064 | 8 November 2026 | Assets (EDAM) | The site's assets live in the Sites author's own DAM (`/content/dam/tcs-corp/`, "AEM Assets: asset picker; images on the media bus", architecture.md); the product owner asked to use a separate AEM Assets environment (EDAM) for the project, including the migrated pages' assets (338 asset packages, 22 GB) | Proposed: a separate AEM Assets as a Cloud Service environment is the source of all site assets (rendering of delivery URLs in the code: `decorateDeliveryImages`, PR "EDAM delivery images"); authors pick from it with the Universal Editor Custom Content Advisor; pages store Dynamic Media with OpenAPI delivery URLs (images through the Edge Delivery media bus); the migration uploads the assets there as EDAM zips with a Metadata Import file (content packages cannot be installed on EDAM) and rebuilds the page packages (1.3.0) with the delivery URLs (E-D1 answered 8 November 2026: Dynamic Media with OpenAPI is licensed). **Decided 8 November 2026 (product owner): not adopted.** The delivery host given for E-D2 is the Sites environment's own delivery tier: there is no separate AEM Assets environment. The assets stay in the Sites author's DAM and the pages reference them by DAM path (the alternative "Keep the Sites author's DAM"): the migration installs `consolidated/packages/` (asset packages 1.2.0, page packages 1.3.0) with `install-packages.*`; no approval, asset ID export, delivery URL rewrite or Custom Content Advisor (the asset picker PR #195 closed unmerged). The rendering of delivery URLs (PR #193) stays: it changes nothing for DAM paths and renders a delivery URL an author pastes; E-D3 to E-D5 no longer apply | Keep the Sites author's DAM (today); Dynamic Media on the Sites author | New integration: Custom Content Advisor extension and configuration, 20 asset fields in 15 models, decorators for delivery URLs, sheet image columns; order: tags and EDAM zips with metadata on EDAM, approval, asset ID export, page packages 1.3.0 on Sites | remote-assets-edam.md | Not adopted (product owner, 8 November 2026) |

## 8. Recommendations (not approved)

Recommendations recorded in the handover (Part 16 recommendations, 11.7, 12.3) and those derived for this knowledge base. None is a decision: each needs an owner's approval and a decision log entry before implementation.

| Recommendation | Source | Status |
|---|---|---|
| Integrate Adobe Tags with the Web SDK and the ACDL extension in `scripts/delayed.js`, with consent | Handover 11.7; F-06; O-07 (blocked: measurement plan, Tags property, datastream, consent platform) | Recommendation (not approved) |
| Run `npm test` in the Build workflow; tag releases; replace the boilerplate values of `package.json`, the PR template and the Sidekick project name | F-37; handover 12.3 | Recommendation (not approved) |
| Configure a CDN or root redirect for `/` → `/en/` and apply the per-locale redirects sheets | F-36; O-03 | Specified in redirects-cdn.md (KB-D-055, Proposed) |
| Add hreflang alternates and a language switcher | F-39 | Recommendation (not approved) |
| Connect a form back end before publishing any form, or hide forms | F-01; O-08 | Recommendation (not approved) |
| Align the handover's Part 6 component chapters with the block READMEs | F-107 | Recommendation (not approved) |
| Reference agent-instructions.md from the root AGENTS.md and place copilot-instructions.md in `.github/` | Derived for this knowledge base | Recommendation (not approved) |
| Remove the language master review code and paths after the rollout to /en (KB-D-010) | F-88 | Recommendation (not approved): planned, the timing isn't recorded |

## 9. Related

- Knowledge base index
- Agent guidelines
- Agent instructions (for AGENTS.md)
- Copilot instructions
- Repository inventory
- Topic files: architecture, content architecture, Universal Editor, personalization, deployment and release
- Project handover (Revision History; Part 16 — Handover Findings and Open Items Register)
- AGENTS.md, docs/proposals/README.md, docs/audit/README.md

## 10. Screenshots

No screenshot belongs to this document (it describes code, configuration or process). Evidence screenshots are listed in the screenshot register.

## 11. Known limitations

- None recorded for this document.

## 12. Open items

- None.

## 13. Change history

| Version | Date | Author | Change |
|---|---|---|---|
| 1.0 | 8 November 2026 | Delivery team (documentation program) | First version of the role-specific documentation (master documentation prompt), generated from the repository and the published reference pages |

# Architecture Risks and Open Items

| Attribute | Value |
|---|---|
| Document ID | ARC-010 |
| Audience | Solution architects, technical leads, developers, AI agents |
| Purpose | Risks and open architecture items with their status |
| Scope | — |
| Prerequisites | — |
| Owner | Solution Architect (proposed) |
| Approver | Product Owner (to be confirmed) |
| Version | 1.0 |
| Status | Draft for review |
| Last updated | 8 November 2026 |
| Last validated | 8 November 2026 |
| Source files | docs/handover/knowledge-base/decision-log.md, docs/handover/knowledge-base/reports/discrepancy-report.md |
| Related documents | — |
| Related AEM paths | — |

## 1. Open architecture items

| ID | Item | Status | Owner (proposed) |
|---|---|---|---|
| AR-01 | Redirects: per-locale sheets aren't applied by the platform; / to /en/ needs a CDN rule (KB-D-055) | Planned (on hold) | Solution Architect |
| AR-02 | Dynamic Media for all assets (KB-D-067): reprocessing, approval, asset IDs, page rewrite and reinstall pending; picker PR #200 needs the Extension Manager set-up | Partially implemented | AEM Administrator |
| AR-03 | Adobe Tags / Web SDK / consent not configured: data layer only | Not implemented | Product Owner |
| AR-04 | Forms have no submission back end (F-01) | Not implemented | Product Owner |
| AR-05 | MSM or separate regional content for the language master rollout | To be confirmed | Product Owner |
| AR-06 | /en locale content was unpublished and deleted from Edge Delivery on 8 November 2026 16:41–16:48 UTC (author-side action) | Open: to confirm intent | Content Lead |
| AR-07 | /design-block-master/en content links to /language-master/en (1,281 links) and to unmapped /content/tcs-corp/design-block-master/english paths (358 links) | Open | Content Lead |
| AR-08 | Production domain, CDN, monitoring not recorded | To be confirmed | Solution Architect |

## 2. Recommendations (not approved)

## 3. Recommendations (not approved)

Recommendations recorded in the handover (Part 16 recommendations, 11.7, 12.3) and those derived for this knowledge base. None is a decision: each needs an owner's approval and a decision log entry before implementation.

| Recommendation | Source | Status |
|---|---|---|
| Integrate Adobe Tags with the Web SDK and the ACDL extension in `scripts/delayed.js`, with consent | Handover 11.7; F-06; O-07 (blocked: measurement plan, Tags property, datastream, consent platform) | Recommendation (not approved) |
| Run `npm test` in the Build workflow; tag releases; replace the boilerplate values of `package.json`, the PR template and the Sidekick project name | F-37; handover 12.3 | Recommendation (not approved) |
| Configure a CDN or root redirect for `/` → `/en/` and apply the per-locale redirects sheets | F-36; O-03 | Specified in redirects-cdn.md (KB-D-055, Proposed) |
| Add hreflang alternates and a language switcher | F-39 | Recommendation (not approved) |
| Connect a form back end before publishing any form, or hide forms | F-01; O-08 | Recommendation (not approved) |
| Align the handover's Part 6 component chapters with the block READMEs | F-107 | Recommendation (not approved) |
| Reference agent-instructions.md from the root AGENTS.md and place copilot-instructions.md in `.github/` | Derived for this knowledge base | Recommendation (not approved) |
| Remove the language master review code and paths after the rollout to /en (KB-D-010) | F-88 | Recommendation (not approved): planned, the timing isn't recorded |

## 4. Screenshots

No screenshot belongs to this document (it describes code, configuration or process). Evidence screenshots are listed in the screenshot register.

## 5. Known limitations

- None recorded for this document.

## 6. Open items

- None.

## 7. Change history

| Version | Date | Author | Change |
|---|---|---|---|
| 1.0 | 8 November 2026 | Delivery team (documentation program) | First version of the role-specific documentation (master documentation prompt), generated from the repository and the published reference pages |

# Content Architecture

| Attribute | Value |
|---|---|
| Document ID | ARC-002 |
| Audience | Solution architects, technical leads, developers, AI agents |
| Purpose | The site and locale hierarchy, review roots, page structure, metadata, taxonomy, placeholders, references, data sheets, redirects and publishing dependencies |
| Scope | — |
| Prerequisites | — |
| Owner | Solution Architect (proposed) |
| Approver | Product Owner (to be confirmed) |
| Version | 1.0 |
| Status | Draft for review |
| Last updated | 8 November 2026 |
| Last validated | 8 November 2026 |
| Source files | docs/handover/knowledge-base/content-architecture.md, utils/site.js, paths.json, helix-query.yaml |
| Related documents | — |
| Related AEM paths | /content/tcs-corp/language-master/en, /content/tcs-corp/language-masters/en, /content/tcs-corp/design-block-master/en |

## 1. The three review roots (KB-D-068)

| Author path | Published at | Purpose |
|---|---|---|
| /content/tcs-corp/language-master/en | /language-master/en/ | Testing page functionality with dummy content, content limitations and other test aspects |
| /content/tcs-corp/language-masters/en | /language-masters/en/ | The language master: the 8,432 migrated tcs.com pages, validated with real sheet data; later the source of the global site and the regions (MSM or separate content to be decided once the content is final) |
| /content/tcs-corp/design-block-master/en | /design-block-master/en/ | The block library and the curated design pages |

Each root has its own references (nav, footer), error pages and sheets and lists its own pages from the query index (`REVIEW_ROOTS`, `REVIEW_SHEETS`, `REVIEW_PAGES` in utils/site.js).

## 2. Site and locale hierarchy

**Purpose.** One Edge Delivery site serves 13 locale sites by path. Each locale owns its pages and its own references, error pages, sheets, data and block library.

**Structure.** The AEM tree `/content/tcs-corp/{country}/{language}` is published at a language-only path. The locales are defined once in `LOCALES` (`utils/site.js`); the first entry is the default (`DEFAULT_LOCALE`, `en`).

| AEM folder | Published root | lang | | AEM folder | Published root | lang |
|---|---|---|---|---|---|---|
| us/en | /en | en-US | | se/sv-se | /sv-se | sv-SE |
| de/de-de | /de-de | de-DE | | dk/da-dk | /da-dk | da-DK |
| fr/fr-fr | /fr-fr | fr-FR | | no/no-no | /no-no | no-NO |
| ca/fr-ca | /fr-ca | fr-CA | | cn/zh-cn | /zh-cn | zh-CN |
| es/es-es | /es-es | es-ES | | jp/ja-jp | /ja-jp | ja-JP |
| pt/pt-pt | /pt-pt | pt-PT | | kr/ko-kr | /ko-kr | ko-KR |
| nl/nl-nl | /nl-nl | nl-NL | | | | |

```
/content/tcs-corp/
├── {country}/{language}/   a locale site; the locale page itself is Home (published as /{locale}/)
│   ├── …                   content pages
│   ├── references/         nav, footer, alternative navs / footers, footer Canvas page, modals,
fragments
│   ├── error-pages/        404 (used by 404.html); 403 / 500 for a CDN
│   ├── block-library/      block validation pages; excluded from publishing
│   ├── placeholders        sheet: Key, Text, Description, Category
│   ├── metadata            sheet: bulk metadata
│   ├── taxonomy            taxonomy page (tag, title)
│   ├── redirects           sheet: Source, Destination
│   └── data/               sheets: authors, locations, partners, social-links, media-contacts
├── language-masters/en/     temporary: published at /language-masters/en for review
└── site-global-templates/  template pages authors copy; excluded from publishing
```

**Path mapping (`paths.json`, generated by `npm run build:paths` from `LOCALES` and `REVIEW_ROOTS`).** One generic set per site root: the folder mapping (`/content/tcs-corp/de/de-de/:/de-de/`: child pages and sheets), the Home page mapping (`/content/tcs-corp/de/de-de:/de-de/`) and the taxonomy page (`…/taxonomy:/de-de/taxonomy.json`): 39 mappings for the locales, 3 temporarily for the language master. The sheets (`placeholders`, `metadata`, `redirects`, `data/*`) need no mapping of their own: the folder mapping publishes them at the same path and spreadsheets get `.json` appended; the former 112 per-sheet mappings are kept commented in the generator. `includes`: the 13 locale pages, `/content/dam/tcs-corp/` and `/content/tcs-corp/language-masters/en`; `excludes`: each locale's `block-library` and `block-library/**`, and `site-global-templates` with its children. AEM applies the site configuration's `public.json` → `paths`, not the repository file (AGENTS.md "Notes").

**Locale resolution (code).** `getLocale(pathname)` reads the AEM path when editing (`/content/tcs-corp/de/de-de/…`) and the published path on the site (`/de-de/…`); pages outside a locale (Site Global Templates, drafts) get `en`. Helpers: `localePagePath`, `resolvePagePath`, `localeSheetPath`, `toPublishedPath`, `toAemPath`, `pageLink`, `getParentPages`, `sheetRows`.

**Review roots (KB-D-068).** There are three review roots, each published as a site of its own (KB-D-068): `/language-master/en` (testing page functionality with dummy content, content limitations), `/language-masters/en` (the language master: the migrated tcs.com content, validated with its real sheet data, later the source of the global site and the regions and languages; MSM or not is open until the content is final) and `/design-block-master/en` (the block library and the curated design pages). Pages below each (`REVIEW_ROOTS`) list their own pages from the query index (`indexRoot`), read their own placeholders, redirects, taxonomy and data sheets (`REVIEW_SHEETS`), use their own `references/` and `error-pages/` (`REVIEW_PAGES`) and inherit up to their root page. The temporary review of the language master ends with its rollout (handover Part 3.9, finding F-88).

**Language master URLs (validated 23 October 2026).** The mappings follow Adobe's path mapping rules (https://www.aem.live/developer/authoring-path-mapping): the folder mapping `/content/tcs-corp/language-masters/en/:/language-masters/en/` covers every child page, and the page mapping `/content/tcs-corp/language-masters/en:/language-masters/en/` is the special case that publishes the root page as `/language-masters/en/index`, served at `/language-masters/en/` (the same pattern as `/en/`). No other mapping

is needed. The mappings of every language master come from `REVIEW_ROOTS` (`utils/site.js`): `tools/paths/generate-paths.mjs` writes this set for each root, so another language master (e.g. `/language-masters/de`) is added to the code and to `paths.json` with one entry and `npm run build:paths`; `test/site/paths.test.mjs` checks every locale and language master is mapped this way and that `paths.json` is in step.

| AEM page | Preview | Live |
|---|---|---|
| `/content/tcs-corp/language-masters/en` (Home) | `https://main--tcs-corp--sradha-tcs.aem.page/language-master/en/` | `https://main--tcs-corp--sradha-tcs.aem.live/language-master/en/` |
| `/content/tcs-corp/language-masters/en/<path>` (child pages) | `https://main--tcs-corp--sradha-tcs.aem.page/language-master/en/<path>` | `https://main--tcs-corp--sradha-tcs.aem.live/language-master/en/<path>` |

The Home URL ends with `/` (`/language-masters/en` without it is a 404, as `/en`). On 23 October 2026 the child pages, sheets, references and error pages resolved, but the Home page was a 404 and wasn't in the query index: it has to be previewed and published from AEM (Manage Publication on `/content/tcs-corp/language-masters/en`). If it still returns a 404 after publishing, check that the site configuration's `public.json` → `paths` has the page mapping (AEM applies that configuration, not the repository file).

**Rules.**

- Add or change a locale only in `LOCALES`, then run `npm run build:paths` and commit both files; the site administrator re-applies `public.json` and `metadata.source`.
- Nothing is published at `/`: a `/` → `/en/` redirect needs another mechanism (handover F-36).

**Common mistakes.** Editing `paths.json` by hand (it is overwritten by the generator); expecting a repository `paths.json` change to take effect without the site configuration; creating pages directly under `/content/tcs-corp` (not published).

## 3. Page structure

**Purpose.** A page is a sequence of sections; each section holds default content (text, headings, images, links) and blocks. The structure is the contract between author and code (AGENTS.md "Blocks").

**Page properties** (`models/_page.json`, model `page-metadata`):

| Field | Name | Effect |
|---|---|---|
| Title | `jcr:title` | `<title>`, `og:title` |
| Description | `jcr:description` | `description` meta |
| Keywords | `keywords` | `keywords` meta |
| Page Type | `page-type` | Default (empty), Home `home`, Article `article`, Blog `blog`, Press Release `press-release`: body class and template |
| Navigation | `nav` | Header page, inherited by child pages |
| Footer | `footer` | Footer page, inherited by child pages |
| 404 Page | `error-page` | 404 page of the subtree, inherited |

Page tags are not a field of this model; they come from the standard AEM page properties (`cq:tags`, published as `cq-tags`) or a `tags` column of the bulk metadata sheet (handover Part 7.1.1).

**Section properties** (`models/_section.json`), applied by `decorateSectionsWithOptions` in `scripts/scripts.js`:

| Field | Name | Delivered as | Result |
|---|---|---|---|
| Section Name | `name` | — | Not one of the section options read by `scripts.js` |
| Style | `style` | section classes | Default, Grey `bg-grey`, Dark `bg-dark`, Dark Green `bg-dark-green` (colour only; dark ones apply the theme contract in `styles/styles.css`) |
| Width Type | `sectionWidth` | `data-sectionwidth` | `fixed-width` class (full width is the default) |
| Rounded edge | `roundedEdge` | `data-roundededge` | `rounded` class |
| Background Image / Image alt text | `backgroundImage`, `backgroundImageAlt` | data attributes | `.section-background` with a responsive picture, `has-background` |
| Anchor name | `anchorName` | `data-anchorname` | section `id` (suffixed `-2`, `-3` when taken) |

Documents that use a `section-metadata` block are read the same way (rows `Section Width`, `Rounded Edge`, `Background Image`, `Background Image Alt`, `Anchor Name`).

**Container sections.** Gen Carousel, Tabs Container and Columns Container are sections whose content `scripts.js` wraps in a synthetic block (`gen-carousel`, `tabs`, `columns-container`). Sections don't get `classes_*` fields delivered, so their Additional tab fields are `container*` data attributes turned into block classes by `applySectionBlockOptions` (AGENTS.md "AEM delivery rules"; columns-container, tabs, gen-carousel).

**Blocks and default content.** The `section` filter allows 26 components (`component-filters.json`); the inventory and each block's content contract are in blocks/README.md. Repeating content is item children added with Add, not composite multifields (AGENTS.md "Item children").

**Common mistakes.** Putting a section inside a Tabs Container (a section can't hold a section: use Column Breaks); relying on a section `classes_*` field (not delivered); adding sample text to templates or model defaults (stored on every new component).

## 4. Templates

**Purpose.** A Page Type gives recurring page kinds their structure. Template pages that authors copy live in `/content/tcs-corp/site-global-templates/` (not published).

**Structure.** `loadTemplate` (`scripts/scripts.js`) loads `templates/{name}/{name}.js` and `.css` when the Page Type (`page-type`, else `template` metadata) is in `TEMPLATES`:

| Page Type | Adds | Data |
|---|---|---|
| `home` | Layout styles only | — |
| `article` | A Page Header from the page metadata (unless authored), "About the author" | Authors sheet (`author-id`) |
| `blog` | The article structure and styles | Authors sheet |
| `press-release` | A Page Header (unless authored) | Page metadata (the media contacts are the Media Contacts block, KB-D-054) |

Source: templates/README.md (including how to add a template).

**Rules.** The Page Type page property wins over the metadata sheet's `page-type` column; Page Type also becomes a body class. Changing a template page doesn't change pages already copied from it (handover

Part 4.8).

**Common mistakes.** A `page-type` value that isn't in `TEMPLATES` (only the body class is added); expecting template content in the query index (templates build DOM in the browser).

## 5. Metadata model

**Purpose.** Page-level values (SEO, social, card content, page type, author, dates, flags) that code, the query index and the data layer read. Every value is a `<meta>` in the server-delivered head.

### Sources and precedence.

| Source | Where authored | Published as |
|---|---|---|
| Page properties (`models/_page.json`) | Universal Editor page properties | `<title>`, `description`, `keywords`, `page-type`, `nav`, `footer`, `error-page` metas |
| Standard AEM page properties | Sites console (e.g. Tags `cq:tags`) | `cq-tags` (one comma-separated meta) |
| Bulk metadata sheet | `/content/tcs-corp/{country}/{language}/metadata` | `/{locale}/metadata.json`; each column becomes `<meta name="{column}">` on pages matching the row's `url` pattern |

Page-level values win over bulk metadata; bulk rows apply top to bottom (handover Part 8.1.2). The bulk sheets apply only once all 13 are listed in the site configuration's `metadata.source`, and each sheet's URL patterns must start with its locale (`/de-de/**`) (AGENTS.md "Notes").

**Bulk metadata columns** (AGENTS.md, 41 columns): `url`, `page-type`, `meta-title`, `meta-description`, `meta-keywords`, `author-id`, `publish-date`, `modified-date`, `reading-time`, `og-title`, `og-description`, `og-image`, `og-type`, `twitter-card`, `twitter-title`, `twitter-description`, `twitter-image`, `canonical-url`, `robots`, `region`, `country`, `language`, `persona`, `thumbnail-image`, `featured-image`, `campaign-id`, `content-type`, `featured`, `priority`, `listen-link`, `author`, `card-title`, `card-description`, `card-image`, `card-image-alt`, `image-alt`, `eyebrow`, `category`, `hide-in-nav`, `card-style`, `card-theme`.

### Consumers.

| Consumer | Reads | Source |
|---|---|---|
| SEO decorator (eager) | `meta-title` → `<title>`; `meta-description`, `meta-keywords`; `og-*` (falling back to `meta-*`); `twitter-*` (falling back to `og-*`, `meta-*`); `canonical-url` → `<link rel="canonical">` and `og:url`; `language` → `<html lang>` | `decorators/seo-decorator.js` |
| `getPageMetadata()` | title, description, image (`featured-image`, else `og:image`), `page-type` (else `template`), `content-type`, `category`, tags, `author-id`, `author`, `publish-date` (else `publication-date`), `modified-date`, `reading-time`, `listen-link`, `thumbnail-image`, `featured-image`, `image-alt`, `region`, `country` (else the locale's), language, `persona`, `campaign-id`, `featured` (yes / true / 1 / on / y), `priority` (number) | `utils/metadata.js` |
| `getPageTags()` | `article:tag` metas and `cq-tags`, else a comma-separated `tags` meta | `utils/metadata.js` |
| `getPageLanguage()` | `language` when it is a language code, else the locale's `lang` | `utils/metadata.js` |
| Page Header, templates, data layer page entry | `getPageMetadata()` | `page-header`, `decorators/data-layer-decorator.js` |
| Query index | the server head (see indexing-and-search.md) | `helix-query.yaml` |

**Rules.** The query index reads the server-delivered head, so values set by JavaScript (the SEO decorator's) aren't indexed (handover Part 8.1.3). `robots` is a standard name and works server side. Use `card-*`, `eyebrow`, `hide-in-nav`, `card-style`, `card-theme` for List cards (List README "Page metadata contract").

**Example.**

```
url                   | page-type      | hide-in-nav
/de-de/newsroom/**    | press-release  |
/de-de/insights/**    | article        |
/de-de/legal/imprint  |                | yes
```

**Common mistakes.** A sheet row pattern without the locale prefix; adding columns to a sheet that isn't listed in `metadata.source` (no effect); expecting a metadata change to reach lists before the page is republished and indexed. Field reference per column: handover Part 8 — Metadata Documentation (8.2, 8.4).

## 6. Taxonomy

**Purpose.** Tag IDs on pages and in block fields are turned into localised labels at runtime.

**Structure.** Tags are maintained in AEM Tagging; a taxonomy page per locale (`/content/tcs-corp/{country}/{language}/taxonomy`) is published as `/{locale}/taxonomy.json` with columns `tag` and `title`. With translated titles the JSON is multi-sheet: `loadTaxonomy()` reads the sheet named after the locale's ISO 639-1 language (`de`), else `default`, else the first sheet (`sheetRows`).

**Consumers.** `resolveTags(values)` (`utils/taxonomy.js`): a value with a taxonomy entry gets its title; a value containing `:` or `/` without an entry gets a label derived from the ID (`parseTag`, `utils/tags.js`); any other value is shown as written. Used by Teaser, Page Header, List, Cards, Filter List and Conversation Canvas.

**Rules.** The published namespace is `tcs-corp:` (handover Part 7.1.2; finding F-08: some help texts and examples use `tcs:`). Pick tags with the tag picker; the `aem-tag` fields don't restrict the namespace (handover Part 7.1.1). A selected tag also matches its child tags in List (`hasTag`, handover Part 7.2.2).

**Example.** `tcs-corp:region/europe` → `Europe` from `/en/taxonomy.json`; without a taxonomy entry, an ID-derived label.

**Common mistakes.** Typing tag IDs by hand; not publishing the locale's taxonomy page (labels fall back to IDs; handover F-09 records that only `/en/taxonomy.json` was published when checked); translating tag titles without republishing the taxonomy page. Governance and lifecycle: handover Part 7 — Taxonomy Documentation.

## 7. Placeholders

**Purpose.** All user-facing UI text in code is authorable and translated per locale (AGENTS.md "Accessibility": all UI text comes from placeholders).

**Structure.** `/{locale}/placeholders.json` with columns Key, Text, Description, Category; code uses Key and Text. Keys are compared without case and separators (`Pause Slides`, `pause-slides` and `pauseSlides` match `pauseSlides`). `{name}` variables are filled by `fillPlaceholder`.

**Consumers.** `const t = await getPlaceholders(); t('nextSlide', 'Next slide')` in every block with UI text, the header, footer and `scripts.js` (404 "Go back", key `goBack`). Source: `utils/placeholders.js`; key lists per block in the block READMEs ("For Site Administrators").

**Rules.** Code always passes an English fallback. There is no fallback from a locale's sheet to the English sheet: a key missing in `/de-de/placeholders.json` shows the code's English text (handover Part 4.9.3).

**Common mistakes.** Editing the sheet without publishing it; adding a new key in code without adding it to the 13 sheets. Full key list: handover Part 4.5.3.

## 8. References, fragments and inherited metadata

**Purpose.** Pages that aren't content pages but are included in other pages: the header navigation, the footer, alternatives per site area, the footer Canvas page, modals and fragments.

**Structure** (in each locale's `references/`):

| Page | Used by |
|---|---|
| `nav` | Header (default Navigation of the locale) |
| `footer` | Footer (default Footer of the locale) |
| `nav-{name}`, `footer-{name}` | Header / Footer when selected in a page's Navigation / Footer (or the `nav` / `footer` sheet column) |
| Footer Canvas page | The Footer block's Conversation Canvas Page |
| `modals/{name}` | Links whose path contains `/modals/` open as a modal (`autolinkModals`); the Modal block |
| Fragments | Fragment block; Tab fragments |

**Inheritance** (`getInheritedMetadata(name)`, `utils/inherited-metadata.js`, for `nav`, `footer`, `error-page`):

1. The page's own value (page property or bulk sheet column).
2. Else the nearest parent page's value up to the locale's Home page (`getParentPages`; for the language master up to `/language-masters/en`): the parents' published HTML heads are fetched in parallel.
3. Else the default: `references/nav`, `references/footer`, or for 404 `error-pages/404`.

Values may be AEM paths, published paths or paths relative to the locale root (`references/nav-careers`); `resolvePagePath` returns the AEM path in the editor and the published path on the site. Parent values are cached in `sessionStorage` (`tcs-inherited-metadata`) for the browser session.

**Consumers.** header (falls back to the locale's `references/nav` when a configured page can't load; with none published the header stays empty), footer (adds the Social Links sheet), `loadErrorPage` (404), fragment, modal, tabs.

**Rules.** Keep reference pages in `references/` (excluded from the query index and the sitemap); `fragments/` and `modals/` folders anywhere are excluded too. A fragment change shows on every including page once the fragment is published. `loadFragment(path, from)` fetches only paths starting with `/` and stops recursion by the chain of including fragments.

**Example.** Navigation = `references/nav-careers` on `/en/careers` gives every page below it the Careers header unless a page sets its own.

**Common mistakes.** Testing a changed Navigation in the same browser session (cached); leaving `nav` unpublished in a new locale (empty header); placing a fragment outside `references/`, `fragments/` or `modals/` (it gets indexed and listed).

## 9. Data sheets

**Purpose.** Shared structured data per locale, maintained by authors as AEM spreadsheets.

**Structure and consumers** (`utils/data.js`; each sheet loaded once per page view, values trimmed, missing sheets give empty lists):

| Sheet (/{locale}/data/…json) | Columns | Read by |
|---|---|---|
| authors | author-id, name, designation, bio, image, linkedin, region | Page Header (getAuthor, case-insensitive id), article / blog templates, List feature card and Filter List spotlights (getAuthors) |
| social-links | platform, url, icon, sort-order | Footer (in sort-order, rows without one last) |
| media-contacts | id, name, designation, email, phone, region | Media Contacts block (blocks/media-contacts: inRegion by the page's region, else all; grouped by designation and region) |
| locations | location-id, location-name, country, region, latitude, longitude | getLocations() only; no component reads it (handover F-40) |
| partners | partner-id, partner-name, logo, website, tier, description | getPartners() only; no component reads it (handover F-40) |

**Rules.** The query index excludes /*/data/**. Sheet JSON is public: the Authors and Media Contacts sheets publish names, e-mail addresses and phone numbers by design. Column reference and authoring steps: handover Part 9 — Data Sheets Documentation.

**Common mistakes.** An author-id on a page that differs from the sheet in case (Filter List and List match exactly; Page Header case-insensitively, handover F-30); unpublished sheets in non-English locales.

## 10. Shared content

| Mechanism | What is shared | Included by |
|---|---|---|
| Header / Footer | references/nav, references/footer or inherited alternatives | Every page, automatically |
| Fragment block | A reference page | Fragment block Reference field |
| Footer Canvas | A page with a Conversation Canvas block | Footer block field |
| Tabs | A fragment per tab | Tab Fragment field |
| Modals | Pages under a /modals/ path | Any link, or the Modal block |
| Placeholders, taxonomy, data sheets | UI text, tag labels, structured data | Code, per locale |
| Block library | Every block variation per locale | Not included in pages; excluded from publishing (the language master's is published for the review, handover Part 3.9) |
| Site Global Templates | Starter pages | Copied by authors; not published |

All Conversation Canvases of a page (header, body, footer) share one floating journey (AGENTS.md "Floating Canvas", conversation-canvas).

## 11. Navigation model

| Area | Model | Source |
|---|---|---|
| Header | The Navigation page (inherited) has the brand (TCS logo), the sections (the menu, any depth) and the tools (TATA logo). Desktop (1200px and wider) shows the logo bar without navigation; tablet and mobile open the menu full screen one level at a time | AGENTS.md "Header", blocks/header/README.md |

| Area | Model | Source |
|---|---|---|
| Site navigation on desktop | The Conversation Canvas: its pop-up menu and the Primary dock list pages from the query index below a Menu root path | conversation-canvas README, indexing-and-search.md |
| Footer | The footer page's Footer block: Conversation Canvas Page, Copyright Text (`{year}`), Footer Link items; social icons from the Social Links sheet | blocks/footer/README.md |
| In page | Anchor List links to section Anchor names and block Anchor IDs (`anchor-…`); the URL hash scrolls after the sections load; `tabs:reveal` opens the tab holding a target | anchor-list, `scripts/scripts.js` |
| Per site area | Set Navigation / Footer on the area's top page | inheritance above |

Not implemented: a breadcrumb, a language selector and `hreflang` alternates (handover Parts 4.9.6 and 16.4).

## 12. Error pages

**Purpose.** An authorable, locale-specific 404 page.

**Structure.** Edge Delivery serves `404.html` for a missing path; `loadErrorPage` (`scripts/scripts.js`) tries, in order: the inherited **404 Page** (`error-page`) of the missing page's nearest parent; the locale's `/{locale}/error-pages/404`; `/en/error-pages/404`. A page without content (no text, image, picture, video or iframe) is skipped; when none loads the built-in content ("404", "Page Not Found", "Go home" to the locale's Home) stays. Content from another locale gets its `lang` on `<main>`; "Go back" (placeholder `goBack`) is added for same-origin referrers.

**Rules.** 403 and 500 aren't served by Edge Delivery; the authored `error-pages/403` and `500` are for a CDN (not configured: Requires validation). Error pages are excluded from the query index (`/*/error-pages/**`). Details: handover Part 10 — Error Pages.

**Common mistakes.** An empty `error-pages/404` (skipped, the next candidate shows); expecting the 404 Page property of the missing page itself (the lookup starts at its nearest existing parent).

## 13. Redirects

**Purpose.** Per-locale lists of moved pages.

**Structure.** `/{locale}/redirects.json` with Source and Destination columns (either case).

**Rules and status.**

- The platform applies only a root `/redirects.json`; the per-locale sheets are **not applied** automatically. Redirects, including `/` → `/en/`, need another mechanism such as CDN rules (AGENTS.md "Notes"; handover Part 3.8; findings F-36, 16.4). Status: Requires validation (no mechanism configured).
- In code only the List block uses the sheet: `resolveRedirect` points card links to the final target, at most 5 hops, stopping at loops and external destinations (`utils/redirects.js`). Filter List, navigation and running text links aren't rewritten.
- Sources must start with `/` (rows without a destination are ignored); paths are compared without query, hash, `.html` and trailing slash.

**Common mistakes.** Expecting a sheet row to redirect visitors; chains of redirects (point sources at final destinations).

## 14. Localization

| Topic | Implementation | Status |
|---|---|---|
| Locale structure | 13 separate page trees; locale resolved from the path | Active |
| `<html lang>` | `language` metadata when valid, else the locale's tag | Active |
| UI text | Placeholders sheet per locale; English fallback in code | Active |
| Tag labels | Taxonomy page per locale, language sheet when translated | Active (only `/en` taxonomy published when checked, F-09) |
| Bulk metadata | One sheet per locale; all 13 in `metadata.source` | Requires validation (F-11, open item O-04) |
| Data sheets | One set per locale | Active |
| Lists | Each locale lists only its own pages (`indexRoot`) | Active |
| Dates | `utils/dates.js` `formatDate`; English month abbreviations in every locale (handover Part 4.9.7) | Active |
| Translation process, MSM / language copies, language selector, `hreflang` | Not in code | Not recorded |

## 15. Content governance

Verified rules (enforced by code or configuration):

- Locale resources sit at fixed locations (`references/nav`, `references/footer`, `error-pages/404`, the sheets, `data/*`): code resolves them per locale.
- Navigation, Footer and 404 Page are inherited from parent pages up to Home.
- `hide-in-nav = yes` keeps a page out of List results (not Filter List or the Canvas menu: F-30, F-33).
- Reference, error, data, nav / footer, fragments and modals paths are excluded from the query index.
- Sheets and reference pages take effect only once published.
- No sample content in templates or model defaults (AEM stores definition template values on every new component).

Ownership, review cadences and approval flows: Not recorded in the repository (handover Parts 7.4, 14 and 17 describe proposed processes and open owners).

## 16. Publishing and indexing dependencies

| Change | Must also be published / done | Effect |
|---|---|---|
| New or changed page | Publish the page | Index row and sitemap update; lists show it (see indexing-and-search.md) |
| Page metadata used by cards | Republish the listed page | List / Filter List cards change; the List page itself needs no republish |
| Sheet (placeholders, metadata, taxonomy, redirects, data) | Publish the sheet | Next page load (each loaded once per page view) |
| Bulk metadata | Sheet listed in `metadata.source` | Metas on matching pages |
| Navigation / footer / fragment | Publish the reference page | Every including page; inherited values in a new browser session |
| Non-image DAM assets (Lottie JSON, videos, captions) | Publish the assets | Served only once published; images travel with the page (AGENTS.md) |
| `helix-query.yaml` | Reindex (Admin API index operation) or republish | Existing rows unchanged until then |
| `LOCALES` | `npm run build:paths`, re-apply `public.json` and `metadata.source` | New paths published |

## 17. Related

- architecture.md — runtime, delivery, integrations, caching
- indexing-and-search.md — query index, listings, sitemap, reindexing
- blocks.md, project-overview.md, repository-structure.md — related knowledge base files
- AGENTS.md — "Site structure and site data", "AEM delivery rules"
- utils/README.md, templates/README.md, blocks/README.md
- Handover (TCS-CORP-HANDOVER.md): Part 3 — Site Structure, Part 4 — Content Architecture, Part 7 — Taxonomy Documentation, Part 8 — Metadata Documentation, Part 9 — Data Sheets Documentation, Part 10 — Error Pages

## 18. Screenshots

No screenshot belongs to this document (it describes code, configuration or process). Evidence screenshots are listed in the screenshot register.

## 19. Known limitations

- None recorded for this document.

## 20. Open items

- /design-block-master/en content still links to /language-master/en and to /content/tcs-corp/design-block-master/english (see the curated design page audit).

## 21. Change history

| Version | Date | Author | Change |
|---|---|---|---|
| 1.0 | 8 November 2026 | Delivery team (documentation program) | First version of the role-specific documentation (master documentation prompt), generated from the repository and the published reference pages |

# Deployment Architecture

| Attribute | Value |
|---|---|
| Document ID | ARC-006 |
| Audience | Solution architects, technical leads, developers, AI agents |
| Purpose | Delivery streams, environments, promotion, content packages, publishing, reindexing and rollback |
| Scope | — |
| Prerequisites | — |
| Owner | Solution Architect (proposed) |
| Approver | Product Owner (to be confirmed) |
| Version | 1.0 |
| Status | Draft for review |
| Last updated | 8 November 2026 |
| Last validated | 8 November 2026 |
| Source files | docs/handover/knowledge-base/deployment-and-release.md, .github/workflows/main.yaml |
| Related documents | — |
| Related AEM paths | — |

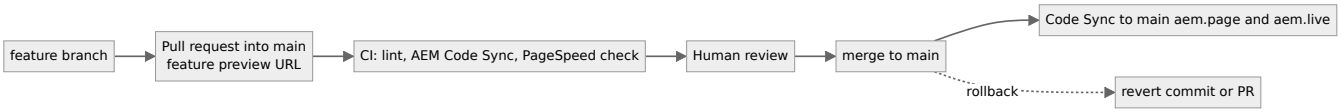
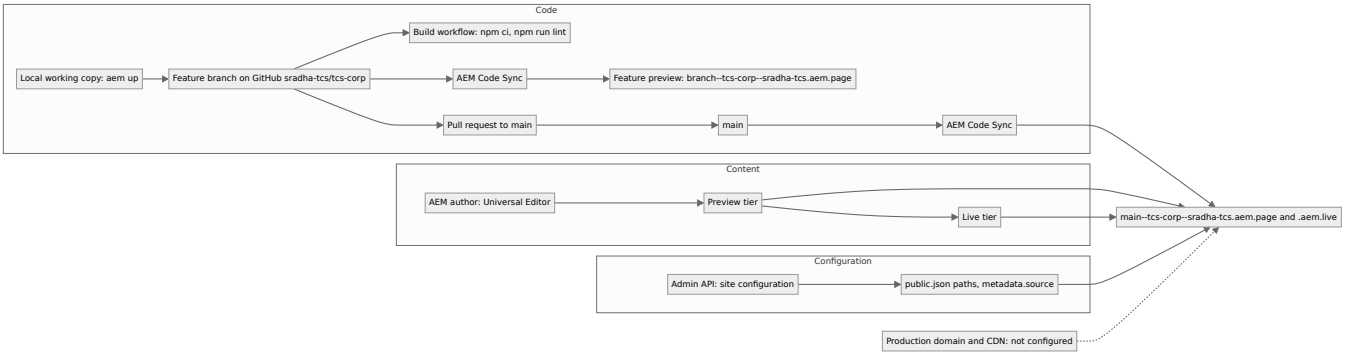



*Diagram: code release flow (source: diagrams/release-flow.mmd).*

## 1. Delivery streams

Content, code and configuration are released independently: a code merge doesn't publish content and publishing content doesn't deploy code. A release that needs both goes **code first, then content** (handover Part 12, introduction).




## 2. Local development

| Task | Command | Notes |
|---|---|---|
| Install dependencies | `npm install` | Also installs the Husky pre-commit hook (`prepare: husky`) |
| Start the dev server | `npx -y @adobe/aem-cli up --no-open --forward-browser-logs` (or `aem up` with the CLI installed globally) | `http://localhost:3000`: local code, previewed content |
| Serve static test content | Add `--html-folder drafts` | Files in `drafts/` (`.html` or `.plain.html`); 14 files are tracked |
| Lint | `npm run lint` (`lint:js`: ESLint on `.json`, `.js`, `.mjs`; `lint:css`: Stylelint on `blocks/**/*.css`, `styles/*.css`) | `npm run lint:fix` fixes what it can |
| Unit tests | `npm test` (`node --test 'test/**/*.test.mjs'`) | Pure modules, model layout, docs, module graph |
| Rebuild the Universal Editor files | `npm run build:json` | `component-models.json`, `component-definition.json`, `component-filters.json` from the partials |
| Regenerate the path mapping | `npm run build:paths` | After changing LOCALES in utils/site.js; commit both |
| Regenerate the module graph | `npm run build:modules` | tools/modules/generate-module-graph.mjs, used for module preloading |
| Regenerate block READMEs | `npm run build:docs` | Generated sections of the block READMEs (blocks/README.md) |

Inspect delivered content with `curl http://localhost:3000/{path}`, `.md` and `.plain.html` (AGENTS.md). Node.js: README.md states 20 or newer; the workflows install Node 24. The local preview renderer used by the audits (`.home-work/render-lib.mjs`, named in AGENTS.md) is not tracked in Git.

## 3. Branches and feature previews

- `main` is production. Changes go through short-lived branches merged with pull requests (130 merge commits on `main` at 2d06316, latest #129).
- Branch prefixes in use: `feature/`, `feat/`, `fix/`, `perf/`, `docs/` (for example `fix/block-reaudit`, `perf/module-preload`, `docs/block-readmes`). The handover's `feature/tcs-<topic>` convention (12.2.2) is not what recent history uses; no naming rule is enforced.
- AEM Code Sync serves every pushed branch at `https://{branch}--tcs-corp--sradha-tcs.aem.page/` (slashes become hyphens; keep names short and lower case: branch, repository and owner form the host name).

## 4. Pull requests

The publishing process of AGENTS.md:

1. Push the feature branch; Code Sync makes it available on the feature preview.
2. Run PageSpeed Insights on the feature preview URL and fix issues; target a score of 100.
3. Open a pull request to `main`. The description **must** link `https://{branch}--tcs-corp--sradha-tcs.aem.page/{path}` showing the change, or the PR is rejected. Without a page, create test content in `drafts/` and ask an author to create a CMS page to link.
4. Check the status of code sync, linting and performance tests (`gh pr checks`).
5. A human reviewer reviews the code, opens the URL and merges; Code Sync updates `main`.

.github/pull_request_template.md asks for the GitHub issue (`Fix #<gh-issue-id>`) and Before / After test URLs. It still contains the boilerplate placeholders `{repo}` and `{owner}`: replace them with `tcs-corp` and `sradha-tcs` (the `cleanup-on-create` workflow that would replace them only runs on repository creation or manual dispatch).

Approval rules: the number of required reviews, required checks and branch protection on `main` are not recorded (handover 12.3; open item O-05).

## 5. Automated checks

| Check | Defined in | When | What it does | Status at 2d06316 |
|---|---|---|---|---|
| Husky pre-commit | .husky/pre-commit → .husky/pre-commit.mjs | Every local commit | If staged files include a model partial (`_*.json`), runs `npm run build:json` and stages the three aggregated files | Aggregated files match the partials: Passed |
| Build | .github/workflows/main.yaml | Every push, every branch (`on: [push]`) | `npm ci`, then `npm run lint` (ESLint with `airbnb-base`, `plugin:json`, `plugin:xwalk/recommended`; Stylelint `stylelint-config-standard`) | `npm run lint` locally: Passed |
| Unit tests | package.json `test` | Manual only: not in any workflow | 311 tests in 53 suites (`test/`) | `npm test` locally: Passed (311 / 311) |
| cleanup-on-create | .github/workflows/cleanup-on-create.yaml | Repository creation or manual dispatch | Boilerplate clean-up; no role in releases | Not applicable |
| Renovate | .renovaterc.json | Renovate schedule | Dependency PRs; devDependency PRs labelled `ignore-psi-check` and auto-merged when tests pass; ESLint pinned to `^8.0.0` | Not applicable |
| AEM Code Sync | GitHub app (not a repository file) | Every push | Syncs the branch to the code bus and reports a check | Not verified here (no GitHub access) |
| PageSpeed (`aem-psi-check`) | AEM GitHub app (not a repository file) | Pull requests | PageSpeed Insights on the URLs in the PR description | Not verified here |

The CI gate is lint only. Run `npm test` before every PR; adding it to `main.yaml` is a recommendation of the handover (14.9 #19, F-37), not a decision.

## 6. Content packages

The block library, the curated design pages and sample pages are delivered to AEM as content packages built **outside the repository**:

- The build and migration tooling lives in the delivery workspace (`.home-work/bl/`, for example `bl-migrate.mjs`, which maps old fields to new ones when the pages and the package are built); it is not tracked in Git (BLOCK-ECOSYSTEM-AUDIT.md, section 11).
- The packages are distributed on the branch `home-content-package` of the repository (files only, no site code; its README describes each package). For example `block-library/tcs-corp-block-library-content-1.0.13.zip` (current), the curated design pages, Home and Columns Container packages and the matching sheet rows (placeholders, metadata, taxonomy, data CSVs).
- Install with AEM **Package Manager** (Upload Package, Install). Installing replaces each page of the package: changes made in AEM to those pages since the last install are overwritten; pages outside the package are kept.
- Tags `home-content-package_<date>` mark earlier states of that branch ("Downloads branch before the handover PDF …"); they are not code releases.

What the handover records (revision history):

| Package | Change | Publication recorded |
|---|---|---|
| 1.0.3, 1.0.4 | Block library Animation folder | Published with the language master review (2.20) |
| 1.0.7 | Library migrated after the block audit | Not recorded (2.27) |
| 1.0.8 | Columns → Columns Containers / Column Breaks, buttons → Button blocks | Not recorded (2.28) |
| 1.0.9 | Re-audit fixes; language-master Placeholders (109) and Metadata (311 rows) imported | 312 / 312 previewed and published; defect R-L-1 found on live (2.30) |
| 1.0.10 | Container sections' `container*` fields | 312 / 312 previewed, published and reindexed (2.31) |
| 1.0.12 | Section Style colours (with the 1.0.11 Teaser and Columns Container options) | 313 / 313 published and reindexed (2.35) |
| 1.0.13 | List Item display options, dark-background theme | 315 / 315 previewed, published and reindexed (2.35) |

## 7. Environment promotion

| Step | From → to | How | Owner |
|---|---|---|---|
| 1 | Local → feature preview | Push the branch; Code Sync | Developer |
| 2 | Feature preview → `main` (preview and live code) | Reviewed PR merged; Code Sync updates `main--tcs-corp--sradha-tcs.aem.page` and `.aem.live` | Reviewer |
| 3 | Universal Editor | Authors reload the editor to get new definitions, models and filters | Content team |
| 4 | Configuration | Apply `public.json` paths, `metadata.source`, sitemap or CDN settings through the configuration Admin API, saving the current JSON first | Site Administrator |
| 5 | Content: author → preview → live | Publish sheets, then fragments, then pages | Content team |

There is no separate staging environment in the repository: feature previews and the `main` preview tier play that role. The author environment reads the code of `main` (fstab.yaml mount point). The production domain and CDN are not configured (open item O-03).

## 8. Installation validation

After a code release or a package install, validate on the preview tier first, then on live:

- ☐ Code Sync check green on the merge commit; a changed file on `main--tcs-corp--sradha-tcs.aem.page/blocks/…` matches the commit (Part 17, R-06).
- ☐ The Universal Editor shows the new fields and components after a reload (R-08).
- ☐ After a package install: every package page previewed, published and, where listed, reindexed, with the count recorded (as in 312 / 312 for 1.0.10).
- ☐ Changed pages and one page per changed block render without `failed to load module for …` or `failed to load template …` (V-02).
- ☐ Header, footer and 404 page correct per locale; sheets reachable (V-03, V-08).
- ☐ Non-image DAM assets used by the pages published (Lottie, video, captions).
- ☐ `aem.page` compared with the local preview block by block (docs/audit/tools/compare.mjs): this comparison found R-L-1 and R-L-2 after 1.0.9.

## 9. Publishing

Content publishing (Universal Editor, Sites console, sheets, references, assets, scheduling) is described in authoring.md; Admin API operations in administration-and-operations.md. In a release, publish in this order: sheets (placeholders, taxonomy, data, metadata) → fragments (nav, footer, other references) → pages (handover 12.4.2, Part 17 R-09). Content that depends on new code is prepared in AEM but published only after Code Sync of the merge.

## 10. Reindexing

- Publishing updates the page's row in `/query-index.json` automatically.
- After a change to helix-query.yaml, rows are rewritten only when pages are republished or re-indexed: `POST https://admin.hlx.page/index/sradha-tcs/tcs-corp/main/{path}` for the affected paths (bulk paths agreed with the site owner), then check the columns (Part 17, R-10, W-01).
- The packages 1.0.10, 1.0.12 and 1.0.13 were reindexed page by page (Admin API `preview`, `live`, `index` per page); after 1.0.10 the query index listed 307 language-master pages (handover revision 2.31).
- The sitemap (helix-sitemap.yaml) is built from the index.

## 11. Smoke and regression testing

| Test | Command / tool | Scope | Result at 2d06316 |
|---|---|---|---|
| Unit tests | `npm test` | `test/` (cards, conversation-canvas, docs, items, list, media, models, modules, motion, privacy, separator, site) | Passed (311 / 311) |
| Lint | `npm run lint` | JS, JSON, models (xwalk rules), CSS | Passed |
| Generated files | `npm run build:json` then check that nothing changed (Part 17, R-03) | Aggregated component files | Passed (compared with a fresh merge) |
| Runtime harness | docs/audit/tools/harness.mjs | Every block library and curated page of the local preview at 1440, 1024, 768, 600 and 390 px: console and page errors, failed requests, block status, overflow, duplicate ids, broken images, data layer, axe (WCAG A / AA) at 1440 and 390, a click test | Not run for this page |
| Published vs local | docs/audit/tools/compare.mjs | Block-by-block signature of `aem.page` against the local preview | Not run for this page |
| Inventory | docs/audit/tools/inventory.mjs | Definitions, models, filters, fields, options and their usage in the content package | Not run for this page |
| Defect register | docs/audit/tools/register.mjs | The audit's defects and outcomes | Not applicable |
| Manual | Part 17 V-01 … V-09 | Content, code, configuration, data layer, SEO, accessibility, locale launch, cache | Not recorded |

The audit tools hard-code paths of the delivery workspace (a Playwright install, an axe copy under `(a local workspace path)`, `drafts/en/block-library`, the unpacked content package under `.home-work/`), which are not in Git: re-running them from a clean clone is Blocked until those inputs are provided. Their recorded runs: 940 page loads before and after the first audit, and the re-audit runs (BLOCK-ECOSYSTEM-AUDIT.md, REAUDIT-2026-10-20.md, data/runtime-before-after.csv). The block libraries in each locale (`block-library/`) are the manual regression pages for every variation.

## 12. Rollback

No rollback has been recorded; the procedures below are the handover's (Part 12, 12.6) and Require validation.

| Stream | Procedure |
|---|---|
| Code | Revert the merged PR in GitHub (a revert PR, checks, merge) or `git revert -m 1 <merge-commit>` on a branch, newest first; Code Sync deploys the previous code. Never force-push or rewrite `main` |
| Models | The revert restores partials and aggregated files; authors reload the editor; values saved with removed fields stay stored but aren't shown |
| Content | Sites console **Timeline > Revert to this Version** (or Restore Version), check, publish again; sheets the same way; unpublish when no good version exists |
| Configuration | Re-apply the saved JSON (`public.json` paths from a previous `paths.json`: `git show <commit>:paths.json`; the previous `metadata.source` list); repository configuration files by reverting the PR, then re-index or regenerate the sitemap |
| Content package | Reinstall the previous package version from the `home-content-package` branch (installing replaces the package's pages) |

Emergency change approval and the hotfix path are not recorded (handover 12.6.1, Part 17 R-15).

## 13. Release evidence

| Evidence | Where | State |
|---|---|---|
| Change history | Merge commits of the pull requests on `main`; the handover revision history (versions 0.1 to 2.37) | Available |
| Validation reports | docs/audit/ (audit, re-audit, dialog options; CSV registers and runtime before / after) | Available |
| Publication records | Handover revisions 2.31 and 2.35 (counts of pages previewed, published, reindexed) | Available |
| Release tags | None on `main`; `package.json` keeps the boilerplate version `1.3.0`; no `CHANGELOG.md` | Not recorded |
| Change records | Change management process | Not recorded |
| Release checklists | Handover 12.4 and Part 17, 17.5 (R-01 … R-15) define what to attach | Defined, no completed record in the repository |

## 14. Production readiness

| Item | State |
|---|---|
| Code on `main`: lint, unit tests, generated files | Passed at 2d06316 |
| Preview and live on `*.aem.page` / `*.aem.live` | Active |
| Production domain, CDN, push invalidation, robots.txt | Blocked: not configured (O-03) |
| Sitemap origin | `.aem.live` host until the production domain exists |
| Redirects (per-locale sheets, `/` → `/en/`) | Blocked: no mechanism applies them |
| Site configuration `public.json` paths and `metadata.source` | Requires validation (O-04) |
| Language master review (`REVIEW_*`, `/language-masters/en`) | Temporary; remove after the rollout to `/en` |
| Adobe Tags / Web SDK, consent | Blocked: measurement plan and consent model |
| Form back end, search service | Blocked (O-08, O-09) |
| Non-image DAM assets published | Requires validation (re-audit operations action) |
| Branch protection, required checks, support model and SLAs | Not recorded (O-05, O-13) |
| Universal Editor dialog checks on the real author | Blocked: author access (audit section 9) |

## 15. Related

- authoring.md, administration-and-operations.md
- architecture.md, repository-structure.md, indexing-and-search.md, decision-log.md
- AGENTS.md: environments and publishing process
- package.json, .github/workflows/main.yaml, .github/pull_request_template.md, .husky/pre-commit.mjs, .renovaterc.json
- docs/audit/README.md and its tools
- Handover (TCS-CORP-HANDOVER.md): Part 12 — Publishing, Deployment & Release, Part 13 — Global Troubleshooting Guide (13.15 lint and PageSpeed failures, 13.22 Code Sync), Part 17 — Operations Guide (17.5 Release Activities, 17.6 Validation Activities)

## 16. Screenshots

No screenshot belongs to this document (it describes code, configuration or process). Evidence screenshots are listed in the screenshot register.

## 17. Known limitations

- None recorded for this document.

## 18. Open items

- None.

## 19. Change history

| Version | Date | Author | Change |
|---|---|---|---|
| 1.0 | 8 November 2026 | Delivery team (documentation program) | First version of the role-specific documentation (master documentation prompt), generated from the repository and the published reference pages |

# Integration Architecture

| Attribute | Value |
|---|---|
| Document ID | ARC-004 |
| Audience | Solution architects, technical leads, developers, AI agents |
| Purpose | Integration points: query index and search, Adobe Client Data Layer, Dynamic Media, forms, personalization, external embeds |
| Scope | — |
| Prerequisites | — |
| Owner | Solution Architect (proposed) |
| Approver | Product Owner (to be confirmed) |
| Version | 1.0 |
| Status | Draft for review |
| Last updated | 8 November 2026 |
| Last validated | 8 November 2026 |
| Source files | docs/handover/knowledge-base/indexing-and-search.md, docs/handover/knowledge-base/personalization.md, docs/handover/knowledge-base/data-layer-and-analytics.md, helix-query.yaml |
| Related documents | — |
| Related AEM paths | — |

## 1. Integration status

| Integration | Status |
|---|---|
| Query index and listing (List, Filter List, Cards from metadata) | Implemented |
| Adobe Client Data Layer (page, components, interactions) | Implemented |
| Adobe Tags, Web SDK, Analytics, CJA, Target, consent platform | Not implemented |
| Dynamic Media with OpenAPI delivery of all assets (KB-D-067) | Partially implemented: rendering Implemented, page rewrite Configured but not validated, asset picker Planned (PR #200 open) |
| Forms (Form Variations) submission back end | Not implemented (no back end connected) |
| Personalization | Not implemented |
| CDN redirects / root redirect / to /en/ | Planned (on hold) |
| Production custom domain and Adobe Managed CDN | To be confirmed |

## 2. Query index and search (section)

# 3. Query index definition

`helix-query.yaml` defines **one index**, `pages`, published as `/query-index.json`, for all 13 locales (and, temporarily, the language master under review).

**Included:** `/**`.

**Excluded:**

| Pattern | Why (from the file's comments) |
|---|---|
| `/**.json` | Sheets and other JSON resources |
| `/*/references/**` | Fragments, nav and footer pages, modals of each locale |
| `/*/error-pages/**` | Error pages |
| `/*/data/**` | Data sheets |
| `/*/nav`, `/*/footer` | Nav and footer pages at the locale root |
| `/*/block-library`, `/*/block-library/**` | Block libraries (also excluded from publishing in `paths.json`; safeguard) |
| `/site-global-templates/**` | Template pages (not published; safeguard) |
| `/language-masters/*/references/**`, `/language-masters/*/error-pages/**`, `/language-masters/*/data/**`, `/language-masters/*/nav`, `/language-masters/*/footer`, `/language-masters/*/block-library`, `/language-masters/*/block-library/**` | The same for the language master under review, a folder level deeper (#131, F-108; to remove with the REVIEW settings after the rollout) |
| `/**/fragments/**`, `/**/modals/**` | Fragment and modal folders anywhere |

The `/*/…` patterns name one path segment before the folder (e.g. `/en/block-library`); the language master (`/language-masters/en/…`) is a level deeper and has its own patterns. Until 21 October 2026 it had none: 166 block library pages, 6 reference pages and the 404 page of the language master were in the live index (F-108). After the fix (#131) a reindex reports "requested path does not match index configuration" for these pages but doesn't remove their rows: they were removed with the Admin API index DELETE (173 / 173; the pages stay published). The index went from 354 to 181 rows (137 language-master content pages, 44 `/en` pages), and the Lists and Filter Lists of the library and the curated Insights page still filled.

**Columns** (37 properties, all read from the server-delivered `<head>`):

| Group | Index columns (meta source) |
|---|---|
| Platform / standard | `lastModified` (`last-modified` header, parsed timestamp), `robots`, `title` (`og:title`), `description`, `image` (`og:image`, path only), `tags` (all `article:tag`, multi-value), `cqTags` (`cq-tags`, AEM page Tags as one comma-separated meta) |
| SEO | `metaTitle`, `metaDescription` |
| Classification | `pageType`, `contentType`, `category`, `eyebrow` |
| Author and dates | `author`, `authorId`, `publishDate`, `modifiedDate`, `readingTime` |
| Images | `thumbnailImage`, `featuredImage`, `imageAlt` |
| Audience | `region`, `country`, `language`, `persona`, `campaignId` |
| Featuring | `featured`, `priority`, `featureCard` and `featuredPriority` (earlier names, read as fallbacks) |
| List cards | `cardTitle`, `cardDescription`, `cardImage`, `cardImageAlt`, `cardStyle`, `cardTheme`, `hideInNav` |

Rules:

- Values set by JavaScript (e.g. the SEO decorator) aren't indexed: only the delivered head counts (handover Part 8.1.3).
- `/query-index.json` is public: don't put internal-only values in metadata (handover Part 2.10.7).
- A new metadata column used by a block needs a property here, then a reindex (List README "Extending the Block", Filter List README step 3).

## 4. Query index use in code

`utils/query-index.js` is the only reader of `/query-index.json`.

| Export | Behaviour |
|---|---|
| `getPages()` | Fetches `/query-index.json?offset={n}&limit=500` until `total` is reached (or a request fails or returns no rows); keeps one promise per page view (module variable); filters rows to `indexRoot()`; maps each row to a typed page (`path`, `title`, `description`, `image`, `tags`, `pageType`, `contentType`, `publishDate`, `readingTime`, `thumbnailImage`, `featuredImage`, `author`, `authorId`, `featured`, `priority`, `lastModified`, `row`). A failed load gives `[]` |
| `tagsOf(row)` | Merges `tags` (JSON list or comma list) and `cqTags`, without duplicates (`test/list/query-index-tags.test.mjs`) |
| `columnOf(page, name)` | A column by its metadata name (`featured-image` → `featuredImage`) |
| `listOf(value)` | A multi-value column: JSON array or comma-separated |
| `getPagesAt(paths)` | Pages the index doesn't have (AEM author, or not indexed yet), read from their own HTML in parallel, once per path per page view (`rowOfDocument`); failures are left out |

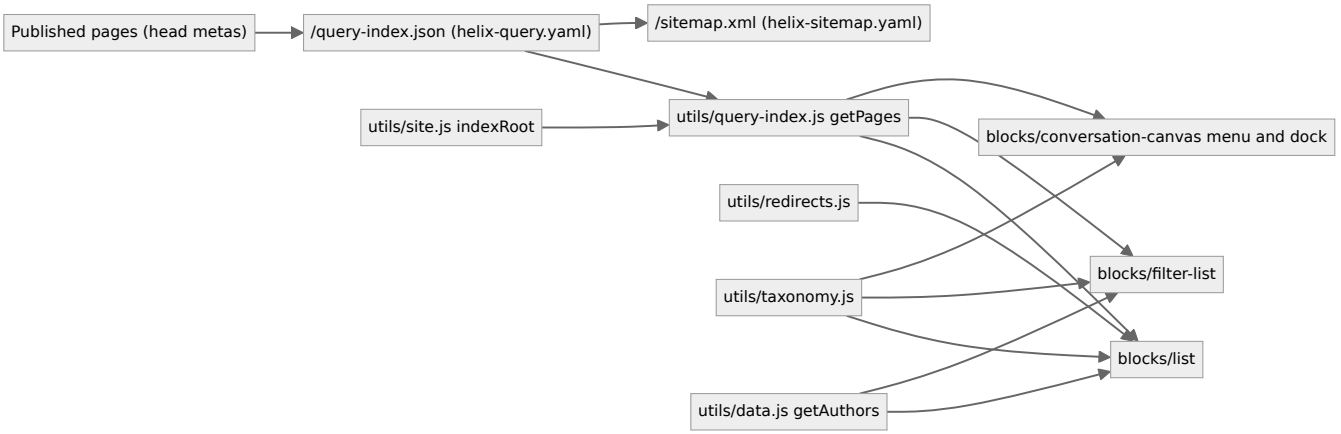
Typed values: `featured` is true for `yes`, `true`, `1`, `on`, `y` in `featured` or `featureCard`; `priority` is the number in `priority`, else `featuredPriority`, else `null`.

**Locale scoping.** `indexRoot()` (`utils/site.js`) returns the review root (`/language-master/en`, `/language-masters/en` or `/design-block-master/en`, `REVIEW_ROOTS`, KB-D-068) for pages below it, else the page's locale root (`/de-de`). `getPages()` keeps rows whose path is that root or below it (`test/site/index-root.test.mjs`). Every page view still downloads the rows of all locales.

**AEM author.** AEM author has no query index (`isAuthor()` in `utils/environment.js`: the path starts with `/content/`); `getPages()` then returns `[]` and blocks show editor hints (below). Fixed-list pages are read with `getPagesAt`.

## 5. Consumers

| Consumer | Uses the index for | Source |
|---|---|---|
| List | Child pages, Tagged pages, card metadata of Fixed list pages | blocks/list/README.md |
| Filter List | All candidate pages (Query Index source) or the Manual List pages' metadata | blocks/filter-list/README.md |
| Conversation Canvas | The pop-up site menu and the Primary dock navigation (`path`, `title`) | blocks/conversation-canvas/README.md |
| Sitemap | `/sitemap.xml` source | `helix-sitemap.yaml` |
| Cards | **Not** a consumer: authored fields only (tag labels from the taxonomy, UI text from placeholders) | blocks/cards/README.md |




### List

- **Sources:** Child pages (below the Parent page or the current page, direct children or down to Child depth), Tagged pages (below the Search root, Any / All of the Tags, never the current page; a tag matches its child tags, `hasTag`), Fixed list (List Items; pages, paths or external URLs).
- **Data rules:** pages with `hide-in-nav = yes` are not listed (`isListable`); card links point to redirect targets (`utils/redirects.js`, at most 5 hops); card values follow the page metadata contract (`card-title` → `og:title`, `card-description` → description, `card-image` → `og:image` / `thumbnail-image` / `featured-image`, …) and a List Item's authored fields win field by field.
- **Sorting:** Order by Manual (authored order for Fixed lists, path order for pages), Title (`Intl.Collator`, numeric, page language) or Last modified (`modified-date`, else last published); Sort order ascending / descending; ties by path. Deduplication by destination path; Maximum items applies last.

### Filter List

- **Sources:** Query Index (pages below Root path, of the selected Page types and Content types, never the current page) or Manual List (Filter Page items in authored order; Sort by ignored).
- **Sorting:** `latest` (`publishDate`, then `lastModified`), `featured`, `priority` (lower first, then latest), `alphabetical` (`Intl.Collator` with the page language).
- **Filtering:** filter pills from Filter Tag items (single or multi select), an advanced filter panel of categories below taxonomy roots (AND / OR), Clear all and Reset filters; filters run in memory and aren't in the URL.
- **Spotlights:** hero (Latest, Featured, Manual) and featured cards (Marked as featured, Priority 1 to 3, Manual).
- **Pagination:** Cards per page (default 9), Load more adds Load more count (default 6) and moves focus to the first new card; "Showing {n} of {total}".
- **Known differences from List:** `hide-in-nav` and redirects aren't applied; the Authors lookup is exact (handover F-30; audit D-A-29 deferred).

### Conversation Canvas

- **Menu:** pages below the Menu root path (default: the locale root) up to Menu depth levels, sorted by level then path (alphabetical), titled by `title`, else the last path segment (`loadMenuTree`). `hide-in-nav` isn't applied (F-33); the menu order follows the path, not a navigation order column (F-78).
- **Search:** submitting dispatches the bubbling, cancelable `conversation-canvas:search` event with `detail.query`; an integration that handles it calls `preventDefault()`; else the authored Search page opens with `?q={query}`; else a polite status "Search isn't available yet." (`searchUnavailable`). Voice input uses the browser's `SpeechRecognition` where available. The data layer records `cmp:search` with the term passed through `redactSearchTerm` (`utils/privacy.js`).
- **No search index or results page exists in the repository:** No verified search service was identified.

## 6. Empty, editor and error states

| Block | Situation | Shown |
|---|---|---|
| List | No cards on the site | Empty behaviour: `message` (default; the authored message, else `listEmpty` "There are no items to show."), `hide` (block hidden; not in the editor or on AEM author), `fallback` (the List Items as a Fixed list) |
| List | AEM author / Universal Editor, Child or Tagged pages | `listEditorPages` "The pages show on the published page (from the query index)." |
| List | Configuration incomplete | Editor hint (warnings `tags-missing`, `parent-missing`, `root-missing`, `items-missing`) |
| Filter List | No matching pages or failed index | `noResources` "No resources found.", hint and Reset filters |
| Filter List | AEM author / Universal Editor | `filterListEditor` "The results show on the published page, from the pages' metadata."; items as chips |
| Conversation Canvas | No pages (including the editor) | `noPagesFound` "No pages found" in the menu |
| All | Index request fails | `getPages()` resolves to `[]`; the block renders its empty state; a failing deferred render is logged and the skeleton removed |

## 7. Performance

- **Prefetch:** `prefetchData(main)` (`scripts/scripts.js`) starts `getPages()` and `loadTaxonomy()` when the lazy phase begins on pages with a List, Filter List or Conversation Canvas, and `loadRedirects()` with a List; not in the eager phase, so it doesn't compete with the LCP image. The blocks reuse the same cached promises.
- **Skeletons:** List and Filter List render through `renderWhenLoaded` (`decorators/skeleton-decorator.js`): data within 150ms renders directly; later data shows a block-shaped skeleton (at least 250ms) so the section and those below aren't held back. The Canvas menus use `skeletonWhile`.
- **Transfer:** every page view with one of these blocks downloads the whole index (all locales) in pages of 500 rows and filters it in the browser; cost grows with the number of published pages (handover Part 2.10.7 and F-30; per-locale indexes are a recommendation, not implemented).
- **Placement:** avoid index-based blocks in the first section (LCP, handover Part 13.16).

## 8. Sitemap

`helix-sitemap.yaml` defines one sitemap, `default`: source `/query-index.json`, destination `/sitemap.xml`, `lastmod: YYYY-MM-DD`, origin `https://main--tcs-corp--sradha-tcs.aem.live` (to be replaced by the production domain once `cdn.prod.host` is configured, per the file's comment).

- It covers every indexed page of all locales; there are no per-locale sitemaps and no `hreflang` alternates (handover Part 4.9.6).
- Handover Part 2.10.7 states that pages with `robots noindex` are left out of the sitemap while they stay in the index and in List / Filter List (no code checks `robots`); the sitemap behaviour is platform behaviour (Requires validation).
- The sitemap is regenerated when the index changes; the Admin API `sitemap` operation regenerates it on demand (handover Part 12.1.9).

## 9. Reindex dependencies

| Event | Index effect | Source |
|---|---|---|
| Page published | Row added or updated | handover Part 12.1.7 (platform) |
| Page unpublished | Row removed | handover Part 12.1.7 (platform) |
| `helix-query.yaml` changed (new column, changed exclude) | Existing rows unchanged until the pages are republished or reindexed | handover Part 12.1.7 |
| Page metadata changed (card title, tags, featured, priority, content type) | Lists change after the page is republished (and reindexed) | List and Filter List READMEs |
| Bulk metadata | Indexed only if the sheet applies (`metadata.source`) and the page is republished | handover Part 8 |

**Verified reindex operation.** The Admin API index operation `POST https://admin.hlx.page/index/{org}/{site}/{ref}/{path}` (org `sradha-tcs`, site `tcs-corp`, ref `main`) re-indexes a published path (handover Part 12.1.9). It was used per page, after `preview` and `live`, for the language master packages (handover revisions 2.31: 312 / 312 pages; 2.35: 313 / 313 and 315 / 315 pages). Calls need an authorised user or API key; no credentials are recorded. Bulk (`/*`) operations act on all locales and must be agreed with the site owner (handover Part 2.10.13).

**Excluded by design.** Block library pages stay out of the query index (handover revision 2.35; `helix-query.yaml` and `paths.json` excludes for the locale block libraries).

**Open finding.** F-10 (handover Part 16): the live index didn't match `helix-query.yaml` (card, eyebrow, category, hide-in-nav, author and feature columns missing) until the site is reindexed. The register still lists it as open; the language master pages were reindexed in revisions 2.31 and 2.35. Whether the 13 locale trees were reindexed is Not recorded.

## 10. Post-reindex validation

A checklist derived from the configuration and code (results of a full run are Not recorded; revision 2.31 recorded 307 language-master pages in the index, 306 with an image):

1. `GET /query-index.json?offset=0&limit=500` on the live host: `total` matches the expected number of published, non-excluded pages; page through until `total`.
2. A row of each locale has the 37 columns of `helix-query.yaml` (e.g. `cardTitle`, `hideInNav`, `cardStyle`, `cardTheme` present, even if empty).
3. No row below `references/`, `error-pages/`, `data/`, `block-library`, `fragments/`, `modals/`, `site-global-templates`, and no `.json` path.
4. Tagged pages have `tags` or `cqTags` values (F-07: AEM may publish page Tags only as `cq-tags`; code merges both).
5. On preview / live (not AEM author): a List (Child pages and Tagged pages), a Filter List and a Conversation Canvas menu of each changed locale show the expected pages; a page with `hide-in-nav = yes` is absent from the List.
6. `/sitemap.xml` lists the pages with the configured origin.

## 11. Related

- architecture.md — runtime phases, caching, integrations
- content-architecture.md — metadata model, taxonomy, locales, references
- blocks.md, data-layer-and-analytics.md — related knowledge base files
- blocks/list/README.md, blocks/filter-list/README.md, blocks/conversation-canvas/README.md, blocks/cards/README.md
- utils/README.md, decorators/README.md, docs/animation/README.md (skeletons)
- AGENTS.md — "Site structure and site data" notes on the query index

- Handover (TCS-CORP-HANDOVER.md): Part 2 — Solution Architecture (2.10.7, 2.11.4), Part 12 — Publishing, Deployment & Release (12.1.7, 12.1.9), Part 13 — Global Troubleshooting Guide (13.9–13.11, 13.25), Part 16 — Handover Findings and Open Items Register

## 12. Personalization (section)

## 13. Verified state

| Item | State | Evidence |
|---|---|---|
| Adobe Target (at.js, Web SDK decisions, mboxes) | Not implemented | No Target, `alloy` or decision code in `blocks/`, `decorators/`, `utils/`, `scripts/`, `templates/` |
| Adobe Journey Optimizer (web channel, offers) | Not implemented | No AJO code |
| AEP Web SDK / Edge Network | Not configured (blocked: measurement plan) | `scripts/delayed.js` holds only a comment; see Data layer and analytics |
| Audiences, segments, visitor profiles | None | No audience source, no identity, no consent (Security, privacy and consent) |
| Decision scopes, activities, offers | None | — |
| Edge or server-side personalization | None | Edge Delivery serves the same cached HTML to every visitor |
| Experimentation (A/B tests) | None | No experimentation plugin or code |
| Personalized impressions in the data layer | None | No personalization events or fields in `decorators/data-layer-decorator.js` |
| Personalization fields in block dialogs | None | No `personalization` field in `models/` or `blocks/*/_*.json` |

## 14. Decisions found

| Decision | Source |
|---|---|
| The Teaser's Personalization option (a class `personalization` and `data-personalization="true"` that nothing read) was removed from the dialog and the code; library content was migrated; Teasers authored with it still render | Handover revision 2.27 (19 October 2026), audit defect D-B-3 (`docs/audit/BLOCK-ECOSYSTEM-AUDIT.md`), re-audit section 1 (`docs/audit/REAUDIT-2026-10-20.md`), `blocks/teaser/README.md` |
| Every block is classified "Not personalizable"; personalized rendering, fallback, flicker, cache-key and analytics tests are blocked until an approved personalization design exists | Audit section 8 |
| Area-specific navigation and footers are content choices (inherited Navigation / Footer pages), not audience personalization | `blocks/header/README.md`, `blocks/footer/README.md` |

Several handover chapters written before 2.27 (for example 1.6.2, 2.9.8, the Teaser chapter 6.11 Variation 66 and 16.4) still describe the Teaser marker; the code and revision 2.27 are authoritative.

## 15. Block support

No block supports personalization: every block (see the inventory in `blocks/README.md`) renders its authored default content, and each block README states "This block does not currently expose verified personalization behavior".

Script hooks that exist but have no caller in the repository:

| Hook | Block | Default without a caller |
|---|---|---|
| `setConversationCanvasSuggestions(block, suggestions)` | Conversation Canvas (Primary, sticky) | The page's tag labels as suggestion pills (`blocks/conversation-canvas/README.md`) |
| `setConversationCanvasFilters(block, tags)` | Conversation Canvas (Secondary) | No filter tags (the list stays hidden) |
| `conversation-canvas:search` DOM event (cancelable) | Conversation Canvas | The Search page with ?q=, else a "not available" status |
| `form-variations:submit` DOM event with `detail.respondWith(promise)` | Form Variations | POST to the Submit action, else "not connected" |

These are integration points (code comments name a personalisation service as one possible caller), not personalization features.

## 16. What a future implementation can build on

| Foundation | What it offers | Source |
|---|---|---|
| Adobe Client Data Layer | Page entry (language, Page Type, tags, path), one component entry per block with a stable id (Anchor ID / Component ID, else `{block}-{n}`), `cmp:*` events | Data layer and analytics |
| Default content of every block | The authored content renders for everyone and is the natural fallback | Block READMEs; Edge Delivery cached HTML |
| Additional tab | Anchor ID (a stable block id and component id for targeting), Hide block, spacing and animation classes on every block model except header and footer | `AGENTS.md` (Additional tab), `decorators/block-options-decorator.js` |
| Page metadata | Page properties and the bulk metadata columns, read by `getPageMetadata()` | `utils/metadata.js` |
| Query index | Metadata columns indexed per page (below) | `helix-query.yaml`, `utils/query-index.js` |
| Delayed phase | `scripts/delayed.js`, imported 3 seconds after load, where the planned Tags / Web SDK embed goes | `scripts/scripts.js` `loadDelayed` |
| Shared DOM events | `tabs:change`, `tabs:reveal`, `header:menu`, `conversation-canvas:search`, `form-variations:submit` | `AGENTS.md` (Shared events and helpers) |

## 17. Data that could feed personalization

| Data | Where it is available today | Current use |
|---|---|---|
| `persona` | Bulk metadata column, `getPageMetadata().persona`, query index `persona` | None in blocks (handover 14.9, item 17) |
| `region`, `country`, `language` | Metadata columns, query index; `xdm:language` in the data layer page entry | `region` selects the contacts of the Media Contacts block (`blocks/media-contacts/media-contacts.js`, `inRegion`): page metadata, not the visitor |
| `campaign-id` | Metadata column, `getPageMetadata().campaignId`, query index `campaignId` | None |
| `page-type`, `content-type`, `category`, tags | Metadata, query index, data layer (`xdm:template`, `xdm:tags`) | Templates, Page Header, List, Filter List |
| `featured`, `priority` | Metadata, query index | Filter List spotlight and featured cards |

| Data | Where it is available today | Current use |
|---|---|---|
| Data sheets (`authors`, `locations`, `partners`, `social-links`, `media-contacts`; `region` columns) | `utils/data.js` | Page Header, footer, Press Release |

All of these describe the page, not the visitor. No visitor attribute (segment, behaviour, geolocation, login) is collected.

## 18. Requirements for any future personalization

Requirements recorded for a future design; none is implemented, and the design itself is a blocked dependency (audit section 9: activities, audiences, the personalization design).

| Area | Requirement | Source |
|---|---|---|
| Default content and fallback | The authored content stays the default and renders when no decision arrives, the service fails or consent is denied | Handover 2.9.8 (failure handling), audit section 7 (Personalization: denied → default content) |
| Consent | No decision request before consent; the consent model and categories must be approved first | Handover 11.7.6, audit section 7 |
| Identity | An identity strategy (Web SDK ECID or other) must be decided; none exists | Not recorded (blocked) |
| Caching | Edge Delivery serves the same cached default content to every visitor; the design must define where variants are applied (browser or edge) and how they interact with the cache (cache keys) | Audit section 8 (cache-key tests blocked) |
| Flicker and performance | Personalizing above-the-fold content conflicts with the eager LCP phase; any pre-hiding needs a performance review; keep the Core Web Vitals targets | Handover 2.9.8 and 11.7.7 |
| Security | Offer content must respect the CSP (`strict-dynamic`) and be inserted with DOM APIs | Handover 2.9.8 |
| Accessibility | Personalized content must meet the same WCAG 2.1 AA rules as the block it replaces (headings, names, focus, placeholders for UI text) | `AGENTS.md` (Accessibility) |
| Analytics | Personalized impressions and decisions must be measurable through the existing data layer (no parallel tracking) | Data layer and analytics |
| Authoring and QA modes | Nothing personalized in Universal Editor edit mode; preview, live and a forced-variant QA mode need defining | Blocked (personalization design) |

## 19. Validation status

| Test | Status |
|---|---|
| Default content of every block renders (no personalization) | Passed (audit runtime runs: 940 page loads, no block not loaded) |
| Personalized rendering, fallback, flicker, cache key, personalized analytics | Blocked: approved personalization design, Adobe Target / AJO access |

## 20. Related

- Architecture, Blocks, Indexing and search, Decision log
- Data layer and analytics
- Security, privacy and consent
- `blocks/README.md` and the block READMEs (Personalization sections)
- `AGENTS.md` (Site structure and site data: metadata columns)
- `docs/audit/BLOCK-ECOSYSTEM-AUDIT.md` (sections 7, 8, 9)

- Handover Part 2 — Solution Architecture (2.9.8), Part 11 — Analytics & Data Layer (11.7.7), Part 16 — Handover Findings and Open Items Register (`TCS-CORP-HANDOVER.md`)

## 21. Screenshots

No screenshot belongs to this document (it describes code, configuration or process). Evidence screenshots are listed in the screenshot register.

## 22. Known limitations

- Adobe Tags, Web SDK, Analytics, Target and a consent platform are not configured; nothing in this document implies they are.

## 23. Open items

- None.

## 24. Change history

| Version | Date | Author | Change |
|---|---|---|---|
| 1.0 | 8 November 2026 | Delivery team (documentation program) | First version of the role-specific documentation (master documentation prompt), generated from the repository and the published reference pages |

# Localization Architecture

| Attribute | Value |
|---|---|
| Document ID | ARC-008 |
| Audience | Solution architects, technical leads, developers, AI agents |
| Purpose | The locales, their published paths, the language master and the review roots, and how code resolves the locale |
| Scope | — |
| Prerequisites | — |
| Owner | Solution Architect (proposed) |
| Approver | Product Owner (to be confirmed) |
| Version | 1.0 |
| Status | Draft for review |
| Last updated | 8 November 2026 |
| Last validated | 8 November 2026 |
| Source files | utils/site.js, paths.json, tools/paths/generate-paths.mjs |
| Related documents | — |
| Related AEM paths | — |

## 1. Locales

| AEM path | Published | lang |
|---|---|---|
| /content/tcs-corp/us/en | /en | en-US |
| /content/tcs-corp/de/de-de | /de-de | de-DE |
| /content/tcs-corp/fr/fr-fr | /fr-fr | fr-FR |
| /content/tcs-corp/ca/fr-ca | /fr-ca | fr-CA |
| /content/tcs-corp/es/es-es | /es-es | es-ES |
| /content/tcs-corp/pt/pt-pt | /pt-pt | pt-PT |
| /content/tcs-corp/nl/nl-nl | /nl-nl | nl-NL |
| /content/tcs-corp/se/sv-se | /sv-se | sv-SE |
| /content/tcs-corp/dk/da-dk | /da-dk | da-DK |
| /content/tcs-corp/no/no-no | /no-no | no-NO |
| /content/tcs-corp/cn/zh-cn | /zh-cn | zh-CN |
| /content/tcs-corp/jp/ja-jp | /ja-jp | ja-JP |
| /content/tcs-corp/kr/ko-kr | /ko-kr | ko-KR |

## 2. Language master and review roots

The language master `/content/tcs-corp/language-masters/en` is the future source of the global site and the regions; whether the regions are created with MSM or carry their own content is open until the content is final (tcs.com's regions don't carry language master content today). The testing root `/content/tcs-corp/language-master/en` and the design root `/content/tcs-corp/design-block-master/en` are not rollout sources.

# 3. Site and locale hierarchy

**Purpose.** One Edge Delivery site serves 13 locale sites by path. Each locale owns its pages and its own references, error pages, sheets, data and block library.

**Structure.** The AEM tree `/content/tcs-corp/{country}/{language}` is published at a language-only path. The locales are defined once in `LOCALES` (`utils/site.js`); the first entry is the default (`DEFAULT_LOCALE`, `en`).

| AEM folder | Published root | lang | | AEM folder | Published root | lang |
|---|---|---|---|---|---|---|
| us/en | /en | en-US | | se/sv-se | /sv-se | sv-SE |
| de/de-de | /de-de | de-DE | | dk/da-dk | /da-dk | da-DK |
| fr/fr-fr | /fr-fr | fr-FR | | no/no-no | /no-no | no-NO |
| ca/fr-ca | /fr-ca | fr-CA | | cn/zh-cn | /zh-cn | zh-CN |
| es/es-es | /es-es | es-ES | | jp/ja-jp | /ja-jp | ja-JP |
| pt/pt-pt | /pt-pt | pt-PT | | kr/ko-kr | /ko-kr | ko-KR |
| nl/nl-nl | /nl-nl | nl-NL | | | | |

```
/content/tcs-corp/
├── {country}/{language}/   a locale site; the locale page itself is Home (published as /{locale}/)
│   ├── …                   content pages
│   ├── references/         nav, footer, alternative navs / footers, footer Canvas page, modals,
fragments
│   ├── error-pages/        404 (used by 404.html); 403 / 500 for a CDN
│   ├── block-library/      block validation pages; excluded from publishing
│   ├── placeholders        sheet: Key, Text, Description, Category
│   ├── metadata            sheet: bulk metadata
│   ├── taxonomy            taxonomy page (tag, title)
│   ├── redirects           sheet: Source, Destination
│   └── data/               sheets: authors, locations, partners, social-links, media-contacts
├── language-masters/en/     temporary: published at /language-masters/en for review
└── site-global-templates/  template pages authors copy; excluded from publishing
```

**Path mapping (`paths.json`, generated by `npm run build:paths` from `LOCALES` and `REVIEW_ROOTS`).** One generic set per site root: the folder mapping (`/content/tcs-corp/de/de-de/:/de-de/`: child pages and sheets), the Home page mapping (`/content/tcs-corp/de/de-de:/de-de/`) and the taxonomy page (`…/taxonomy:/de-de/taxonomy.json`): 39 mappings for the locales, 3 temporarily for the language master. The sheets (`placeholders`, `metadata`, `redirects`, `data/*`) need no mapping of their own: the folder mapping publishes them at the same path and spreadsheets get `.json` appended; the former 112 per-sheet mappings are kept commented in the generator. `includes`: the 13 locale pages, `/content/dam/tcs-corp/` and `/content/tcs-corp/language-masters/en`; `excludes`: each locale's `block-library` and `block-library/**`, and `site-global-templates` with its children. AEM applies the site configuration's `public.json` → `paths`, not the repository file (AGENTS.md "Notes").

**Locale resolution (code).** `getLocale(pathname)` reads the AEM path when editing (`/content/tcs-corp/de/de-de/…`) and the published path on the site (`/de-de/…`); pages outside a locale (Site Global Templates, drafts) get `en`. Helpers: `localePagePath`, `resolvePagePath`, `localeSheetPath`, `toPublishedPath`, `toAemPath`, `pageLink`, `getParentPages`, `sheetRows`.

**Review roots (KB-D-068).** There are three review roots, each published as a site of its own (KB-D-068): `/language-master/en` (testing page functionality with dummy content, content limitations), `/language-masters/en` (the language master: the migrated tcs.com content, validated with its real sheet data, later the source of the global site and the regions and languages; MSM or not is open until the content is final) and `/design-block-master/en` (the block library and the curated design pages). Pages below each (`REVIEW_ROOTS`) list their own pages from the query index (`indexRoot`), read their own placeholders, redirects, taxonomy and data sheets (`REVIEW_SHEETS`), use their own `references/` and `error-pages/` (`REVIEW_PAGES`) and inherit up to their

root page. The temporary review of the language master ends with its rollout (handover Part 3.9, finding F-88).

**Language master URLs (validated 23 October 2026).** The mappings follow Adobe's path mapping rules (https://www.aem.live/developer/authoring-path-mapping): the folder mapping `/content/tcs-corp/language-masters/en/:/language-masters/en/` covers every child page, and the page mapping `/content/tcs-corp/language-masters/en:/language-masters/en/` is the special case that publishes the root page as `/language-masters/en/index`, served at `/language-masters/en/` (the same pattern as `/en/`). No other mapping is needed. The mappings of every language master come from `REVIEW_ROOTS` (`utils/site.js`): `tools/paths/generate-paths.mjs` writes this set for each root, so another language master (e.g. `/language-masters/de`) is added to the code and to `paths.json` with one entry and `npm run build:paths`; `test/site/paths.test.mjs` checks every locale and language master is mapped this way and that `paths.json` is in step.

| AEM page | Preview | Live |
|---|---|---|
| `/content/tcs-corp/language-masters/en` (Home) | `https://main--tcs-corp--sradha-tcs.aem.page/language-master/en/` | `https://main--tcs-corp--sradha-tcs.aem.live/language-master/en/` |
| `/content/tcs-corp/language-masters/en/<path>` (child pages) | `https://main--tcs-corp--sradha-tcs.aem.page/language-master/en/<path>` | `https://main--tcs-corp--sradha-tcs.aem.live/language-master/en/<path>` |

The Home URL ends with `/` (`/language-masters/en` without it is a 404, as `/en`). On 23 October 2026 the child pages, sheets, references and error pages resolved, but the Home page was a 404 and wasn't in the query index: it has to be previewed and published from AEM (Manage Publication on `/content/tcs-corp/language-masters/en`). If it still returns a 404 after publishing, check that the site configuration's `public.json` → `paths` has the page mapping (AEM applies that configuration, not the repository file).

**Rules.**

- Add or change a locale only in `LOCALES`, then run `npm run build:paths` and commit both files; the site administrator re-applies `public.json` and `metadata.source`.
- Nothing is published at `/`: a `/` → `/en/` redirect needs another mechanism (handover F-36).

**Common mistakes.** Editing `paths.json` by hand (it is overwritten by the generator); expecting a repository `paths.json` change to take effect without the site configuration; creating pages directly under `/content/tcs-corp` (not published).

## 4. Localization

| Topic | Implementation | Status |
|---|---|---|
| Locale structure | 13 separate page trees; locale resolved from the path | Active |
| `<html lang>` | `language` metadata when valid, else the locale's tag | Active |
| UI text | Placeholders sheet per locale; English fallback in code | Active |
| Tag labels | Taxonomy page per locale, language sheet when translated | Active (only `/en` taxonomy published when checked, F-09) |
| Bulk metadata | One sheet per locale; all 13 in `metadata.source` | Requires validation (F-11, open item O-04) |
| Data sheets | One set per locale | Active |
| Lists | Each locale lists only its own pages (`indexRoot`) | Active |
| Dates | `utils/dates.js` `formatDate`; English month abbreviations in every locale (handover Part 4.9.7) | Active |
| Translation process, MSM / language copies, language selector, `hreflang` | Not in code | Not recorded |

## 5. Screenshots

No screenshot belongs to this document (it describes code, configuration or process). Evidence screenshots are listed in the screenshot register.

## 6. Known limitations

- None recorded for this document.

## 7. Open items

- Decide MSM or separate regional content once the language master content is final (owner: Product Owner).

## 8. Change history

| Version | Date | Author | Change |
|---|---|---|---|
| 1.0 | 8 November 2026 | Delivery team (documentation program) | First version of the role-specific documentation (master documentation prompt), generated from the repository and the published reference pages |

# Performance Architecture

| Attribute | Value |
|---|---|
| Document ID | ARC-007 |
| Audience | Solution architects, technical leads, developers, AI agents |
| Purpose | Core Web Vitals requirements, loading strategy, images, fonts, layout shift prevention and budgets |
| Scope | — |
| Prerequisites | — |
| Owner | Solution Architect (proposed) |
| Approver | Product Owner (to be confirmed) |
| Version | 1.0 |
| Status | Draft for review |
| Last updated | 8 November 2026 |
| Last validated | 8 November 2026 |
| Source files | docs/handover/knowledge-base/performance.md, scripts/scripts.js, styles/styles.css |
| Related documents | — |
| Related AEM paths | — |

## 1. Requirements (Core Web Vitals and NFRs)

| Requirement | Value | Source | Enforcement |
|---|---|---|---|
| PageSpeed Insights / Lighthouse score | 100, mobile and desktop, on the feature preview URL before merge | AGENTS.md "Publishing Process" step 3, handover 1.7.1 | Process: the aem-psi-check app on pull requests; CI (`.github/workflows/main.yaml`) runs lint only |
| Largest Contentful Paint | < 2.5 s (RUM, p75) | handover 13.16 (escalation thresholds), revision 2.36 ("the 2.5 s NFR") | Code: eager phase loads only the first section and its first image |
| Cumulative Layout Shift | ≤ 0.1 (< 0.1 at p75 in RUM) | handover 13.16, revision 2.36 ("NFR ≤ 0.1") | Code: header space reservation, footer hidden until sections load, metric-matched fallback fonts |
| Interaction to Next Paint | escalate above 200 ms at p75 | handover 13.16 | No specific code rule; long tasks avoided while decorating |
| Three-phase loading, per-block code splitting, module preloading, no forced layout while decorating, image optimisation, no third-party scripts on load, shared data loaded once and early, RUM | as implemented below | handover 1.7.1 table | Code (module graph also checked by `npm test`) |

## 2. Measured evidence

Recorded in handover revision **2.36** (20 October 2026) and finding **F-106** (closed in 2.36):

| Measurement | Before | After |
|---|---|---|
| Lighthouse 12 on aem.live (curated Home, Insights, What we do, a block library page), mobile / desktop | 96–100 / 100; LCP 1.2–1.4 s mobile; CLS ≤ 0.011; Total Blocking Time up to 220 ms | 99–100 mobile, 100 desktop (Home and What we do) |
| Local preview, 4× CPU and mobile network | LCP 1.9–2.6 s; all sections 4.4–5.4 s; blocking 60–80 ms | LCP 1.7–1.95 s; all sections 3.5–3.8 s; blocking 25–48 ms |
| aem.live, branch against main, same throttling, 3 runs: Home | LCP 1.83–1.88 s; header 3.30–3.43 s; all sections 3.89–4.04 s | LCP 1.12–1.61 s; header 2.05–2.51 s; all sections 2.34–2.77 s (about 37 % sooner) |
| aem.live, same: What we do | LCP 2.36–2.43 s; all sections 3.42–3.50 s | LCP 0.94–1.00 s; all sections 1.60–1.68 s (about 52 % sooner) |
| Font swap on a block library page with a long intro (mobile, Android / Linux metrics) | CLS 0.33 (paragraph 230 → 205 px) | CLS 0.003 (205 → 205 px) |
| Regression | — | 955 loads: 0 errors, 0 axe violations, 0 overflow; 791 / 791 test clicks |

Causes found (F-106): no build step, so the 21 modules of `scripts.js` loaded over 4 network round trips before the first section and each block's (up to 42 modules) over up to 7 more; one 225 ms forced layout per List scroller; the Canvas controller rewrote `<html>` custom properties on every resize during loading (100–120 ms tasks); header block stylesheets were requested only after its fragments were decorated. Earlier layout-shift fixes: F-90 (footer, sticky Anchor List, CLS 0.35, 2.13), F-93 (header, CLS 0.14–0.25 → below 0.01, 2.15).

## 3. Lab and field data

| Kind | Tool | Status |
|---|---|---|
| Lab | Lighthouse / PageSpeed Insights (pull requests, manual runs), throttled browser runs recorded in 2.36 | Recorded for the pages above |
| Field (RUM) | `sampleRUM` in `scripts/aem.js` (Edge Delivery Operational Telemetry): about 1 in 100 page views (`weight` 100), checkpoints to `ot.aem.live`, the RUM enhancer loaded for sampled views after the first section (`sampleRUM.enhance()` in `loadMainSections`); `404.html` sends a `404` checkpoint | Implemented; **no field results are recorded** in the repository or handover; dashboard access and domain key "[To be confirmed]" (handover 2.10.14) |

Lab figures don't replace field data: low-traffic locales give few samples (handover 2.10.14). Read RUM before and after a release when it becomes available.

## 4. JavaScript loading




| Phase | What runs (`scripts/scripts.js`) |
|---|---|
| Eager (`loadEager`) | `decorateHead` (SEO tags, `<html lang>`), `trackInteractions` (data-layer listeners and page entry), template / theme / page-type classes, the 404 content or the Page Type template (`templates/{name}`), `decorateMain`, `reserveHeaderSpace`, `body.appear`, `preloadBlockModules(first section)`, then the first section with `prepareMotion` and `waitForFirstImage` (sets the first image `loading="eager"` and waits for it); `fonts.css` only on wide viewports or when cached |
| Lazy (`loadLazy`) | Modal auto-links, `preloadBlockModules` for the rest of `main` and the chrome blocks (`header`, `footer`, `fragment`, `tcs-header`, `conversation-canvas`), `prefetchData`, `loadHeader`, `loadFooter` (hidden), `loadMainSections` (each section in order, animations prepared before it shows), then footer shown, `trackBlocks(main)`, Go back on 404 pages, hash scroll, `lazy-styles.css`, `loadFonts` |

| Phase | What runs (scripts/scripts.js) |
|---|---|
| Delayed (loadDelayed) | After 3 s: `trackBlocks(document)` (blocks added later, such as header, footer, modals) and `import('./delayed.js')`, which is empty ("add delayed functionality here") |

- **Per-block code splitting**: `loadBlock` (`scripts/aem.js`) loads a block's CSS and JS in parallel only when the block is on the page; blocks of a section load one after another, and sections load in order.
- **Module preloading**: there is no build step, so each import level is a network round trip. `head.html` has 20 `modulepreload` links (the modules `scripts.js` imports); `preloadBlockModules` adds `modulepreload` for each block's modules and a `preload as="style"` for its stylesheet from the generated `scripts/module-graph.js`. Stylesheet preloads go in the **body**, because `loadCSS` skips any stylesheet whose href is already linked in the head. Run `npm run build:modules` after any import change; `npm test` (`test/modules/module-graph.test.mjs`) fails when the graph or the head links are stale (AGENTS.md "Setup Commands").
- **Dynamic imports**: on-demand modules are imported only when needed: `utils/inherited-metadata.js` (404 pages), `utils/placeholders.js` (Go back), `utils/query-index.js`, `utils/taxonomy.js`, `utils/redirects.js` (`prefetchData`), `blocks/modal/modal.js` (modal links and the video pop-up), template modules; the Lottie player (`scripts/lib/lottie`, vendored) loads when an animation is near the viewport.

## 5. CSS loading

| Stylesheet | When it loads |
|---|---|
| `styles/styles.css` | Render-blocking `<link>` in `head.html` (tokens, global styles, sections, buttons, theme contract, fallback font faces, skeletons, Additional tab classes) |
| `blocks/<name>/<name>.css` | With the block (`loadBlock`), preloaded by `preloadBlockModules` (`blockStylesheet`) |
| `styles/card-list.css` | `@import` from `blocks/list/list.css` and `blocks/cards/cards.css` (a chained request after the block stylesheet; not in the preload list) |
| `templates/<name>/<name>.css` | With the Page Type template in the eager phase (`templates/blog/blog.css` imports `article.css`) |
| `styles/motion.css` | Only on pages with an animated block (`prepareMotion`) |
| `styles/fonts.css` | Eager on viewports ≥ 900px or when the session flag is set, else end of the lazy phase |
| `styles/lazy-styles.css` | End of the lazy phase (currently only the Universal Editor item placeholder styles) |

## 6. Images

- Authored images are optimised `/media_…` renditions served as `<picture>` with WebP and fallback sources (`createOptimizedPicture`: 2000px from 600px, 750px below; `format=webply&optimize=medium`).
- The first image of the first section is the LCP candidate: `waitForFirstImage` sets `loading="eager"` on it and waits for it before the lazy phase. The Hero sets `loading="eager"` and `fetchPriority="high"` on its image when it opens the page; the Page Header does the same for its featured image; List / Cards "Image loading" Eager gives only the first image a high fetch priority (`decorators/card-list-decorator.js`).
- `buildImage` (`decorators/responsive-image-decorator.js`) optimises same-origin media URLs and lazy-loads other URLs; section backgrounds use 2000px / 750px. Open: a featured image that isn't site media stays lazy (handover F-14).
- Icons are lazy `<img>` (16 × 16 attributes). Images committed to git must be optimised (AGENTS.md "Performance"); the only font file is 48,256 bytes.

## 7. Fonts

- One self-hosted variable font (`fonts/inter-variable.woff2`, `font-display: swap`, Latin `unicode-range`); no external font service.
- Eager only when `window.innerWidth >= 900` or `sessionStorage` `fonts-loaded` is set (set after the first load, except on localhost); otherwise loaded at the end of the lazy phase, so the font doesn't compete with the LCP image on small screens.
- Text first renders in a fallback face scaled to Inter: `inter-fallback` (Arial, Liberation Sans), `inter-fallback-roboto` (Android), `inter-fallback-dejavu` (Linux), in `styles/styles.css` (values in design-system.md). Keep the faces when the font changes, or text reflows when Inter arrives (2.36: CLS 0.33 → 0.003).

## 8. Layout shift prevention

| Source of shift | Prevention |
|---|---|
| Header (lazy phase) | `header:not(:has(.nav-wrapper))` keeps `--header-reserved` (the logo bar `--nav-height`: 4rem, 7.75rem from 1200px) or the height remembered for the breakpoint (`utils/header-space.js`, `localStorage` key `header-height-v2-<breakpoint>`); from 1200px the header lies over a light first hero out of the flow; the header Canvas lies on the hero and takes no header space |
| Header / footer content | `header .header` and `footer .footer` stay `visibility: hidden` until `data-block-status="loaded"` |
| Footer | `display: none` until all main sections have loaded |
| Fonts | Metric-matched fallback faces |
| Data blocks (List, Filter List) | Skeletons shaped like the block (`renderWhenLoaded`) when data takes longer than 150 ms; shown at least 250 ms so they don't flash |
| Sticky Anchor List | Doesn't pin while its section is hidden (F-90) |
| Animations | Only `opacity`, `transform`, `clip-path`; first frames applied only on screen; Background animation on the first hero delays the LCP image paint (F-94: use it below the fold) |

## 9. DOM size, listeners and observers

- DOM size: no measurement or limit is recorded. Skeletons add about 20–40 small spans per block, removed with the skeleton; animations add no wrapper elements (animation report, section I).
- Data-layer listeners are delegated: one capturing listener per event type on `document` (`click`, `toggle`, `close`, `play` / `pause` / `ended`, `submit`, `change`), registered once (`trackInteractions`).
- Observers: one IntersectionObserver for all animations (`decorators/motion-decorator.js`); `utils/viewport.js` `onVisible` / `watchVisibility` for deferred work (Embed, Video, Gen Carousel, Lottie in `decorators/animation-decorator.js`); own IntersectionObservers in the Separator (motion) and the Modal (inside the editor); ResizeObserver in scrollers, masonry, Table, Anchor List, Header and the Canvas controller (resizes measured once they settle).
- Scroll listeners are passive and scheduled (Anchor List, Canvas controller); the Canvas never measures while scrolling.
- Layout reads are batched: scrollers measure in one shared animation frame, reads before writes; the Canvas controller writes `<html>` custom properties only when they change (AGENTS.md module preloading paragraph).

## 10. Query and data requests

| Data | Loading |
|---|---|
| Query index (`/query-index.json`) | `utils/query-index.js` `getPages()`: fetched once per page view in pages of 500 rows (`offset` / `limit`), kept as an in-memory promise, filtered to the locale (or the review root) |
| Taxonomy, redirects, placeholders, data sheets | Fetched once per page view, cached in memory |
| Inherited metadata (Navigation, Footer, 404 Page) | Parent pages' heads loaded in parallel, cached per browser session (`sessionStorage`) |
| Prefetch | `prefetchData` starts the query index and taxonomy (and, with a List, the redirects) when the lazy phase begins on pages with a List, Filter List or Conversation Canvas; not in the eager phase, so it doesn't compete with the LCP image |
| Late data | `renderWhenLoaded`: data within 150 ms renders directly; else a skeleton, so the section is shown and the sections below aren't held back (a 1.5 s index showed the sections below a List at 0.5 s instead of after the index) |

## 11. Analytics dispatch

The Adobe Client Data Layer (`window.adobeDataLayer`) is filled by `decorators/data-layer-decorator.js`: the page entry and interaction listeners in the eager phase (`trackInteractions`), the blocks of `main` once its sections have loaded (`trackBlocks(main)`), and the rest (header, footer, later blocks) in the delayed phase after 3 s (`trackBlocks(document)`). No analytics library consumes it: Adobe Tags and the Web SDK are **not configured** (blocked: measurement plan), and `scripts/delayed.js` is empty, so there is no tag-manager cost today. When Tags is added, load it from the delayed phase (`scripts/delayed.js`) and re-measure.

## 12. Personalization

None: no Adobe Target, AJO or edge personalization runtime exists, so there is no flicker, anti-flicker snippet or personalization request. Every page is the cached default content.

## 13. Third-party code

| Code | When |
|---|---|
| `head.html` | No external script; a CSP with `nonce-aem` and `strict-dynamic` |
| X (Twitter) widgets | `platform.twitter.com/widgets.js` only when an X embed is rendered (`blocks/embed/embed.js`) |
| YouTube / Vimeo | Behind a placeholder with a Play button when a placeholder image is authored, else loaded when the block is visible (`onVisible`); privacy-enhanced domains |
| Lottie player | Vendored (`scripts/lib/lottie/lottie_light.min.js`), loaded when an animation is near the viewport |
| RUM enhancer | Loaded for sampled views only |
| DOMPurify | `scripts/dompurify.min.js`, used by the Universal Editor support code |

## 14. Budgets

**No verified performance budget** (bytes, requests, long tasks) is defined in the repository, CI or handover. The working targets are the NFRs above (PSI 100, LCP < 2.5 s, CLS ≤ 0.1, INP ≤ 200 ms at p75). Proposed budgets are a recommendation, not a decision.

## 15. Testing tools and evidence

| Tool | Use | Where recorded |
|---|---|---|
| PageSpeed Insights (aem-psi-check on pull requests) | Score and diagnostics for the preview URL in the PR description; a missing or wrong URL fails it | handover 13.15; PR descriptions |
| Lighthouse (local, Chrome) | Lab scores with mobile / desktop presets | handover revision 2.36 |
| Throttled browser runs (4× CPU, mobile network) | LCP, header time, all-sections time, blocking time on the local preview and on aem.live (branch vs main, 3 runs) | handover revision 2.36 |
| Performance panel recording | Long tasks, forced layouts, the block that decorates last in the first section | handover 13.16 diagnosis |
| Audit harness (`docs/audit/tools/harness.mjs`) | Every block library and curated page at 1440 / 1024 / 768 / 600 / 390px: console errors, failed requests, block status, overflow, broken images, axe, data layer, clicks | audit section 2, re-audit section 2; uses delivery-workspace paths (Requires validation before re-use) |
| `npm test` | `test/modules/module-graph.test.mjs` keeps the preload graph and `head.html` current | repository |
| RUM (`sampleRUM`) | Field Core Web Vitals | No results recorded |

No Lighthouse or performance script is committed to the repository; the 2.36 runs are documented in the handover only.

## 16. Required performance validation

Validate before merging (AGENTS.md "Publishing Process"; handover 12.4 release checklist and 13.16):

| Change | Validate |
|---|---|
| A block's JS or CSS | PageSpeed Insights 100 on the feature preview of a page using the block (mobile and desktop); `npm run build:modules` if imports changed, then `npm test`; no forced layout while decorating (reads before writes in one frame); no new eager work in the first section; the block README's "Performance Considerations" updated |
| A dialog option or combination | The block library page of the option, first section and below the fold; options that load media or data (Image loading Eager / Lazy, Background animation, Lotto, autoplay video, List sources) checked for LCP and CLS; the dialog options report "Responsive and performance" section is the model (CSS-only options: no new requests, listeners or blocking resources) |
| Block library pages | The harness run (0 errors, 0 failed requests, 0 overflow) and Lighthouse on a representative library page |
| Curated design pages and key templates (Home, Insights, What we do, article) | Lighthouse mobile / desktop and a throttled run against main on aem.live; CLS from the font swap on Android-like metrics (long intro paragraphs) |
| A new third-party script | Loaded from `scripts/delayed.js` or on interaction / visibility, never in `head.html`; CSP compatible |
| Fonts | Keep the fallback faces; measure CLS on mobile |

Author guidance that affects performance: keep the LCP image in the first block (Hero or Page Header); set List / Cards "Image loading" to Eager only for lists in the first section; avoid page lists from the query index and Background animation as the first block (handover 13.16, F-94).

## 17. Gaps and recommendations

| Gap | Status | Proposed next step (not an approved decision) |
|---|---|---|
| Field data (RUM) results and dashboard access | Requires validation: RUM domain key and dashboard access "[To be confirmed]" | Review p75 LCP / CLS / INP per locale after each release |
| Performance budgets | Not recorded | Define budgets (JS / CSS bytes, requests, long tasks) per template |
| PSI is a process check, not enforced by CI | Recorded (handover 1.7.1) | Keep the PR preview URL mandatory |
| `styles/card-list.css` is an `@import` of the List and Cards stylesheets, outside the preload graph | Observation | Consider preloading it with the block stylesheet |
| Featured image from a non-site URL stays lazy (F-14) | Open | Apply loading attributes to plain `img` too |
| Harness and performance scripts tied to the delivery workspace | Requires validation | Commit runnable versions under `tools/` |

## 18. Related

- design-system.md: fonts and fallback faces, tokens
- accessibility.md: skeleton and loading states, reduced motion
- Other knowledge base pages: architecture.md, testing-and-quality.md, troubleshooting.md, data-layer-and-analytics.md, security-privacy-consent.md, deployment-and-release.md
- AGENTS.md "Three-Phase Page Loading" and Performance
- `scripts/scripts.js`, `head.html`, `scripts/module-graph.js`, `tools/modules/generate-module-graph.mjs`, `decorators/skeleton-decorator.js`, `utils/query-index.js`
- Each block README's "Performance Considerations" (`blocks/README.md`)
- Animation report (I. Performance), block ecosystem audit
- Handover: Part 1 — 1.7.1 Performance, 2.10.14 Operational Telemetry (RUM), 13.15 and 13.16 (troubleshooting), Part 16 — F-106, revision 2.36

## 19. Screenshots

No screenshot belongs to this document (it describes code, configuration or process). Evidence screenshots are listed in the screenshot register.

## 20. Known limitations

- No PageSpeed or field measurement was taken in this program (Not recorded).

## 21. Open items

- None.

## 22. Change history

| Version | Date | Author | Change |
|---|---|---|---|
| 1.0 | 8 November 2026 | Delivery team (documentation program) | First version of the role-specific documentation (master documentation prompt), generated from the repository and the published reference pages |

# Security Architecture

| Attribute | Value |
|---|---|
| Document ID | ARC-005 |
| Audience | Solution architects, technical leads, developers, AI agents |
| Purpose | Security boundaries, sanitisation, link safety, CSP, secrets, storage, data minimisation and consent dependencies |
| Scope | — |
| Prerequisites | — |
| Owner | Solution Architect (proposed) |
| Approver | Product Owner (to be confirmed) |
| Version | 1.0 |
| Status | Draft for review |
| Last updated | 8 November 2026 |
| Last validated | 8 November 2026 |
| Source files | docs/handover/knowledge-base/security-privacy-consent.md, utils/privacy.js |
| Related documents | — |
| Related AEM paths | — |

This file summarizes the security and privacy measures implemented in the front-end code of the TCS Corporate Website (content sanitization, link safety, external resources, the Content Security Policy, storage, cookies, query strings, logging and data minimisation) and records what is not implemented (consent, identity). It describes the code's behaviour only: no legal classifications, retention rules or approvals are recorded here, and decisions that need privacy or legal input are listed as blocked.

## 1. Status at a glance

| Area | Status | Where |
|---|---|---|
| Universal Editor update sanitization (DOMPurify) | Active | `scripts/editor-support.js` |
| Same-origin fragment and error page loading | Active | `blocks/fragment/fragment.js`, `scripts/scripts.js` `loadPageContent` |
| Safe URL check for card links | Active | `utils/card-list-config.js` `isSafeUrl` (List, Cards) |
| New-tab links with `noopener noreferrer` and a screen reader hint | Active | `decorators/link-decorator.js` |
| Privacy-enhanced video players | Active | Embed (always), Video (Privacy mode, default on) |
| Content Security Policy (nonce, `strict-dynamic`) | Active | `head.html`, `404.html` |
| Secrets in the repository | None found | — |
| Search term redaction, field names only in form events | Active | `utils/privacy.js`, `decorators/data-layer-decorator.js` |
| Cookies set by project code | None | — |

| Area | Status | Where |
|---|---|---|
| Consent management, visitor identity | Not implemented (blocked: consent model) | — |
| Generic iframe embeds: host allow-list, `sandbox` | Not implemented (known limitation) | `decorators/media-decorator.js` `createFrame` |

## 2. Content sanitization

| Path | What happens | Code |
|---|---|---|
| Universal Editor updates | The HTML returned by the editor after an edit is sanitized with DOMPurify (`USE_PROFILES: { html: true }`) before it is parsed and re-decorated; DOMPurify is loaded on demand with `loadScript` | `scripts/editor-support.js` `applyChanges` |
| Rich text in the editor | `decorateRichtext` groups rich text instrumentation into editable wrappers; it moves attributes and does not sanitize | `scripts/editor-support-rte.js` |
| Fragments (header, footer, Modal, Tabs, Fragment block) | `loadFragment` fetches only paths that start with `/` and not `//` (`{path}.plain.html`, same origin) and sets the response as `innerHTML`; a fragment that includes itself or the current page isn't loaded again | `blocks/fragment/fragment.js` |
| Error pages | `loadPageContent` fetches same-origin `.plain.html` paths (the inherited 404 Page, else the locale's `error-pages/404`) into `main` | `scripts/scripts.js` |
| Block rendering | Blocks build elements with DOM APIs; players are built without writing authored URLs into HTML strings | `decorators/media-decorator.js` |

DOMPurify is the only user of `scripts/dompurify.min.js`; it is not loaded on the public site. The `innerHTML` uses in `fragment.js` and `scripts.js` rely on Edge Delivery `.plain.html` responses from the same origin (authored content). `scripts/aem.js` (never modified) also uses `innerHTML` in `buildBlock` when a block is built from strings (auto-blocking).

## 3. Link safety

| Measure | Behaviour | Code |
|---|---|---|
| `isSafeUrl(href)` | Allows `http:`, `https:`, `mailto:`, `tel:` and relative paths (`/`, `./`, `../`, `#`, `?`, a word); any other scheme (`javascript:`, `data:`, …) is rejected and the card is not rendered | `utils/card-list-config.js`, used by `blocks/list/list-data.js` and `blocks/cards/cards-data.js` |
| New tab | `setLinkTarget` sets `target="_blank"` with `rel="noopener noreferrer"`; `announceNewTab` adds the "opens in a new tab" hint from the placeholders | `decorators/link-decorator.js` |
| Logo links | A TCS Header logo link typed as a bare domain (`www.example.com`) gets `https://` so it isn't a relative link | `blocks/tcs-header/tcs-header.js` `buildLogo` |
| Fragment paths | Only same-origin absolute paths (`/…`, not `//…`) | `loadFragment` |
| AEM paths | `toSitePath` maps same-origin AEM paths to published paths; other origins pass through unchanged | `utils/urls.js` |

Links outside List and Cards are rendered as AEM delivers them; no other block runs a scheme check (Requires validation: how AEM delivers a `javascript:` link authored in other fields).

## 4. External resources

| Resource | Behaviour | Code |
|---|---|---|
| YouTube (Embed) | Always the privacy-enhanced host `www.youtube-nocookie.com`; only the video id is passed on (`rel=0`, `enablejsapi=1`) | `blocks/embed/embed.js`, `utils/media.js` |
| YouTube (Video) | `youtube-nocookie.com` when the Privacy mode field is on (default on), else `www.youtube.com` | `blocks/video/video.js` |
| Vimeo | `player.vimeo.com` with `dnt=1` (Embed always; Video with Privacy mode); the video id is URL-encoded | `utils/media.js`, `blocks/embed/embed.js`, `blocks/video/video.js` |
| X (Twitter) posts | `https://platform.twitter.com/widgets.js` is loaded only when an X embed renders; `x.com` links are rewritten to `twitter.com` | `blocks/embed/embed.js` `embedTwitter` |
| Any other Embed URL | A generic lazy iframe (`allow="encrypted-media"`, fullscreen allowed) with no host allow-list and no `sandbox` attribute: a known limitation; only approved, framable URLs should be authored | `decorators/media-decorator.js` `createFrame`, `blocks/embed/README.md` |
| Embed loading | With a placeholder image the player loads only after the visitor presses Play; without one, when the block scrolls into view | `blocks/embed/embed.js` |
| Form submissions | A valid Form Variations submission goes to a page integration (`form-variations:submit`), else is POSTed as JSON to the authored Submit action URL (no allow-list), else the visitor is told the form isn't connected | `blocks/form-variations/form-variations.js` |
| RUM | `sampleRUM` sends sampled checkpoints with `navigator.sendBeacon` to the RUM collector (`ot.aem.live`) and loads the RUM enhancer | `scripts/aem.js` |
| Fonts, Lottie player, DOMPurify | Served from the repository (no font or script CDN) | `fonts/`, `scripts/lib/lottie/`, `scripts/dompurify.min.js` |

Third-party players and widgets set cookies and collect data as their providers do; no consent gate exists before they load (handover open item O-16: approved embed domains and consent for third-party players).

## 5. Content Security Policy

`head.html` (all pages) and `404.html` declare:

```
script-src 'nonce-aem' 'strict-dynamic' 'unsafe-inline' http: https:; base-uri 'self'; object-src 'none';
```

| Directive | Effect |
|---|---|
| `'nonce-aem'` with `'strict-dynamic'` | Only the `<script nonce="aem">` tags (and `modulepreload` links with the nonce) and the scripts they load run; `'unsafe-inline' http: https:` are fallbacks ignored by browsers that support `strict-dynamic` |
| `base-uri 'self'` | Prevents <base> injection |
| `object-src 'none'` | Blocks plugins |
| `move-to-http-header="true"` | Edge Delivery sends the policy as a response header |
| Not set | `style-src`, `img-src`, `frame-src`, `connect-src` (embeds, form POSTs and RUM aren't restricted by the policy) |

A future tag manager must be loaded by project code (`scripts/delayed.js`) so `strict-dynamic` trusts it (handover 2.7.1, 13.23 Content Security Policy Blocking Scripts).

## 6. Secrets and environment configuration

- No secrets were found in the repository (no API keys, tokens or passwords in the code or configuration); everything in it is public client-side code. Keep it that way: an integration that needs credentials needs a server-side or Adobe-managed component.
- Configuration is code and site configuration, not environment variables: `fstab.yaml` (the content source), `paths.json` (generated from `utils/site.js`), `helix-query.yaml`, `helix-sitemap.yaml`, and the site configuration in the Edge Delivery admin service.
- Runtime context checks: `isAuthor()` (path starts with `/content/`: AEM author, no query index), `isEditing()` (Universal Editor edit mode, `adobe-ue-edit`), `isInEditor(el)` (`utils/environment.js`); `localhost` skips the `fonts-loaded` flag.
- `.hlxignore` keeps dot-folders, `*.md`, `package.json`, `package-lock.json`, `test/*`, `_*` (model partials), `docs/*`, `LICENSE` and `karma.config.js` off the code bus; other repository folders are served. Block libraries and Site Global Templates are excluded from publishing in `paths.json`.
- Author access, Admin API roles and whether preview / live hosts are access-restricted are not recorded in the repository (handover 2.7.3: To be confirmed).

## 7. Storage and cookies

| Key | Storage | Written by | Content |
|---|---|---|---|
| `header-height-v2-{mobile\|tablet\|desktop}` | `localStorage` | `utils/header-space.js` | The loaded header height in rem, so the next page reserves it (no layout shift) |
| `header-overlay-v1` | `localStorage` | `utils/header-space.js` `setHeaderOverlay` (from the TCS Header) | The nav page's Header over the first block mode (`always` / `never`), so the next page lays out for it; no personal data |
| `fonts-loaded` | `sessionStorage` | `scripts/scripts.js` `loadFonts` (not on `localhost`) | `true` once `fonts.css` has loaded |
| `tcs-inherited-metadata` | `sessionStorage` | `utils/inherited-metadata.js` | The Navigation, Footer and 404 Page values of parent pages, by URL |

All storage access is wrapped in `try / catch` (private mode works). None of these values identify a visitor. Project code sets no cookies (`document.cookie` is not used), and `scripts/aem.js` (RUM) sets none. Third-party players (X widget, generic iframes, YouTube and Vimeo once played) may set their own cookies. The data layer code writes no storage.

## 8. Query strings and debug flags

| Parameter | Read or written by | Effect |
|---|---|---|
| `?q=` | Written by the Conversation Canvas when a Search page is set | The raw (unredacted) query in the Search page URL; no code in the repository reads it (re-audit R-D-12: blocked until Tags exists) |
| `?debug=list`, `?debug=separator` | `blocks/list/list.js`, `blocks/separator/separator.js` | `console.debug` of the block's configuration and data |
| `?rum=`, `?optel=` | `scripts/aem.js` | RUM sampling rate (`on`, `off`, `high`, `low`) |
| `?lighthouse=on` | `scripts/aem.js` | Sets `window.hlx.lighthouse` |

Other query strings pass through links unchanged (`toSitePath` keeps `search` and `hash`), and link URLs with their query strings are pushed to the data layer as `xdm:linkURL`.

## 9. Logging and debugging safeguards

- `console.error` only for failures (template, auto-blocking, skeleton render, motion, Lottie, Canvas navigation); `console.warn` in the editor rich text support; `console.debug` only behind the `?debug=` flags above. No visitor input is logged. No remote logging exists besides RUM.
- Editor-only hints (empty Fragment, Form without Submit action, list placeholders) render only in Universal Editor (`isEditing()` / `isAuthor()`).
- Analytics never breaks the page: data layer pushes are wrapped in `try / catch` (`utils/analytics.js`).

## 10. Data minimisation and personal information

| Data | Handling | Code |
|---|---|---|
| Search terms | Redacted before the data layer: e-mail addresses → `[email]`, phone-like numbers → `[number]`, trimmed, at most 100 characters; empty searches not pushed (unit tested) | `utils/privacy.js`, `test/privacy/privacy.test.mjs` |
| Form values | Never in the data layer (`cmp:submit` carries field names only); sent only to the page integration or the authored Submit action | `decorators/data-layer-decorator.js`, `blocks/form-variations/form-variations.js` |
| Labels | `dc:title` values truncated to 100 characters | `decorators/data-layer-decorator.js` |
| `mailto:` / `tel:` links | Pushed in full as `xdm:linkURL` (exit links): open finding F-05 | `decorators/data-layer-decorator.js` |
| Authored contact data | The Authors and Media Contacts sheets (names, designations, e-mail, phone, LinkedIn) are published as `/{locale}/data/*.json` by design and excluded from the query index (`/*/data/**`) | `utils/data.js`, `helix-query.yaml` |
| Authored ids and labels | Anchor IDs, Component IDs and Analytics labels are authored and pushed to the data layer: authors must not put personal data in them (handover 11.9) | — |
| 404 RUM checkpoint | `sampleRUM('404', { source: document.referrer })` sends the referrer of sampled 404 views | `404.html` |

## 11. Identity, consent and analytics / personalization dependencies

- **Identity**: no login, no gated content, no visitor id read or set by project code.
- **Consent**: no consent banner, consent management platform or consent check is implemented. The data layer is filled without consent because nothing leaves the page; third-party embeds load without a consent gate. The consent model, categories and per-locale requirements are blocked dependencies (handover 11.7.6 and open item O-07; audit section 7).
- **Analytics**: Adobe Tags and the AEP Web SDK are not configured (blocked: measurement plan); a future embed belongs in `scripts/delayed.js`, never in `head.html` or Universal Editor (Data layer and analytics).
- **Personalization**: no Adobe Target or AJO integration (Personalization).

## 12. Third-party dependencies

| Dependency | Kind | Version | Notes |
|---|---|---|---|
| DOMPurify | Vendored runtime script | 3.2.4 (Apache 2.0 / MPL 2.0, header in the file) | `scripts/dompurify.min.js`; Universal Editor only |
| Lottie (lottie_light) | Vendored runtime script | 5.13.0 (MIT, `LICENSE.md`) | `scripts/lib/lottie/`; loaded by `utils/lottie.js` when an animation renders |

| Dependency | Kind | Version | Notes |
|---|---|---|---|
| npm packages | Development only | See `package.json` devDependencies (ESLint, Stylelint, husky, merge-json-cli, npm-run-all, eslint-plugin-xwalk) | No runtime npm dependencies |
| Renovate | Dependency updates | `.renovaterc.json` | `config:recommended`; dev dependencies automerge; ESLint pinned to v8 |
| CI | GitHub Actions | `.github/workflows/main.yaml` | `npm ci` and `npm run lint` on push (no `npm test`, no dependency audit step) |

Third-party services loaded at runtime: the X widget script, the YouTube / Vimeo players and generic iframe hosts (only when authored), and the RUM collector.

## 13. Known limitations and open items

| Item | Status | Source |
|---|---|---|
| Generic iframes: no host allow-list, no `sandbox` | Known limitation; approved embed domains not recorded | `decorators/media-decorator.js`, Embed README, handover O-16 |
| Third-party players and the X widget load without consent | Blocked: consent model | Audit section 7 |
| Form Submit action accepts any URL; no form back end defined | Requires validation (handover O-08: form back end and privacy notices) | `blocks/form-variations/form-variations.js` |
| `mailto:` / `tel:` links in the data layer | Open (F-05) | Handover Part 16 |
| Raw query in the Search page URL (?q=) | Blocked until Tags / Web SDK exists (R-D-12) | `docs/audit/REAUDIT-2026-10-20.md` |
| CSP leaves `frame-src`, `connect-src`, `img-src`, `style-src` unrestricted | By design for embeds; Requires validation by a security review | `head.html` |
| Preview / live access restriction, Admin API roles | Not recorded | Handover 2.7.3 |

## 14. Related

- Architecture, Blocks, Indexing and search, Decision log
- Data layer and analytics
- Personalization
- `AGENTS.md` (Security Considerations, Accessibility)
- `utils/README.md`, `decorators/README.md`
- `blocks/embed/README.md`, `blocks/video/README.md`, `blocks/form-variations/README.md`
- `docs/audit/BLOCK-ECOSYSTEM-AUDIT.md` (section 7), `docs/audit/REAUDIT-2026-10-20.md`
- Handover Part 2 — Solution Architecture (2.7), Part 11 — Analytics & Data Layer (11.7.6, 11.9), Part 13 — Global Troubleshooting Guide (13.23, 13.24), Part 16 — Handover Findings and Open Items Register (`TCS-CORP-HANDOVER.md`)

## 15. Screenshots

No screenshot belongs to this document (it describes code, configuration or process). Evidence screenshots are listed in the screenshot register.

## 16. Known limitations

- None recorded for this document.

## 17. Open items

- None.

## 18. Change history

| Version | Date | Author | Change |
|---|---|---|---|
| 1.0 | 8 November 2026 | Delivery team (documentation program) | First version of the role-specific documentation (master documentation prompt), generated from the repository and the published reference pages |

# Solution Architecture

| Attribute | Value |
|---|---|
| Document ID | ARC-001 |
| Audience | Solution architects, technical leads, developers, AI agents |
| Purpose | Business and technical context, scope, technology stack and constraints of the solution |
| Scope | — |
| Prerequisites | — |
| Owner | Solution Architect (proposed) |
| Approver | Product Owner (to be confirmed) |
| Version | 1.0 |
| Status | Draft for review |
| Last updated | 8 November 2026 |
| Last validated | 8 November 2026 |
| Source files | docs/handover/knowledge-base/project-overview.md, AGENTS.md |
| Related documents | — |
| Related AEM paths | /content/tcs-corp/language-masters/en |

This page gives anyone joining the TCS Corporate Website project, human or AI agent, a verified summary of what the project is: its objectives and business value, who it serves, the brand, the 13 locale sites and the temporary language master review, the site structure, how content and code are delivered, the technology stack, what is in and out of scope, and the assumptions, dependencies and constraints it works under. It summarizes and links the detailed sources (the handover Part 1, `AGENTS.md`, `utils/site.js`) rather than copying them.

## 1. Project at a Glance

| Attribute | Value |
|---|---|
| Project name | TCS Corporate Website (code name `tcs-corp`) |
| Purpose | The multi-locale corporate web presence of TCS (Tata Consultancy Services) |
| Platform | Adobe Experience Manager (AEM) as a Cloud Service with Edge Delivery Services |
| Authoring | Universal Editor (AEM authoring for Edge Delivery Services, "xwalk") |
| Content source | The AEM as a Cloud Service author, through the xwalk delivery bridge configured in fstab.yaml (type `markup`, suffix `.html`) |
| Code repository | GitHub `sradha-tcs/tcs-corp`, default branch `main` |
| Edge Delivery org / site | `sradha-tcs` / `tcs-corp` |
| Boilerplate | `adobe-rnd/aem-boilerplate-xwalk` (`package.json` still carries the boilerplate name `@adobe/aem-boilerplate` and version `1.3.0`) |
| Sites | 13 locale sites, plus the language master published temporarily for review |
| Hosts in use | `main--tcs-corp--sradha-tcs.aem.page` (preview) and `main--tcs-corp--sradha-tcs.aem.live` (live) |
| Production domain and CDN | Not recorded (handover open item O-03, "To be confirmed") |

| Attribute | Value |
|---|---|
| Handover | TCS-CORP-HANDOVER.md version 2.37 (PDF: TCS-CORP-HANDOVER.pdf), status "Draft for review"; approvals pending |

## 2. Business and Program Objectives

The handover states that its objectives are derived from the delivered implementation; business KPIs and target values are not recorded in the repository (handover 1.2, BO-9 and SC-9 "To be confirmed").

| Id | Objective | How the implementation supports it |
|---|---|---|
| BO-1 | One corporate web platform for all markets | 13 locale sites served by one code base; locale-aware navigation, footer, error pages, UI text, taxonomy and data (utils/site.js) |
| BO-2 | Fast, search-friendly pages | Three-phase loading, image optimisation, self-hosted Inter variable font, module preloading, PageSpeed target of 100 |
| BO-3 | Autonomous, efficient authoring | Universal Editor component models for every block, the Additional tab, Site Global Templates, a block library per locale |
| BO-4 | Consistent brand experience | TCS.com Design System (Figma) patterns implemented as blocks; design tokens in styles/styles.css |
| BO-5 | Content discovery | List, Filter List, Cards and Conversation Canvas over the query index and taxonomy |
| BO-6 | Measurable digital experience | Adobe Client Data Layer for every page and block (decorators/data-layer-decorator.js) |
| BO-7 | Inclusive experience | WCAG 2.1 AA target in blocks and shared decorators |
| BO-8 | Low operational overhead | Serverless delivery, Git-based deployment through AEM Code Sync, no build pipeline |
| BO-9 | Business KPIs | Not recorded |

Program goals (handover 1.3), with their state on `main` at commit 2d06316:

| Id | Goal | State |
|---|---|---|
| PG-1 | Launch on Edge Delivery Services with Universal Editor authoring | Running on the `aem.page` / `aem.live` hosts; production domain not recorded |
| PG-2 | Block library covering the TCS.com Design System patterns | 30 block folders documented in blocks/README.md; Interactive Dashboard is a placeholder (Experimental) |
| PG-3 | 13 locale sites with their own references, error pages, sheets and data | Implemented in code and `paths.json`; content status per locale not recorded |
| PG-4 | Templates for recurring page types | Home, Article, Blog, Press Release (templates/README.md) |
| PG-5 | Analytics readiness | Data layer implemented; Adobe Tags, Web SDK, Analytics, CJA, Target and AJO not configured |
| PG-6 | Accessibility (WCAG 2.1 AA) | Implemented in blocks and decorators; a formal third-party audit is not recorded |
| PG-7 | Performance (Lighthouse / PageSpeed 100) | Handover revision 2.36 records Lighthouse 99–100 mobile and 100 desktop on the measured pages; no PageSpeed check runs in CI |
| PG-8 | Handover to the run organisation (AMS) | The handover document; AMS acceptance not recorded |

## 3. Business Value

Summarized from handover 1.5 (Business Benefits); each benefit is traceable to code:

- **Performance and SEO**: pages are rendered by Edge Delivery Services from authored HTML and served from its CDN; only the code of the blocks on the page loads; the first section is rendered first for LCP (`scripts/scripts.js`, `scripts/aem.js`).
- **Faster time to market**: authors build and publish pages from ready blocks; code ships by merging to `main` (AEM Code Sync), without a build or deployment window.
- **Lower cost of ownership**: no delivery application servers, no front-end framework or build chain, one code base for 13 locales.
- **Global consistency with local autonomy**: shared design and blocks; each locale owns its navigation, footer, 404 page, UI text, taxonomy, metadata, redirects and data sheets.
- **Data-driven optimisation, once tags are deployed**: every block and interaction is described in the Adobe Client Data Layer, so Adobe Tags rules could be configured without block changes.
- **Inclusive reach and reuse**: accessibility built into shared decorators; templates add structure (Page Header, About the author, Media contacts) from metadata and sheets.

**Why Edge Delivery Services, xwalk and Universal Editor.** No verified selection rationale was identified: the handover records the platform as given (Document Control, 1.1) and lists its benefits (1.5), but does not state why it was chosen over alternatives. The handover does state that authors use the Universal Editor and that documents (Google Drive / SharePoint) are not a content source (assumption 5 in 1.8).

## 4. Target Users

| Group | What they do with the solution | Source |
|---|---|---|
| Site visitors | Browse the 13 locale sites | Handover 2.2 (Solution Context) |
| Search engines | Crawl the pages and `/sitemap.xml` | Handover 2.2, helix-sitemap.yaml |
| Content authors (13 locale teams) | Create, edit, preview and publish pages and sheets in Universal Editor | Handover 2.2, Part 5 |
| Developers | Change code through pull requests, preview on feature-branch hosts | Handover 2.2, AGENTS.md |
| Site and AEM platform administrators | Edge Delivery site configuration, Admin API, AEM users and groups | Handover 2.2, Part 14 |
| AMS / support | Run support, troubleshooting, monitoring | Handover Parts 13, 14, 17 |

Visitor segments or personas are not defined: a `persona` metadata column is indexed (helix-query.yaml) and an optional `tcs-corp:audience` tag tree (`cxo`, `it-leaders`, `investors`, `analysts`, `job-seekers`, `media`) is only **proposed** in handover 7.3.2; the published namespace has only `content-type`, `industry` (empty) and `region` (handover 7.3.1).

## 5. Brands

| Brand element | Where it appears | Source |
|---|---|---|
| TCS (Tata Consultancy Services) | The site's brand: the TCS logo in the header brand area, linked to its authored TCS Logo Link (usually the locale Home); accessible name from its alt text, else the `tcsLogo` placeholder ("Tata Consultancy Services") | blocks/tcs-header/tcs-header.js, blocks/tcs-header/README.md |
| TATA Group | The TATA logo in the header tools area (TATA Logo Link, usually `https://www.tata.com/`); accessible name from its alt text, else the `tataLogo` placeholder ("TATA Group") | Same |

| Brand element | Where it appears | Source |
|---|---|---|
| TCS.com Design System | The Figma design the blocks implement ("New Screens from New DS" in the handover revisions) | Handover 1.2 (BO-4), revision history 2.2–2.6; the Figma files are not in the repository |

No multi-brand or sub-brand mechanism exists in the code: one brand, one design token set (styles/styles.css) and one header for all locales.

## 6. Locales

One code base serves every locale; the locale is resolved from the AEM path when editing in Universal Editor, else from the published path (`getLocale` in utils/site.js). Pages outside a locale (Site Global Templates, drafts) use the default locale `en`. paths.json is generated from `LOCALES` by `npm run build:paths` (tools/paths/generate-paths.mjs).

| # | AEM path (below /content/tcs-corp) | Published at | <html lang> |
|---|---|---|---|
| 1 | us/en | /en | en-US |
| 2 | de/de-de | /de-de | de-DE |
| 3 | fr/fr-fr | /fr-fr | fr-FR |
| 4 | ca/fr-ca | /fr-ca | fr-CA |
| 5 | es/es-es | /es-es | es-ES |
| 6 | pt/pt-pt | /pt-pt | pt-PT |
| 7 | nl/nl-nl | /nl-nl | nl-NL |
| 8 | se/sv-se | /sv-se | sv-SE |
| 9 | dk/da-dk | /da-dk | da-DK |
| 10 | no/no-no | /no-no | no-NO |
| 11 | cn/zh-cn | /zh-cn | zh-CN |
| 12 | jp/ja-jp | /ja-jp | ja-JP |
| 13 | kr/ko-kr | /ko-kr | ko-KR |

**Review roots (KB-D-068).** Besides the 13 locales there are three review roots, each published as a site of its own (KB-D-068): `/language-master/en` (testing page functionality with dummy content, content limitations), `/language-masters/en` (the language master: the migrated tcs.com content, validated with its real sheet data, later the source of the global site and the regions and languages; MSM or not is open until the content is final) and `/design-block-master/en` (the block library and the curated design pages); each is published at its own path (`/language-master/en/`, `/language-masters/en/`, `/design-block-master/en/`) as a site of its own. The language master `/content/tcs-corp/language-masters/en`, the rollout source of `/en`, is reviewed this way. Its pages list the language master's own pages from the query index, read its own placeholders, redirects, taxonomy and data sheets and its own references and error pages, and inherit Navigation, Footer and 404 Page up to its root page (`REVIEW_ROOTS`, `REVIEW_SHEETS`, `REVIEW_PAGES`, `indexRoot`, `localeSheetPath`, `localePagePath` in utils/site.js; the `REVIEW` block of tools/paths/generate-paths.mjs: 11 of the 154 mappings and one include). It is to be removed after the rollout (handover 3.9, finding F-88); the rollout date is not recorded. Test: test/site/index-root.test.mjs.

Content status per locale and the launch scope per locale are not recorded (handover PG-3, SC-1 "To be confirmed"). There is no language switcher component and no hreflang (handover 1.6.2).

## 7. Site Structure

Every locale site has the same layout (details: AGENTS.md, handover Part 3 — Site Structure):

```
/content/tcs-corp/                  site root (not published; "/" needs a redirect to /en/)
├── us/en/                          a locale site; the locale page itself is Home (/en/)
│   ├── …                           content pages
│   ├── references/                 nav (header), footer and other fragments
│   ├── error-pages/                404 (used by 404.html); 403 / 500 for a CDN
│   ├── block-library/              every block variation; not published
│   ├── placeholders, metadata, taxonomy, redirects   sheets / taxonomy page
│   └── data/                       authors, locations, partners, social-links, media-contacts
├── de/de-de/ … kr/ko-kr/           the 12 other locale sites, same layout
├── language-masters/en/             published temporarily at /language-masters/en/ for review
└── site-global-templates/          template pages authors copy; not published
/content/dam/tcs-corp/              AEM Assets (included in paths.json, published with pages)
```

Header, footer and the 404 page are inherited like page properties: the page's own value, else the nearest parent page's up to the locale Home, else the locale's `references/nav`, `references/footer` or `error-pages/404` (utils/inherited-metadata.js). One query index (`/query-index.json`) and one sitemap (`/sitemap.xml`) serve all locales (helix-query.yaml, helix-sitemap.yaml).

## 8. Delivery Model

Authoring, code and delivery are separate (handover 2.1, 2.6):




| Environment | Host | Content | Code |
|---|---|---|---|
| Local development | `http://localhost:3000` (`aem up`) | Previewed content, or `drafts/` with `--html-folder drafts` | Local working copy |
| Feature preview | `https://{branch}--tcs-corp--sradha-tcs.aem.page/` | Previewed content | Feature branch |
| Preview (main) | `https://main--tcs-corp--sradha-tcs.aem.page/` | Previewed content | `main` |
| Live (main) | `https://main--tcs-corp--sradha-tcs.aem.live/` | Published content | `main` |
| Production | Not recorded (open item O-03) | Published content | `main` |

Code is trunk-based: short-lived branches merged into `main` through pull requests; Code Sync deploys every push (handover 12.2, 12.3; AGENTS.md). CI (.github/workflows/main.yaml) runs `npm ci` and `npm run lint` on every push; unit tests and PageSpeed checks are not run by CI. Content is previewed and published from AEM. The only Git tags (`home-content-package_*`) mark earlier states of the `home-content-package` downloads branch (deliverables such as the handover PDF), not code releases. Branch protection, approval workflows and SLAs are not recorded (open items O-05, O-06, O-13).

## 9. Technology Stack

| Layer | Technology | Source |
|---|---|---|
| Authoring | AEM as a Cloud Service, Universal Editor, component definitions / models / filters | component-definition.json, component-models.json, component-filters.json, `blocks/*/_*.json`, `models/_*.json` |
| Delivery | Edge Delivery Services (aem.page / aem.live), path mapping, query index, sitemap | fstab.yaml, paths.json, helix-query.yaml, helix-sitemap.yaml |

| Layer | Technology | Source |
|---|---|---|
| Front end | Vanilla JavaScript (ES modules), CSS3, semantic HTML5; no framework, no transpiling, no build step | scripts/scripts.js, styles/styles.css |
| Core library | `scripts/aem.js` (never modified), including `sampleRUM` (RUM) | scripts/aem.js |
| Vendored libraries | DOMPurify 3.2.4 (editor updates), Lottie light player 5.13.0 (Lotto animations) | scripts/dompurify.min.js, scripts/lib/lottie/ |
| Fonts | Self-hosted Inter variable font (Latin `unicode-range`), scaled fallback faces | styles/fonts.css |
| Analytics | Adobe Client Data Layer, populated centrally; Adobe Tags and the Web SDK not configured | decorators/data-layer-decorator.js, utils/analytics.js |
| Tooling | Node.js (README: 20 or newer; CI: 24), npm, AEM CLI, ESLint 8 (Airbnb, `eslint-plugin-xwalk`), Stylelint (standard), Husky, Renovate, Node test runner | package.json, .eslintrc.js, .renovaterc.json |

No runtime npm dependencies exist; only dev dependencies (package.json).

## 10. Scope

**In scope (implemented on `main`).** 13 locale sites and the language master review; 30 block folders (blocks/README.md: 28 Active, Columns Compatibility-only, Interactive Dashboard Experimental); default content through AEM's built-in Text, Title and Image (the built-in Button is "Button (basic)", no longer offered); three section containers (Columns Container, Tabs Container, Gen Carousel); four templates; the Additional tab, animations and skeletons; the data layer; placeholders, metadata, taxonomy, redirects and data sheets per locale; SEO head mapping (decorators/seo-decorator.js); the 404 page with inherited error pages (404.html); WCAG 2.1 AA practices.

**Out of scope or not implemented** (handover 1.6.2 and 16.4, checked against the code):

| Item | State |
|---|---|
| Adobe Tags, AEP Web SDK, Adobe Analytics, CJA | Not configured; scripts/delayed.js holds only a comment (blocked: measurement plan) |
| Adobe Target, Adobe Journey Optimizer | No integration in the repository |
| Consent management | Not implemented |
| Form submission back end | Not implemented: Form Variations dispatches `form-variations:submit` (finding F-01) |
| Search service | Not implemented: Conversation Canvas dispatches `conversation-canvas:search` |
| Production CDN and domain | Not configured; the sitemap origin is the `aem.live` host |
| Per-locale redirects at the edge, `/` → `/en/` | Not implemented (the platform applies only a root `/redirects.json`) |
| 403 / 500 / maintenance pages served by the platform | Not implemented (only `404.html`) |
| hreflang, language switcher, breadcrumb, authentication | Not implemented |
| Interactive Dashboard | Placeholder that renders nothing |

## 11. Assumptions

From handover 1.8 (each still "To be confirmed" where marked there): AEM as a Cloud Service stays the single content source (one mount point in `fstab.yaml`); the site configuration's `public.json` paths match the

repository `paths.json`; bulk metadata applies only once all 13 sheets are listed in `metadata.source`; each locale publishes at least `references/nav`, `references/footer` and `error-pages/404`; authors use Universal Editor, not documents; the site is public and stores no personal data itself; a production domain and CDN will be configured before go-live; analytics, personalisation and form processing come later through the data layer and DOM events; translations are managed in AEM; visitors use current evergreen browsers.

## 12. Dependencies

| Dependency | Used for |
|---|---|
| AEM as a Cloud Service author, Universal Editor, AEM Assets, Adobe IMS / Admin Console | Authoring, assets, sign-in |
| Edge Delivery Services (content, media and code bus; Admin API; configuration service) | Rendering and delivery |
| AEM Code Sync (GitHub App), GitHub `sradha-tcs/tcs-corp` | Code deployment, pull requests, CI |
| YouTube, Vimeo, X (Twitter) | Embeds and videos, only where authored |
| Browser Web Speech API | Conversation Canvas speech input, where supported |
| Node.js, npm, AEM CLI | Developer tooling only |
| Production CDN / DNS | Not recorded (O-03) |

Details and impact if unavailable: handover 1.9.

## 13. Constraints

Verified constraints the code and content work under (rules in AGENTS.md):

- **No build step**: modules load as authored, so `head.html` modulepreload links and `scripts/module-graph.js` must be regenerated (`npm run build:modules`) after import changes; `npm test` checks them.
- **`scripts/aem.js` is never modified** (boilerplate rule).
- **AEM delivery rules**: empty rows per dialog tab, item first cells, collapsed field suffixes, no grouped fields in key-value blocks, no `classes_*` on sections (container sections use `container*` fields), template values stored on new components, item names in class-name form, non-image DAM files served only once published.
- **Universal Editor**: at most three short dialog tabs (the last one Additional); `multi: true` on `aem-content` fields isn't saved; author has no query index (List and Filter List show an author message).
- **Platform**: only a root `/redirects.json` is applied; bulk metadata needs the site configuration's `metadata.source`; only `404.html` is served as an error page.
- **CSS**: sizes in rem, mobile-first `min-width` breakpoints at 600 / 900 / 1200 px, selectors scoped to the block.
- **Tooling**: ESLint pinned to v8 until airbnb-base supports v9 (.renovaterc.json).
- **Typography**: the Inter font covers Latin only; Chinese, Japanese and Korean text uses the system sans-serif font.

## 14. Key Terminology

| Term | Short meaning |
|---|---|
| xwalk | AEM authoring (Universal Editor) for Edge Delivery Services |
| Locale site | One of the 13 country / language sites |
| Language master | `/content/tcs-corp/language-masters/en`, temporarily published at `/language-masters/en` |
| Block / section container / item | A decorated component / a section turned into a block / a row added to a block with Add |

| Term | Short meaning |
|---|---|
| Additional tab | The last dialog tab of every block except header and footer |
| Query index | `/query-index.json`, the published pages and their metadata columns |
| Placeholders | UI text per locale, with English fallbacks in code |
| Data layer | `window.adobeDataLayer` (Adobe Client Data Layer) |

Full definitions: glossary.md.

## 15. Related

- repository-structure.md: folders and files, purpose and rules
- glossary.md: project, Adobe and handover terms
- references.md: repository documents and cited Adobe documentation
- architecture.md, blocks.md, content-architecture.md, data-layer-and-analytics.md, design-system.md, personalization.md: the other knowledge base topics
- AGENTS.md: verified project rules
- blocks/README.md: block inventory
- utils/site.js: locales and the language master review
- Handover Part 1 — Project Overview, Part 3 — Site Structure, Part 16 — Handover Findings and Open Items Register: TCS-CORP-HANDOVER.md

## 16. Screenshots

No screenshot belongs to this document (it describes code, configuration or process). Evidence screenshots are listed in the screenshot register.

## 17. Known limitations

- Business objectives are carried from the project overview; they are not validated with the business in this program.

## 18. Open items

- None.

## 19. Change history

| Version | Date | Author | Change |
|---|---|---|---|
| 1.0 | 8 November 2026 | Delivery team (documentation program) | First version of the role-specific documentation (master documentation prompt), generated from the repository and the published reference pages |

# Technical Architecture

| Attribute | Value |
|---|---|
| Document ID | ARC-003 |
| Audience | Solution architects, technical leads, developers, AI agents |
| Purpose | Runtime, block loading, module preloading, templates, authoring and delivery architecture, environments, data and rendering flows, error handling and caching |
| Scope | — |
| Prerequisites | — |
| Owner | Solution Architect (proposed) |
| Approver | Product Owner (to be confirmed) |
| Version | 1.0 |
| Status | Draft for review |
| Last updated | 8 November 2026 |
| Last validated | 8 November 2026 |
| Source files | docs/handover/knowledge-base/architecture.md, scripts/scripts.js, scripts/aem.js, head.html |
| Related documents | — |
| Related AEM paths | — |

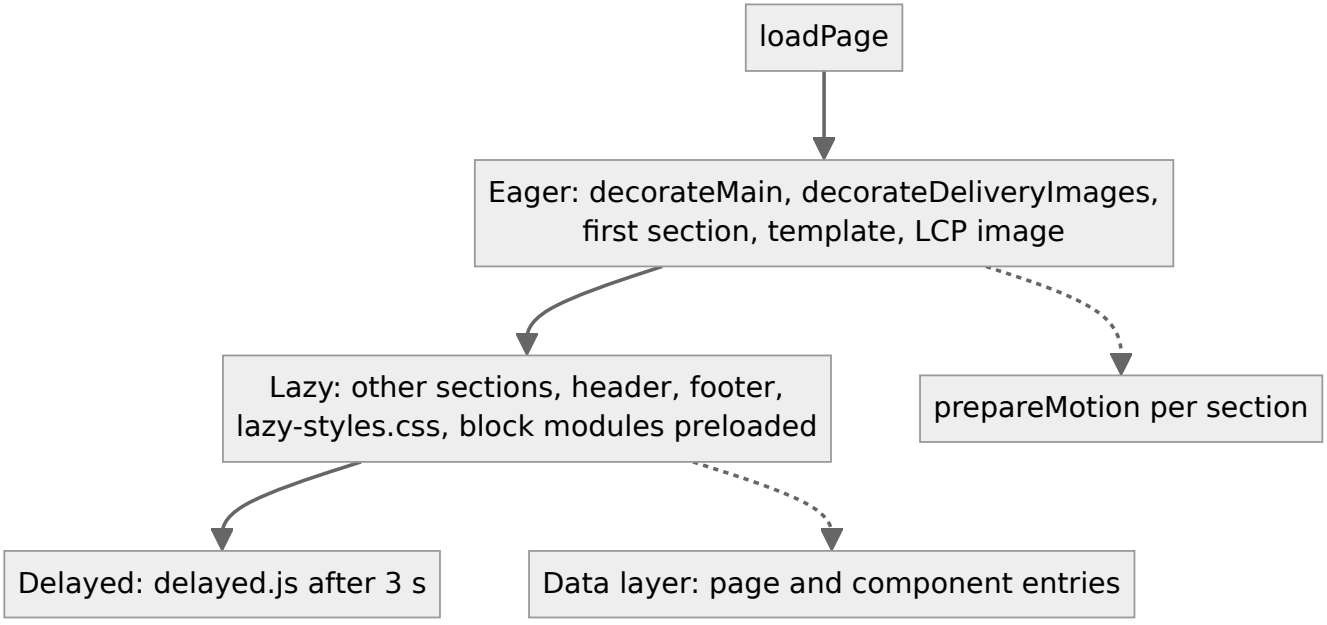



*Diagram: three-phase page loading (source: diagrams/page-loading.mmd).*

## 1. Solution summary

The site separates three concerns (AGENTS.md, handover Part 2.1):

| Concern | System | Repository artefacts |
|---|---|---|
| Authoring | AEM Sites as a Cloud Service author with the Universal Editor (xwalk) | `models/`, `blocks/*/_*.json`, `component-definition.json`, `component-models.json`, `component-filters.json`, `xwalk.json`, `scripts/editor-support.js` |
| Code | GitHub repository `sradha-tcs/tcs-corp` (based on `adobe-rnd/aem-boilerplate-xwalk`), deployed by AEM Code Sync | `blocks/`, `templates/`, `decorators/`, `utils/`, `scripts/`, `styles/`, `head.html`, `404.html` |
| Delivery | Edge Delivery Services (content bus, media bus, code bus; preview `*.aem.page`, live `*.aem.live`) | `fstab.yaml`, `paths.json`, `helix-query.yaml`, `helix-sitemap.yaml`, `.hlxignore` |

Key characteristics verified in the repository:

- **No build step.** Vanilla ES modules and CSS are served as committed; `.hlxignore` keeps dot files, Markdown, `package*.json`, `test/*`, `docs/*`, `_*` model partials, `LICENSE` and `karma.config.js` off the code bus.
- **Client-side decoration.** Edge Delivery delivers semantic HTML; `scripts/scripts.js` decorates sections and blocks and loads each block's JS / CSS on demand.
- **One site, 13 locales by path.** Locales are defined once in `LOCALES` (`utils/site.js`); `paths.json` is generated from them (`npm run build:paths`). See content-architecture.md.
- **Site data from published sheets.** Placeholders, taxonomy, redirects, data sheets and the query index are fetched as JSON by single-purpose utils, cached per page view.
- **Data layer only.** `window.adobeDataLayer` is populated centrally; Adobe Tags and the Web SDK are not configured.

## 2. System context




| Actor / system | Interaction | Status |
|---|---|---|
| Content authors | Edit, preview, publish pages and sheets in AEM with the Universal Editor | Active |
| Developers | Change code through pull requests; feature-branch preview hosts | Active |
| Site administrator | Site configuration (`public.json` paths, `metadata.source`), Admin API operations | Active (roles: Not recorded) |
| Visitors | Browse the 13 locale sites | Active |
| Search engines | Crawl pages and `/sitemap.xml` | Active (sitemap origin is the `main` live host until a production domain exists) |
| Production CDN / domain | Production hostname, redirects, 403 / 500 pages | Requires validation (not configured in the repository; handover open item O-03) |

## 3. Application architecture

The code is layered (decorators/README.md, utils/README.md, handover Part 2.3):

| Layer | Location | Responsibility |
|---|---|---|
| Global head | `head.html` | CSP, viewport, `aem.js`, `scripts.js`, `styles.css`, generated `modulepreload` links |

| Layer | Location | Responsibility |
|---|---|---|
| Core library | `scripts/aem.js` | Edge Delivery boilerplate library: sections, blocks, `loadBlock`, `loadCSS`, `loadScript`, `getMetadata`, `sampleRUM`. **Never modified** (AGENTS.md) |
| Page lifecycle | `scripts/scripts.js` | `loadEager` / `loadLazy` / `loadDelayed`, `decorateMain`, section options and container sections, templates, 404 loading, preloading, prefetch |
| Templates | `templates/{home,article,blog,press-release}/` | Structure and styles per Page Type (templates/README.md) |
| Blocks | `blocks/{name}/` | `{name}.js`, `{name}.css`, `_{name}.json` (models); inventory in blocks/README.md |
| Decorators | `decorators/` | Create or decorate elements (buttons, links, images, media, data layer, SEO, block options, motion, skeletons) |
| Utils | `utils/` | No-UI helpers (site / locale, metadata, placeholders, taxonomy, data sheets, query index, redirects, ids, environment) |
| Styles | `styles/` | `styles.css` (tokens, LCP), `lazy-styles.css`, `fonts.css`, `card-list.css`, `motion.css` |
| Editor support | `scripts/editor-support.js`, `editor-support-rte.js`, `dompurify.min.js` | Universal Editor live updates (editor context only) |

Layering rules (AGENTS.md "Shared code: decorators and utils"): blocks and templates import decorators and utils; decorators may import utils; **utils never import decorators**; one purpose per file; shared behaviour goes into these folders, not into a block.

# 4. Runtime architecture

### Three-phase loading

`loadPage()` in `scripts/scripts.js` runs `loadEager`, then `loadLazy`, then `loadDelayed`.

| Phase | Steps (in code order) |
|---|---|
| Eager (`loadEager`) | `decorateHead()` (SEO head tags and `<html lang>`, `decorators/seo-decorator.js`); `trackInteractions()` (page entry and capture listeners, `decorators/data-layer-decorator.js`); `decorateTemplateAndTheme()`; `decoratePageType()` (Page Type body class); on the 404 page `loadErrorPage(main)`, else `loadTemplate(main)`; `decorateMain(main)`; `reserveHeaderSpace()`; `body.appear`; `preloadBlockModules(first section)`; `loadSection(first)` with `prepareMotion` and `waitForFirstImage` (LCP); `fonts.css` only at 900px and wider or when the session flag `fonts-loaded` is set |
| Lazy (`loadLazy`) | `autolinkModals` (links containing `/modals/` open the Modal block); `preloadBlockModules(main)` and of the chrome blocks (`header`, `footer`, `fragment`, `tcs-header`, `conversation-canvas`); `prefetchData(main)`; `loadHeader`; `loadFooter` (hidden until the sections have loaded); `loadMainSections` (each section with `prepareMotion`, RUM enhancement after the first); footer shown; `trackBlocks(main)`; `addErrorBackButton` on 404 pages; scroll to the URL hash; `lazy-styles.css`; `loadFonts()` |
| Delayed (`loadDelayed`) | After 3 seconds: `trackBlocks(document)` (header, footer, late blocks such as modals) and `import('./delayed.js')`. `scripts/delayed.js` contains only a comment (`// add delayed functionality here`) |

`decorateMain(main)` runs `decorateIcons` → `buildAutoBlocks` (an empty placeholder) → `decorateSectionsWithOptions` (section options, then the container sections Gen Carousel, Tabs Container and Columns Container wrapped in synthetic blocks) → `decorateBlocks` → `applyBlockOptions` (Additional tab) → `decorateButtons`.

`prefetchData(main)` starts the query index (`getPages`) and the taxonomy (`loadTaxonomy`) at the lazy phase only when the page has a `.list`, `.filter-list` or `.conversation-canvas`, and the redirects sheet (`loadRedirects`) only

with a `.list`. It isn't run in the eager phase so it doesn't compete with the LCP image (code comment). See indexing-and-search.md.

### Block loading

From `scripts/aem.js`:

- `decorateBlock(block)` adds `.block`, `data-block-name` (the first class), `data-block-status="initialized"`, wraps text nodes and adds `{name}-wrapper` to the parent.
- `loadBlock(block)` sets `loading`, loads `/blocks/{name}/{name}.css` and imports `/blocks/{name}/{name}.js` in parallel, awaits the default export, catches and logs errors (`failed to load module for {name}`, `failed to load block {name}`) and always ends with `data-block-status="loaded"`.
- `loadSection(section, callback)` loads the section's blocks one after another, runs the callback (here `prepareMotion`), then shows the section.

### Module preloading

There is no bundler, so a module's imports are only discovered after it loads (AGENTS.md "Module preloading"). Two generated lists remove these round trips (`npm run build:modules`, `tools/modules/generate-module-graph.mjs`):

- `head.html` has `modulepreload` links for every module `scripts.js` imports (between `<!-- modulepreload -->` markers).
- `scripts/module-graph.js` lists, per block, the modules it imports beyond `scripts.js` and whether it has a stylesheet; `preloadBlockModules()` adds `modulepreload` links (head) and stylesheet `preload` links (body, because `loadCSS` skips a stylesheet already linked in the head).

`npm test` includes `test/modules/module-graph.test.mjs`; AGENTS.md states the tests fail when the lists are out of date.

### Templates

`loadTemplate(main)` reads the Page Type (`page-type` metadata, else `template`) and, when it is one of `TEMPLATES = ['home', 'article', 'blog', 'press-release']`, imports `templates/{name}/{name}.js` and loads its CSS, then calls the default export with `main` before decoration. Failures are logged (`failed to load template {name}`) and the page continues. What each template adds: templates/README.md and content-architecture.md.

## 5. Authoring architecture

| Element | Implementation | Source |
|---|---|---|
| Editor | AEM Universal Editor on the AEM author; pages under `/content/tcs-corp`, assets under `/content/dam/tcs-corp` | handover Part 2.9.1–2.9.2, Part 5 |
| Component palette | `component-definition.json`: groups Default Content (4), Sections (1), Blocks (57 components) | generated by `npm run build:json` from `models/_component-definition.json` and `blocks/*/_*.json` |
| Dialogs | `component-models.json` (fields per model) | generated from `models/_component-models.json` and partials |
| Allowed children | `component-filters.json`: `main` allows `section`, `gen-carousel`, `tabs`, `columns-container`; `section` allows 26 components | generated from `models/_component-filters.json` and partials |
| Page properties | `models/_page.json` (model `page-metadata`): Title, Description, Keywords, Page Type, Navigation, Footer, 404 Page | content-architecture.md |
| Section properties | `models/_section.json`: Section Name, Style, Width Type, Rounded edge, Background Image, Image alt text, Anchor name | `scripts.js` `decorateSectionsWithOptions` |

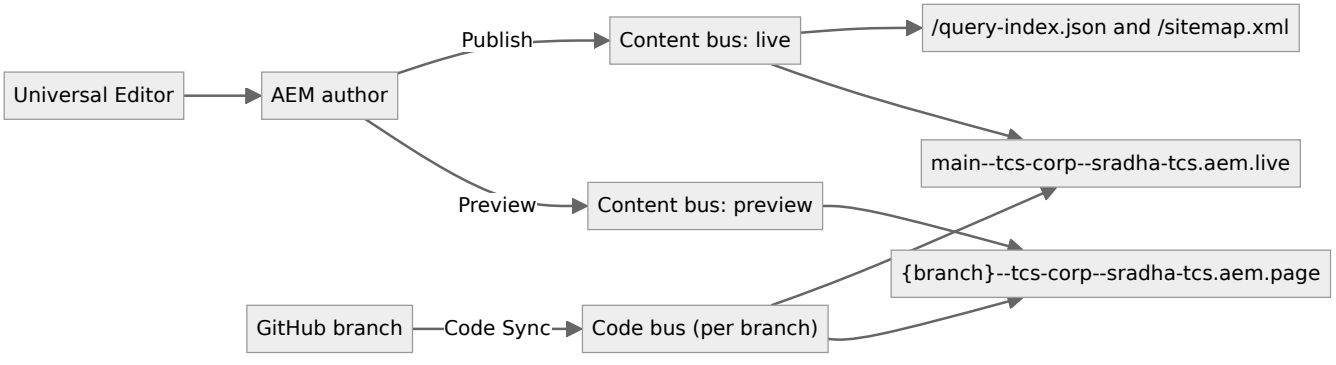
| Element | Implementation | Source |
|---|---|---|
| xwalk features | `xwalk.json`: `public.xwalk.multi-field.enabled: true` | AGENTS.md: repeating content uses item children, not composite multifields |
| Live editing | `scripts/editor-support.js` | below |

`scripts/editor-support.js` listens on `main` for `aue:content-patch`, `-update`, `-add`, `-move`, `-remove` and `-copy`. For each change it loads `scripts/dompurify.min.js`, sanitises the returned HTML (`USE_PROFILES: { html: true }`) and re-decorates only the changed `main`, block, section or default content (`decorateMain`, `decorateBlock`, `applyBlockOptions`, `decorateSectionsWithOptions`, `loadBlock`, `loadSections`); when an update can't be applied in place, the page reloads. `editor-support-rte.js` handles rich-text instrumentation, re-run by a `MutationObserver`. The file isn't referenced by `head.html` or `scripts.js`; handover Part 2.3.6 states it is loaded only in the Universal Editor context (the loading mechanism is outside this repository: Requires validation).

Authoring rules that shape code (block rows, field collapse, item rows, sections without `classes_*`): AGENTS.md "AEM delivery rules", the block READMEs and docs/audit/README.md.

## 6. Publishing and delivery architecture

| Element | What it does | Source |
|---|---|---|
| `fstab.yaml` | One mount point `/` of type `markup`, suffix `.html`, pointing to the AEM author's `bin/franklin.delivery/sradha-tcs/tcs-corp/main` bridge (host not reproduced here) | `fstab.yaml` |
| `paths.json` | 42 mappings (39 for the 13 locales: folder, Home page, taxonomy page each; 3 temporary for the language master under review; sheets are covered by the folder mappings), 15 includes (13 locale pages, `/content/dam/tcs-corp/`, `/content/tcs-corp/language-masters/en`), 28 excludes (13 × 2 block-library entries, `site-global-templates` and its children); generated by `npm run build:paths` | `paths.json`, `tools/paths/generate-paths.mjs` |
| Site configuration | AEM uses the configuration's `public.json` → `paths`, not the repository file; `metadata.source` lists the bulk metadata sheets | AGENTS.md "Notes"; handover Part 2.10.13 |
| Preview / live | Preview writes to the preview partition (`*.aem.page`), publish to the live partition (`*.aem.live`); images go to the media bus (`/media_<hash>`) | handover Part 2.5.2, 2.10.3 |
| Code Sync | Every push of every branch updates the code bus for that branch | handover Part 2.5.1, 2.10.5 |
| CI | `.github/workflows/main.yaml` (Build): `npm ci`, `npm run lint` on every push; unit tests are not run in CI (handover F-37) | `.github/workflows/main.yaml` |




Publishing details (what must be published separately, sheets, rollback): handover Part 12 — Publishing, Deployment & Release.

## 7. Content architecture (summary)

Pages consist of sections holding default content and blocks; page and bulk metadata arrive as `<meta>` tags; site data (placeholders, taxonomy, redirects, data sheets) is published per locale as JSON; header, footer and 404 pages are reference pages resolved by inherited metadata. Full description: content-architecture.md. Query index and listings: indexing-and-search.md.

## 8. Component architecture

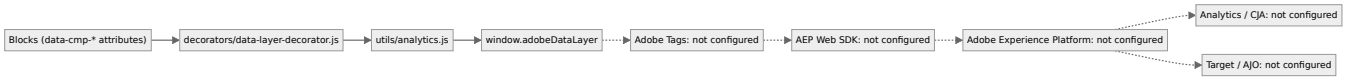
| Kind | Examples | How it runs |
|---|---|---|
| Authored blocks | Hero, Teaser, List, Cards, Filter List, Accordion, Table, Video, Form Variations, Conversation Canvas | `decorateBlocks` + `loadBlock`; inventory and status in blocks/README.md |
| Structural blocks | `header`, `footer` | built by `loadHeader` / `loadFooter` (aem.js) in the lazy phase; content from reference pages (header, footer) |
| Section containers | Gen Carousel, Tabs Container, Columns Container | the section is wrapped in a synthetic block by `buildCarouselSections`, `buildTabsSections`, `buildColumnsSections`; the container fields arrive as section data attributes and `applySectionBlockOptions` turns `container*` fields into block classes |
| Fragments | Fragment block, nav and footer pages, modals, tab fragments | `loadFragment(path, from)` in `blocks/fragment/fragment.js` fetches `{path}.plain.html` and decorates it; recursion stopped by the chain of including fragments (fragment) |
| Additional tab | Anchor ID, Hide block, spacing, animation | `classes_*` fields as block classes; `decorators/block-options-decorator.js`, `decorators/motion-decorator.js`, `styles/styles.css` |
| Shared modules | 28 decorators, 26 utils | decorators/README.md, utils/README.md |

Shared events (AGENTS.md "Shared events and helpers"): `tabs:reveal` (opens the tab holding its target), `header:menu` (header menu state for the Canvas journey), `form-variations:submit` (with `detail.respondWith(promise)`), `conversation-canvas:search` (cancelable search hand-off).

## 9. Integration architecture

| Integration | Mechanism | Status |
|---|---|---|
| AEM author → Edge Delivery | franklin.delivery bridge (`fstab.yaml`), preview / publish | Active |
| GitHub → Edge Delivery | AEM Code Sync | Active |
| AEM Assets | Asset picker; images on the media bus; `/content/dam/tcs-corp/` in `paths.json` includes; non-image DAM files are served only once published (AGENTS.md) | Active |
| AEM Assets, separate environment (EDAM) | Dynamic Media with OpenAPI delivery URLs in the content; standalone image delivery links rendered as responsive pictures (`decorateDeliveryImages`, `decorators/responsive-image-decorator.js`); the Universal Editor Custom Content Advisor for picking (remote-assets-edam.md) | Not adopted (KB-D-064, 8 November 2026): no separate Assets environment, the assets stay in the Sites author's DAM; the rendering of delivery URLs stays in the code (no effect on DAM paths) |
| Query index and sitemap | `helix-query.yaml`, `helix-sitemap.yaml` | Active (indexing-and-search.md) |

| Integration | Mechanism | Status |
|---|---|---|
| Adobe Client Data Layer | `window.adobeDataLayer` populated by `decorators/data-layer-decorator.js` and `utils/analytics.js` (Core Components schema: page, components, `cmp:*` events) | Active |
| Adobe Tags (Data Collection) | Would be embedded from `scripts/delayed.js` | Not configured (blocked: measurement plan) |
| AEP Web SDK | Through Adobe Tags | Not configured (blocked: measurement plan) |
| Adobe Analytics, CJA, Target, AJO | — | No integration exists in the repository |
| Consent | — | No consent mechanism is implemented |
| Form back end | `form-variations:submit` event; no endpoint in code | No verified back end was identified |
| Search service | `conversation-canvas:search` event, else the authored Search page with `?q=` | No verified search service was identified |
| Third-party media | YouTube (incl. `youtube-nocookie.com`), Vimeo, X widgets, only for authored embeds | Active |
| Operational telemetry | `sampleRUM` in `scripts/aem.js`; `404.html` sends a 404 checkpoint | Active |




Details: AGENTS.md "Data layer", handover Part 11 — Analytics & Data Layer.

## 10. Deployment

1. Develop locally with `aem up` (`http://localhost:3000`, local code with previewed content; `--html-folder drafts` for static drafts).
2. Run `npm run lint` and `npm test`; after model changes `npm run build:json`, after import changes `npm run build:modules`, after locale changes `npm run build:paths`.
3. Push a feature branch: Code Sync serves it at `https://{branch}--tcs-corp--sradha-tcs.aem.page/`; the `Build` workflow lints it.
4. Open a pull request with a feature preview URL (AGENTS.md "Publishing Process"); a reviewer merges to `main`.
5. Code Sync updates `main`; preview and live hosts serve the new code (code isn't versioned per content publication).
6. Configuration that lives outside the repository (`public.json` paths, `metadata.source`) is applied by the site administrator.

Rollback: revert the commit on `main` (handover Part 12.6).

## 11. Environment topology

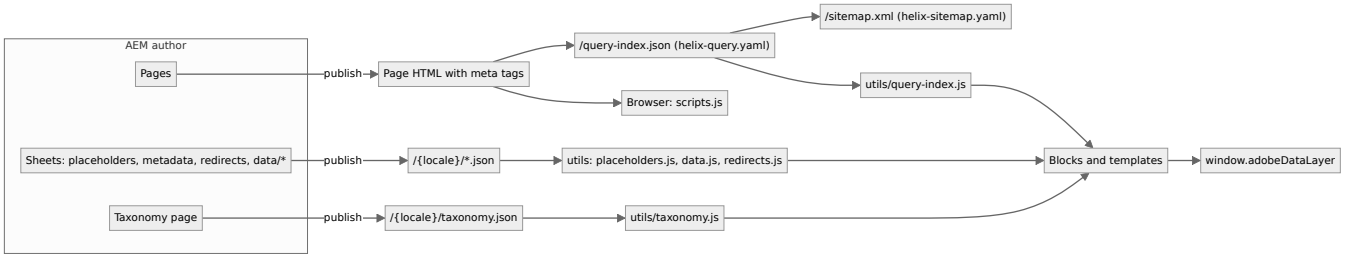
| Environment | Host pattern | Content | Code |
|---|---|---|---|
| Local | `http://localhost:3000` | Previewed content, or `drafts/` | Working copy |

| Environment | Host pattern | Content | Code |
|---|---|---|---|
| Feature preview | `https://{branch}--tcs-corp--sradha-tcs.aem.page/` | Previewed | Feature branch |
| Preview (main) | `https://main--tcs-corp--sradha-tcs.aem.page/` | Previewed | `main` |
| Live (main) | `https://main--tcs-corp--sradha-tcs.aem.live/` | Published | `main` |
| Production domain | Not recorded (not configured; `helix-sitemap.yaml` uses the live host as origin until `cdn.prod.host` is set) | Published | `main` |
| AEM author | Not reproduced (confidential host) | Source content | — |
| Other AEM environments | Not recorded | — | — |

## 12. Security boundaries

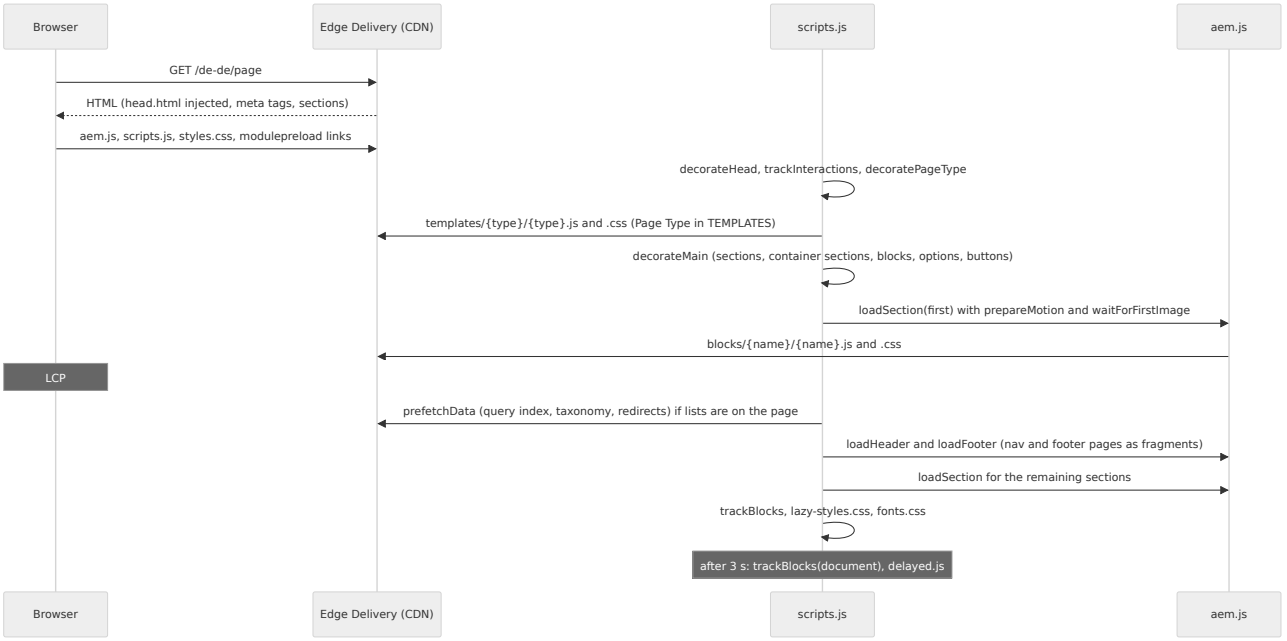
| Boundary | Control in the repository | Source |
|---|---|---|
| Script execution | CSP `script-src 'nonce-aem' 'strict-dynamic' 'unsafe-inline' http: https:; base-uri 'self'; object-src 'none'` with `move-to-http-header="true"` in `head.html` and `404.html`; only nonce scripts and what they load run | `head.html`, `404.html` |
| Not restricted by CSP | `style-src`, `img-src`, `frame-src`, `connect-src` aren't set | `head.html`; handover Part 2.7.1 |
| Public code | Everything not in `.hlxignore` is public on the code bus; no secrets in code (AGENTS.md "Security Considerations") | `.hlxignore` |
| Public data | Every published sheet and `/query-index.json` is public (Authors and Media Contacts sheets hold names, e-mail addresses and phone numbers by design, handover Part 2.10.3) | content-architecture.md |
| Editor updates | Universal Editor HTML is sanitised with DOMPurify before re-rendering; DOMPurify is loaded only by `editor-support.js` | `scripts/editor-support.js` |
| Fragments and error pages | `loadFragment` fetches only paths starting with `/` and not `//`; `loadPageContent` (404) fetches same-origin `.plain.html`; `innerHTML` is used only for those responses | `blocks/fragment/fragment.js`, `scripts/scripts.js` |
| Links | `isSafeUrl` (`utils/card-list-config.js`) allows `http:`, `https:`, `mailto:`, `tel:` and relative paths in List and Cards; new-window links get `rel="noopener noreferrer"` (`decorators/link-decorator.js`) | block READMEs |
| Analytics privacy | `cmp:submit` carries field names only; search terms pass `redactSearchTerm` (`utils/privacy.js`: e-mail and phone patterns replaced, 100 characters) | `decorators/data-layer-decorator.js` |
| Unpublished areas | Block libraries and Site Global Templates are excluded in `paths.json` | `paths.json` |
| Visitor identity | No login, no gated content, no consent mechanism, no cookies set by project code | handover Part 2.7.3 |
| Admin operations | Admin API calls require authorised users; roles are configured outside the repository | Not recorded (handover open item) |
| Preview / live access protection | — | Not recorded |

## 13. Data flow




| Flow | Source → consumer | Detail |
|---|---|---|
| Page metadata | Page properties and bulk metadata → `<meta>` → `getMetadata` / `utils/metadata.js` → SEO decorator, data layer, templates, Page Header | content-architecture.md |
| Inherited metadata | Parent pages' `nav`, `footer`, `error-page` metas → `utils/inherited-metadata.js` → header, footer, 404 | parent heads loaded in parallel, cached in `sessionStorage` |
| UI text | `/{locale}/placeholders.json` → `getPlaceholders()` → every block with UI text | English fallback in code |
| Tag labels | `/{locale}/taxonomy.json` → `resolveTags()` | ID-derived labels as fallback |
| Listings | `/query-index.json` → `getPages()` → List, Filter List, Conversation Canvas menu | indexing-and-search.md |
| Analytics | Blocks' `data-cmp-*` attributes → data layer decorator → `window.adobeDataLayer` | no outbound collection configured |
| Telemetry | `sampleRUM` checkpoints → operational telemetry collector | platform |

## 14. Rendering flow




## 15. Error handling

| Failure | Behaviour | Where |
|---|---|---|
| Block module fails to load or `decorate()` throws | Logged (`failed to load module for {name}`); block marked `loaded`; the section and page continue (block isolation) | `loadBlock`, `scripts/aem.js` |
| Template fails | Logged (`failed to load template {name}`); the page renders without it | `loadTemplate`, `scripts/scripts.js` |
| Missing page | Edge Delivery serves `404.html` (`window.isErrorPage`, `window.errorCode = '404'`); `loadErrorPage` tries the inherited 404 Page, then `/{locale}/error-pages/404`, then `/en/error-pages/404`; a page without content is skipped; else the built-in "404 / Page Not Found / Go home" stays; content from another locale gets its `lang` on `<main>`; "Go back" is added for same-origin referrers | `scripts/scripts.js` `loadErrorPage`, `loadPageContent`, `addErrorBackButton` |
| 403 / 500 | Not served by Edge Delivery; authored `error-pages/403` and `500` exist for a CDN | Requires validation (CDN not configured) |
| Sheet or index request fails | Each util resolves to empty data: placeholders → English fallbacks, taxonomy → ID-derived labels, data sheets → empty lists, redirects → none, query index → no pages | `utils/*.js` (`.catch(() => …)`) |
| Navigation page fails | Falls back to the locale's `references/nav`; none published: header left empty | `blocks/header/header.js` |
| Fragment can't load (unpublished, path not starting with `/`, self-inclusion) | The Fragment block keeps its authored content (the reference link); in Universal Editor edit mode it shows the hint `fragmentEmpty` | `blocks/fragment/fragment.js` |
| Slow data | `renderWhenLoaded` shows a skeleton after 150ms (at least 250ms) and removes it on success or failure | `decorators/skeleton-decorator.js` |
| Universal Editor update can't be applied in place | Page reloads | `scripts/editor-support.js` |
| Analytics push error | Swallowed (`pushToDataLayer` in `try / catch`) | `utils/analytics.js` |
| Storage unavailable | Ignored (`sessionStorage`, `localStorage` in `try / catch`) | `scripts.js`, `utils/inherited-metadata.js`, `utils/header-space.js` |

Troubleshooting by symptom: handover Part 13 — Global Troubleshooting Guide.

## 16. Caching

| Cache | Holds | Lifetime / invalidation | Source |
|---|---|---|---|
| Edge Delivery CDN | HTML, `.plain.html`, sheet JSON, media, code | Purged for the path on preview / publish and on code sync; manual purge with the Admin API cache operation | handover Part 12.1.8 (platform behaviour) |
| Production CDN | Live responses | Not recorded (not configured) | — |
| Query index rows | `/query-index.json`, all locales | One promise per page view (module variable `pages`) | `utils/query-index.js` `getPages` |
| Pages read from HTML | Rows for pages missing from the index (AEM author) | Map per page view | `utils/query-index.js` `getPagesAt` |

| Cache | Holds | Lifetime / invalidation | Source |
|---|---|---|---|
| Placeholders, taxonomy, redirects | Locale sheets | One promise per page view | `utils/placeholders.js`, `utils/taxonomy.js`, `utils/redirects.js` |
| Data sheets | Authors, locations, partners, social links, media contacts | Map per sheet per page view | `utils/data.js` |
| Inherited metadata | `nav`, `footer`, `error-page` of parent pages | In memory per page view and `sessionStorage` key `tcs-inherited-metadata` for the browser session (a changed parent value shows in a new session) | `utils/inherited-metadata.js` |
| Font flag | `fonts-loaded` | `sessionStorage`, browser session; not set on `localhost` | `scripts/scripts.js` |
| Header height | `header-height-v2-{breakpoint}` (rem) | `localStorage`, no expiry; refreshed when the header loads | `utils/header-space.js` |
| Header overlay mode | `header-overlay-v1` (`always` / `never`; removed for Auto) | `localStorage`, no expiry; set when the header's TCS Header loads, applied in the eager phase (`<html data-header-overlay>`) | `utils/header-space.js` |
| Preloads | Module and stylesheet paths already preloaded | Set per page view | `scripts/scripts.js` `preloadedModules` |

## 17. Related

- content-architecture.md — sites, locales, page structure, metadata, sheets, references
- indexing-and-search.md — query index, listings, sitemap, reindexing
- data-layer-and-analytics.md, blocks.md, repository-structure.md, project-overview.md — related knowledge base files
- AGENTS.md — project rules (delivery rules, three-phase loading, module preloading, data layer)
- blocks/README.md, decorators/README.md, utils/README.md, templates/README.md
- docs/audit/README.md, docs/animation/README.md
- Handover (TCS-CORP-HANDOVER.md): Part 2 — Solution Architecture, Part 11 — Analytics & Data Layer, Part 12 — Publishing, Deployment & Release, Part 13 — Global Troubleshooting Guide, Part 16 — Handover Findings and Open Items Register

## 18. Screenshots

No screenshot belongs to this document (it describes code, configuration or process). Evidence screenshots are listed in the screenshot register.

## 19. Known limitations

- None recorded for this document.

## 20. Open items

- None.

## 21. Change history

| Version | Date | Author | Change |
|---|---|---|---|
| 1.0 | 8 November 2026 | Delivery team (documentation program) | First version of the role-specific documentation (master documentation prompt), generated from the repository and the published reference pages |